Power System Fault Location

Impedance Methods and Machine Learning

Academic team project
IEEE 39-bus transmission network

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الملخص

يتناول هذا العمل تحديد موقع الأعطال في شبكات نقل الطاقة الكهربائية بهدف تقليل زمن إعادة الخدمة وآثار الانقطاعات. تقارن الدراسة بين الطرق التقليدية المعتمدة على الممانعة الكهربائية وطرق التعلم الآلي، مثل الشبكات العصبية الاصطناعية، وآلات متجهات الدعم، والغابات العشوائية. كما تعتمد مقاربة هجينة تستعمل التقدير التقليدي كنقطة انطلاق، ثم تصحح الخطأ المتبقي باستعمال نموذج تعلم آلي.

تم توليد المعطيات باستعمال نموذج MATLAB/Simulink لنظام IEEE ذي 39 عقدة لنيو إنجلاند، مع إدماج مصدر طاقة متجددة موصول عبر عاكس. تبين النتائج أن التعلم الآلي يحقق أداء جيدا في المحاكاة، لكن الاعتماد عليه وحده عمليا يحتاج إلى تحقق إضافي. لذلك، يمثل الجمع بين الطرق التقليدية والتعلم الآلي اتجاها أكثر موثوقية، خاصة في الشبكات التي تتضمن مصادر طاقة متجددة.

الكلمات المفتاحية: تحديد موقع الأعطال؛ شبكات القدرة الكهربائية؛ خطوط النقل؛ طرق الممانعة؛ التعلم الآلي؛ الشبكات العصبية الاصطناعية؛ آلات متجهات الدعم؛ الغابات العشوائية؛ الطاقات المتجددة؛ العواكس؛ MATLAB/Simulink.

Résumé

Ce mémoire traite de la localisation des défauts dans les réseaux de transport d'énergie électrique, afin de réduire le temps de rétablissement et les effets des interruptions. Il compare les méthodes conventionnelles fondées sur l'impédance avec des approches d'apprentissage automatique, notamment les réseaux de neurones artificiels, les machines à vecteurs de support et les forêts aléatoires. Une approche hybride est aussi étudiée : l'estimation conventionnelle sert de point de départ, puis un modèle corrige l'erreur résiduelle.

Les données sont générées à partir d'un modèle MATLAB/Simulink du réseau IEEE 39 bus de la Nouvelle-Angleterre, avec une source renouvelable raccordée par onduleur. Les résultats montrent que l'apprentissage automatique offre de bonnes performances en simulation, mais que son utilisation seule reste délicate en pratique. L'approche hybride apparaît donc plus robuste, car elle associe l'interprétabilité des méthodes conventionnelles à la capacité de correction des modèles d'apprentissage automatique.

Mots clés : localisation des défauts; réseaux électriques; lignes de transmission; méthodes d'impédance; apprentissage automatique; réseaux de neurones artificiels; machines à vecteurs de support; forêts aléatoires; énergies renouvelables; onduleurs; MATLAB/Simulink.

Abstract

This thesis investigates fault location in electric transmission networks, with the aim of reducing restoration time and the impact of power interruptions. It compares conventional impedance-based methods with machine-learning approaches, including Artificial Neural Networks, Support Vector Machines, and Random Forests. A hybrid strategy is also studied, where the conventional estimate is used as a starting point and a machine-learning model corrects the remaining error.

The dataset is generated using a MATLAB/Simulink model of the IEEE 39-bus New England system, with an inverter-based renewable source integrated. The results show that machine-learning models can perform well in simulation, but relying on them alone remains difficult for real-world deployment. The hybrid approach therefore offers a more practical direction by combining the interpretability of conventional methods with the correction capability of machine learning.

Keywords: fault location, power systems, transmission lines, impedance-based methods, machine learning, artificial neural networks, support vector machines, random forests, renewable energy integration, inverter, MATLAB/Simulink.

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List of Abbreviations

AC	Alternating Current
ACB	Reverse Phase Sequence
AESC	Avoided Energy Supply Components
AI	Artificial Intelligence
ANN	Artificial Neural Network
CDF	Cumulative Distribution Function
C_{outage}	Estimated Interruption Cost
CSV	Comma Separated Values
DOE	U.S. Department of Energy
EIA	U.S. Energy Information Administration
EIA-930	U.S. Energy Information Administration Form EIA-930
$F0, F1, F2$	Fault-Point Zero-, Positive-, and Negative-Sequence Components
F1	Harmonic Mean of Precision and Recall
FN	False Negative
FP	False Positive
$G0, G1, G2$	Terminal-G Zero-, Positive-, and Negative-Sequence Components
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
$H0, H1, H2$	Terminal-H Zero-, Positive-, and Negative-Sequence Components
IEC	International Electrotechnical Commission
IED	Intelligent Electronic Device
IEEE	Institute of Electrical and Electronics Engineers
ISO	Independent System Operator
kWh	Kilowatt-Hour
LBNL	Lawrence Berkeley National Laboratory
LG	Line-to-Ground Fault
LL	Line to Line Fault
LLG	Double Line-to-Ground Fault
LLL	Three Phase Fault
LLLG	Three Phase-to-Ground Fault
MAE	Mean Absolute Error
MATLAB	Matrix Laboratory
ML	Machine Learning
MLP	Multilayer Perceptron
MSE	Mean Squared Error
MW	Megawatt
MVA	Megavolt-Ampere
OOB	Out-of-Bag
ORNL	Oak Ridge National Laboratory
PCA	Principal Component Analysis
P_{int}	Interrupted Power or Interrupted Load
PI	Proportional Integral
PLL	Phase-Locked Loop
PMU	Phasor Measurement Unit
pu	Per Unit

PSS	Power System Stabilizer
RBF	Radial Basis Function
RF	Random Forest
RFC	Random Forest Classifier
RFR	Random Forest Regressor
RMS	Root Mean Square
RMSE	Root Mean Squared Error
SCADA	Supervisory Control and Data Acquisition
SM	Smart Meter
SLG	Single Line to Ground Fault
SVC	Support Vector Classifier
SVM	Support Vector Machine
SVR	Support Vector Regressor
t	Interruption Duration
$T2$	Terminal-T Negative-Sequence Component
TN	True Negative
TP	True Positive
VoLL	Value of Lost Load
WiMAX	Worldwide Interoperability for Microwave Access

Introduction

The reliability of electric power systems depends strongly on the ability to detect, classify, and locate faults rapidly. When a transmission line is affected by a short circuit, the protection system must isolate the faulted element, but the restoration process still requires a reliable estimate of the fault position. Accurate fault location reduces inspection time, supports maintenance decisions, and helps operators return the network to service with less uncertainty.

Conventional impedance-based methods remain widely used because they are directly linked to the electrical parameters of the line and to measured voltage and current phasors. However, their accuracy may be affected by fault resistance, remote infeed, source impedance, measurement errors, and changes in operating conditions. These limitations motivate the use of data-driven methods that can learn nonlinear relationships from simulated or measured fault data.

This thesis studies both families of approaches. The theoretical part introduces power-system faults, symmetrical components, conventional fault-location techniques, and the main artificial-intelligence methods used in the work. The methodological part then presents the IEEE 39-bus New England system, the MATLAB/Simulink simulation framework, the automated generation of fault cases, the preprocessing of electrical signals, and the validation strategy used for the machine-learning models. The results chapter discusses the performance of the considered methods and compares the conventional and data-driven approaches.

The main objective is to build a coherent simulation and evaluation framework for fault location in transmission systems, while examining how machine learning models can improve estimation accuracy under different fault and operating conditions.

Chapter 1 Power Systems and Fault Location Fundamentals

Summary

This chapter builds the general background needed for the rest of the thesis. It begins with the structure of an electric power system, then introduces power system faults with particular attention to shunt faults, their main categories, and their physical causes. The chapter then explains why fault location matters in practice: it reduces restoration time, supports maintenance and protection analysis, and limits the economic impact of interrupted supply. After that, the main families of fault location methods are introduced, followed by the measurement and communication infrastructure needed to collect useful fault data. The chapter ends by presenting the basic performance indicators used to judge whether a fault location method is accurate and practically useful.

1.1 Power Systems

1.1.1 Introduction

An electric power system comprises an interconnected assembly of generators, transformers, transmission lines, distribution feeders, loads, and associated equipment. Its fundamental purpose is to transport electrical energy from generating stations to end users via the transmission and distribution networks. In modern society, electricity is an indispensable commodity, expected to be available instantaneously for lighting, device charging, and countless daily activities. This expectation, however, is not always fulfilled, particularly when faults occur within the power system [1].

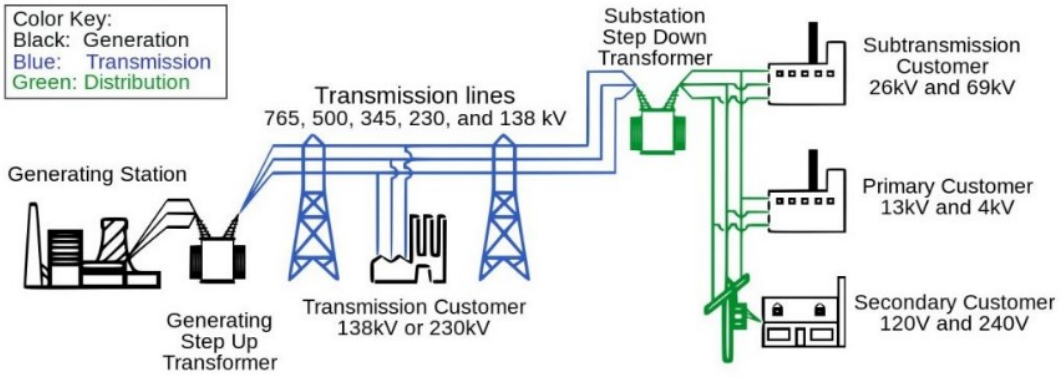


Figure 1: Basic structure of an electric power system [2]

1.2 Power System Faults

Faults in a power system are abnormal operating states that lead to deviations in voltage, current, frequency, and power from their rated values. These disturbances are generally classified into two categories: series faults and shunt faults [1].

1.2.1 Series Faults

Series faults arise from an open circuit in one or two phase conductors under load conditions. Since the open circuit appears in series with the conductor, such disturbances are termed series faults. Common causes include broken jumpers, the failure of one or more circuit-breaker poles to close during manual or automatic operation, and blown fuses, particularly when only one or two high-side fuses operate because of overcurrent. During a series fault, the current in the affected phase decreases as load is lost, while the healthy phases continue to carry load current. Consequently, the phase voltages and currents become unbalanced. Although series faults do not generate high-magnitude fault currents, the resulting negative-sequence currents can impose additional thermal stress on transformers and motors [1].

1.2.2 Shunt Faults

Shunt faults occur when a conductive path is established between one or more phase conductors and ground, or directly between phase conductors. This creates a short circuit condition through an alternative path of lower impedance, resulting in a sharp increase in current on the faulted phases alongside a corresponding drop in voltage.

Shunt faults are typically classified into four types, as shown in Figure 2. A single-line-to-ground fault (or phase-to-ground fault) involves one phase conductor making contact

with ground or grounded equipment. This is the most prevalent type, accounting for 70–80% of all power system faults [3]. A line-to-line fault (or phase-to-phase fault) occurs when two phase conductors come into direct contact. A double-line-to-ground fault (or double-phase-to-ground fault) involves two phases contacting each other while also making contact with ground or grounded equipment. Finally, a three-phase fault involves all three phases, with or without ground involvement. This type is comparatively rare.

Three-phase faults are considered balanced faults because all three phases are symmetrically involved. In contrast, the other three types affect only one or two phases and introduce asymmetry; they are therefore referred to as unbalanced faults [1].

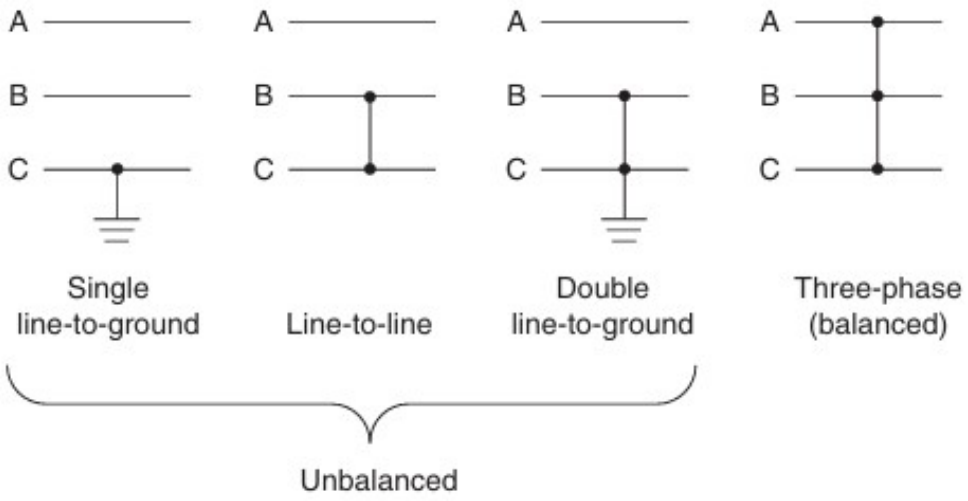


Figure 2: The four types of shunt faults [1]

Shunt faults may be classified as permanent or temporary. Permanent faults involve irreversible damage to power-system equipment and require on-site repairs before the system can be re-energized. Temporary faults, by contrast, arise from transient events such as lightning strikes, animal contact, or wind-borne debris and clear naturally once the fault arc is extinguished. When extinction occurs without protective-device operation, the fault is termed self-clearing. Recurrent temporary faults may indicate deteriorating insulation and may eventually culminate in a permanent fault [4].

Fault arcs can also be interrupted through the coordinated action of relays and circuit breakers. After a deliberate open-time delay that allows deionization of the arc path, the relay issues a reclose command to restore normal operation. The nature of the system strongly influences fault permanence: most faults on underground cables are permanent, whereas those on overhead lines are predominantly temporary [5].

Shunt faults pose considerable thermal risk to power-system components. The thermal energy released during a fault depends on both the magnitude and the duration of the fault current. If this energy exceeds the thermal withstand capability of transformers, motors, or other apparatus, insulation failure and consequent damage may occur. Additionally, high fault currents generate substantial mechanical forces that can fracture or deform equipment. In particular, external through-faults can impose severe mechanical and thermal stress on transformers, which are critical substation assets [6].

Safety implications are equally significant. Fault-induced arcing can ignite forest fires, and faults within oil-filled transformers may provoke fires or explosions, endangering substation personnel. Arc-flash events in switchgear release intense heat, ultraviolet radiation, shrapnel, noise, and blast pressure, creating potentially fatal conditions [7]. From a system-stability perspective, failure to clear a shunt fault before the critical clearing time can precipitate instability and cascading outages [8]. Finally, a shunt fault on one feeder

may cause voltage sags or swells on adjacent healthy feeders. A voltage sag occurs when the root-mean-square voltage falls to between 0.1 and 0.9 per unit for a duration ranging from half a cycle to one minute [1].

1.2.3 Causes of Shunt Faults

Given that this thesis is primarily concerned with shunt faults, the subsequent discussion will be confined to events and conditions that can precipitate such faults.

1.2.3.1 Lightning

Lightning is one of the most frequent natural triggers of shunt faults, particularly on overhead transmission lines. A cloud-to-ground discharge can strike a phase conductor directly or hit a tower or shield wire. In either case, the injected current surge produces an overvoltage. If this overvoltage exceeds the critical flashover voltage of the insulator string, a flashover occurs, creating a temporary phase-to-ground fault. When the strike terminates on a tower, the voltage rise across the tower-footing resistance may cause a backflash across the insulator. Most lightning-induced faults are temporary; the arc self-extinguishes within a few cycles, and the line can be successfully reclosed. Permanent faults arise if the power arc causes significant physical damage to insulators or conductors [3, 4].

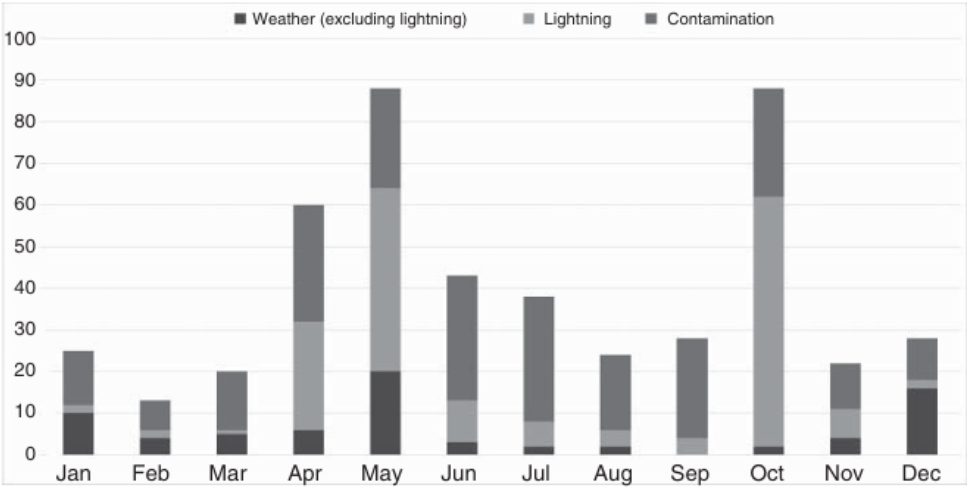


Figure 3: Monthly shunt-fault events on 345 kV Texas lines [1]

The chart illustrates the monthly distribution of disturbances caused by weather (excluding lightning), lightning, and contamination. The highest numbers of incidents occur in May and October, when the total reaches nearly 90 events, mainly because of an increase in lightning-related disturbances. April also shows a high number of incidents, whereas February and March record the lowest values. Weather-related disturbances occur throughout the year, particularly during the summer months, while contamination has a more moderate impact. Overall, the chart indicates a seasonal trend, with lightning being the dominant cause during the peak months.

1.2.3.2 Animal and Bird Activity

Animals and birds are common causes of shunt faults, especially on distribution systems and in substations. Birds may bridge phase conductors or phase-to-ground gaps with their wings or bodies, while their droppings may contaminate insulator surfaces and lead to flashover. Small animals such as squirrels, snakes, and rats can create short circuits by climbing onto energized equipment or chewing through cable insulation. In substations, larger animals may occasionally make simultaneous contact with live parts and grounded

structures. Mitigation measures include animal guards, bird spikes, and periodic insulator washing.

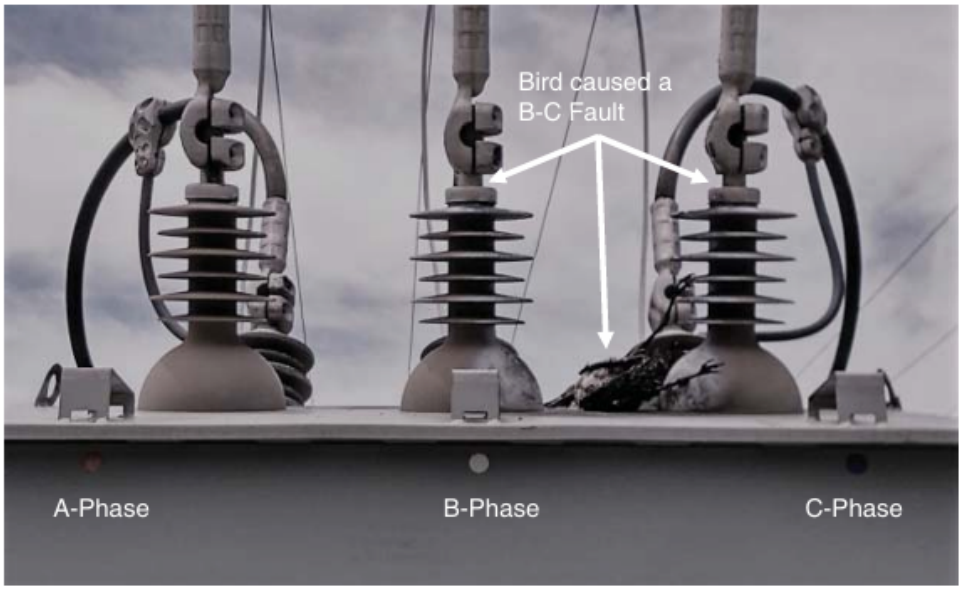


Figure 4: Line-to-line fault on a 22 kV breaker [1]

1.2.3.3 Vegetation and Trees

Contact between power lines and vegetation is a major contributor to shunt faults. Trees or large branches may fall onto lines during storms, overgrown vegetation can bridge phase conductors, and wind-blown limbs can temporarily bring conductors into contact. Line sag under heavy loading may also reduce clearance to underlying trees. A well-known example of the severe consequences of tree-related faults is the 2003 Northeast blackout [9]. Utilities manage this risk through vegetation trimming, increased phase spacing, and tree-resistant conductor configurations.

1.2.3.4 Adverse Weather and Environmental Contamination

Severe weather conditions, including high winds, hurricanes, tornadoes, ice storms, and snow accumulation, can cause conductors to clash, snap, or be brought down by falling debris. Ice and snow loading may also break poles or cause conductors to gallop. In coastal or agricultural areas, airborne contaminants, such as salt spray, dust, pesticides, and fertilizers, accumulate on insulator surfaces. When combined with moisture, this contamination layer becomes conductive and may trigger a flashover at nominal operating voltage. Such outages are often referred to as contamination-induced or pollution-induced faults.

1.2.3.5 Equipment Failure and Aging

Internal failures of power-system apparatus can directly initiate shunt faults. Insulation breakdown due to aging, partial discharge, overvoltage, or manufacturing defects may create a low-impedance path to ground or between phases. Typical examples include failures of cable terminations, bushing flashovers, and the breakdown of solid or liquid insulation in transformers, regulators, and capacitor banks. These faults are often permanent and require equipment repair or replacement.



Figure 5: Internal fault in a substation voltage regulator [1]

Figure 5 illustrates the severity of internal faults in oil-filled substation equipment. Once insulation failure produces an internal arc, the resulting thermal stress can damage the tank, bushings, and nearby apparatus, making rapid protection and accurate post-fault diagnosis essential.

1.2.3.6 Human Activities and Accidents

A variety of human actions, whether accidental or deliberate, can cause shunt faults. Vehicular collisions with poles or substation equipment, intrusion of drones, kites, or metallic balloons into live lines, and inadvertent grounding after maintenance are frequent accidental causes. Vandalism, such as theft of conductors or rifle attacks on insulators, also poses a significant threat to system reliability [10]. Additionally, poorly controlled construction work near power lines may result in temporary or permanent faults.

1.3 Importance of Fault Location

1.3.1 Technical and Operational Importance

The primary aim of fault location is to expedite the restoration of power following a permanent fault and to identify the origin of temporary faults before they develop into permanent ones. When a permanent fault occurs, the affected section is isolated, and downstream loads experience an outage. Unless the network can be reconfigured, the outage duration equals the time required to locate the fault plus the time required to perform repairs. Even when alternative supply paths exist, they may become overloaded, particularly in modern systems operated near their stability limits, potentially triggering cascading trips and widespread blackouts [11]. Rapid fault location therefore directly contributes to system reliability and continuity of supply.

Temporary faults, while not causing sustained outages, also warrant accurate location. Identifying their source allows utilities to take corrective action such as trimming vegetation, replacing weakened insulators, or installing animal deterrents before insulation degradation or recurrent arcing leads to a permanent failure [4]. Faults on transmission

and distribution lines are inherently more difficult to locate than those inside power plants or substations, owing to the extensive geographic span of overhead circuits and underground cables. Traditional manual patrols are labor-intensive, time-consuming, and often impeded by rough terrain, adverse weather, or poor accessibility. Furthermore, not all faults leave conspicuous evidence; a flashed insulator may be invisible from ground level, and many temporary faults leave no physical trace. In such cases, knowing where to concentrate inspection efforts can reduce patrol time from hours to minutes and enable field crews to restore service with minimal delay [1].

Beyond restoration, fault location serves several operational and protection engineering objectives. It provides a means to verify the correctness of protective relay operations, detect wiring errors in current and voltage transformer circuits, and validate relay settings, all of which are essential to avoid undesired trips and ensure selective coordination [1].

1.3.2 Economic Impact of Power Interruptions

The importance of fast and accurate fault location is not limited to the technical restoration of the network. When a transmission or distribution element is disconnected after a fault, part of the system may lose supply or operate under constrained conditions until the faulted section is identified, isolated, repaired, or bypassed. The economic impact of such an event therefore includes the value of the energy that cannot be served, the loss of industrial and commercial production, the disturbance of public services, and the operational cost of emergency switching and field intervention.

In the United States, the economic cost of power interruptions is very high. A Lawrence Berkeley National Laboratory study prepared for the U.S. Department of Energy estimated the annual cost of power interruptions to U.S. electricity consumers at approximately 79–80 billion dollars, with most of the cost carried by commercial and industrial customers [12]. More recent work from Oak Ridge National Laboratory reported that major outages cost U.S. customers an average of more than 67 billion dollars per year between 2018 and 2024, increasing to about 121 billion dollars in 2024 [13]. These values are not the repair cost of one line; they represent the broader customer and economic cost of interrupted electricity service.

For the New England context, ISO New England identifies transmission equipment outages as periods when equipment is out of service and distinguishes planned outages from unplanned outages caused by forced equipment problems. Such outages may last from 15 minutes to several weeks or months, depending on the nature of the equipment condition and the restoration process [14]. This shows why reducing the time required to locate a fault is operationally meaningful: every reduction in search time can reduce the duration of exposure to interrupted or constrained service.

A useful way to express this impact is the value of lost load (VoLL), which represents the economic value customers assign to avoiding an outage. The Avoided Energy Supply Components (AESC) 2021 study for New England projects a VoLL of approximately 73 \$/kWh [15]. If an interruption prevents the delivery of P_{int} MW for t hours, the corresponding cost of unserved energy can be approximated as:

$$C_{\text{outage}} = P_{\text{int}} \times 1000 \times t \times \text{VoLL}, \tag{1.1}$$

where P_{int} is expressed in MW, t in hours, and VoLL in \$/kWh. For example, an interruption of 100 MW lasting one hour corresponds to 100,000 kWh of unserved energy. Using the New England VoLL estimate, the interruption cost is approximately:

$$C_{\text{outage}} = 100 \times 1000 \times 1 \times 73 = 7.3 \times 10^6 \text{ \$} \tag{1.2}$$

This calculation is only an approximate economic indicator, since the real cost depends on customer type, outage duration, time of day, season, and the ability of the network to transfer power through alternative paths. Even with this simplification, it illustrates the practical value of accurate and fast fault location: a method that reduces the patrol area or helps operators identify the faulted section more rapidly can contribute directly to lower interruption costs and improved continuity of supply.

1.4 Fault Location Algorithm Families

The use of electrical signals for fault location has been investigated since the early 1900s [16, 17, 18]. In a broad view, the methods used in the literature can be grouped into two main families: conventional methods, which are derived from electrical circuit models and signal-propagation principles, and machine-learning or artificial-intelligence methods, which learn fault-location relationships from measured or simulated data.

1.4.1 Conventional Methods

Conventional fault-location techniques are based on physical models of the line and on the electrical signals generated during the fault. The most established group is formed by impedance-based methods. These algorithms apply Ohm's law to estimate the distance to a shunt fault. Voltage and current phasors measured at a line terminal during the fault are used to compute the apparent impedance from the measurement point to the fault. This impedance value is then related to the positive-sequence line impedance per unit length, expressed for example in Ω/km, to obtain a distance estimate. Since the calculation relies on power-frequency quantities, impedance-based methods are applicable to both transmission and distribution lines.

Another conventional group is formed by traveling-wave methods. These algorithms use the high-frequency voltage and current surges generated at the instant of fault inception. The surges propagate along the line at a velocity close to the speed of light, and the fault distance is obtained from the measured arrival times of the initial and reflected waves, or from synchronized measurements at both line terminals, together with the known propagation velocity. Traveling-wave methods can be very accurate when the required transient information is captured, but their practical implementation requires high sampling rates, accurate time synchronization, and wide-bandwidth instrumentation.

1.4.2 Machine-Learning and Artificial-Intelligence Methods

Machine-learning and artificial-intelligence methods have become an important research direction for fault detection, classification, and location. Early studies used artificial neural networks to learn nonlinear mappings between electrical measurements and fault position [19]. More recent reviews show that neural networks, support vector machines, random forests, fuzzy systems, expert systems, and deep-learning models have all been investigated for fault-location and fault-diagnosis problems [20].

The state of the art can be divided according to the type of information used by the learning model. Some methods rely mainly on power-frequency voltage and current measurements or phasor-domain quantities. For example, graph-based and deep-learning approaches have been proposed to exploit the spatial structure of distribution networks and the information provided by measurement devices [21, 22]. Other methods use high-frequency or time-frequency features extracted from fault transients, often with wavelet analysis combined with neural or fuzzy models [23, 24]. These approaches are attractive because they can capture nonlinear relationships that are difficult to express with closed-form equations, especially under variations in fault resistance, load level, network topology, and measurement availability.

Despite this progress, AI-based fault-location methods also introduce practical challenges. Their performance depends strongly on the representativeness of the training data, the selected input features, the quality of measurements, and the ability of the model to generalize outside the conditions used during training. For this reason, AI methods are often studied as a complement to conventional protection and fault-location principles rather than as a complete replacement for them.

1.5 Measurement and Communication Infrastructure

A data-gathering system, which may include one or more metering devices, sends recorded fault signals to a network processor. Many recent fault-location methods use measurements collected at multiple locations through a communication system. The communication links connect metering devices installed on power-network elements to a central processing unit or control center, as shown in Figure 6.

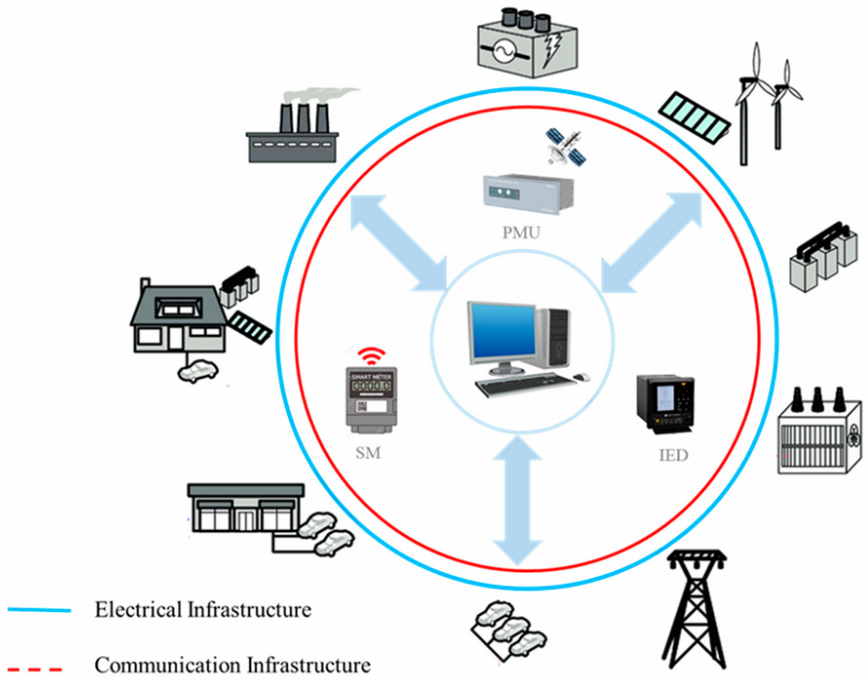


Figure 6: Smart-grid power and communication layers [20]

The communication system may use either wired or wireless links. Wired supervisory control and data acquisition (SCADA) systems are widely used in power-system applications [25]; however, cellular architectures [26, 27] and WiMAX networks [28, 29] have also been proposed. The IEEE 802.16 standard specifies broadband wireless access technologies that can support such communication.

Three important types of data-collection devices in power systems are smart meters (SMs), intelligent electronic devices (IEDs), and phasor measurement units (PMUs). SMs are used mainly for energy-management purposes. Although they do not normally record the high-frequency content of fault signals, their measurements can be used to estimate network loading [30]. IEDs are programmable devices that can record, process, and transmit information over a communication network. Their applications include protection, control, monitoring, and measurement [31]. IEC 61850 defines communication services, including a process bus between process-level equipment and bay-level IEDs, for power-system protection and control. PMUs measure time-synchronized voltage and current phasors using an internal or external time reference, such as GPS [32]. Their reporting rates support wide-area monitoring and selected protection applications [33]. Choi and

Meliopoulos [34] proposed a real-time microgrid protection approach based on distributed dynamic state estimation using PMUs. Biswal et al. [35] developed a PMU-based supervisory protection framework that extracts features for rapid classification. Conventional PMUs are used mainly at transmission-voltage levels, whereas micro-PMUs (μPMUs) have been developed for distribution networks [36, 37].

In terms of their required measurements, fault-location techniques can generally be divided into two classes. The first class, which includes impedance-based methods, relies on fundamental-frequency phasors extracted from recorded voltages and currents. The second class, which includes traveling-wave methods, relies on sampled fault waveforms and extracts high-frequency information. AI-based methods may use either type of input: for example, Refs. [21, 22] rely mainly on power-frequency measurements, whereas Refs. [23, 24] use high-frequency features. The required input measurements are therefore a key factor in determining the infrastructure needed to implement each fault-location method.

1.6 Performance Evaluation of Fault Location Algorithms

The accuracy of fault location algorithms is quantified by the *error percentage*, typically expressed relative to the total line length. The standard formulation is given by [1]:

$$\epsilon = \frac{|d_{\text{est}} - d_{\text{true}}|}{L_{\text{line}}} \times 100\% \tag{1.3}$$

where d_{est} is the estimated distance to the fault, d_{true} is the true fault distance, and L_{line} is the total line length. All quantities must be expressed in consistent units (kilometres or miles). A lower error percentage indicates greater precision, enabling field crews to narrow their search area and thereby reduce outage duration. For example, on a 100 km line, an assumed error tolerance of 5% corresponds to a distance tolerance of 5 km.

In addition to accuracy, computation time has become an important performance metric. Traditionally, fault-location estimates were used mainly for post-event restoration, for which delays of several minutes could be acceptable. However, emerging applications require real-time or near-real-time fault location. One example is adaptive autoreclosing on hybrid lines that contain both overhead and underground sections. Overhead-line faults may be temporary and therefore suitable for reclosing, whereas underground-cable faults are usually permanent. By rapidly determining the fault location, a protective relay can decide whether to initiate reclosing for an overhead-section fault or block it for an underground-section fault. Such applications impose substantially tighter latency requirements than conventional outage-management tasks.

1.7 General Conclusion

This chapter established the basic power system and fault location concepts used throughout the thesis. It showed that faults are abnormal operating conditions that disturb the normal flow of electrical energy and may lead to dangerous overcurrents, voltage deviations, equipment damage, and service interruption. The distinction between series faults and shunt faults is important because each type produces different electrical signatures and therefore requires different analysis and protection considerations.

The chapter also explained that fault location is not only a theoretical calculation but also a practical task that supports restoration, maintenance, protection coordination, and outage reduction. Conventional methods generally rely either on fundamental-frequency phasors, as in impedance-based algorithms, or on high-frequency transient information, as in traveling-wave algorithms. AI-based methods may use phasor-domain, waveform-

domain, or feature-engineered inputs. Therefore, the performance of any method depends not only on the algorithm itself but also on the available measurement devices, communication infrastructure, sampling rate, synchronization, and data quality.

Overall, this chapter provides the general framework for the rest of the thesis. The following chapter develops symmetrical components and conventional impedance-based fault-location methods in greater detail, while later chapters introduce and evaluate AI-based methods for fault classification and fault-distance estimation.

Chapter 2 Symmetrical Components and Fault Location

Summary

This chapter develops the analytical foundation behind conventional impedance-based fault location. It first reviews phasors and symmetrical components, because unbalanced faults are easier to study when phase quantities are separated into positive-, negative-, and zero-sequence components. The chapter then shows how sequence networks are connected for the main shunt-fault types and how transmission-line sequence impedances can be obtained by calculation, measurement, or estimation from operating data. These ideas are then used to present one-ended, two-ended, and three-terminal fault-location methods. By the end of the chapter, the conventional approach is established as the physical and mathematical reference against which the data-driven methods can later be interpreted.

2.1 General Introduction

Accurate fault location in transmission and distribution networks depends on the correct interpretation of voltage and current measurements recorded during abnormal operating conditions. For balanced three-phase faults, the analysis can often be simplified because the three-phase quantities remain separated by 120°. However, most practical faults are unbalanced, such as single-line-to-ground, line-to-line, and double-line-to-ground faults. In these cases, direct phase-domain analysis becomes more difficult because the three phases are no longer symmetrical. This difficulty motivates the use of symmetrical components.

The method of symmetrical components was developed by Charles L. Fortescue for analyzing unbalanced conditions in polyphase power systems [38]. It decomposes an unbalanced set of three-phase quantities into three balanced sets: positive-, negative-, and zero-sequence components. These components make it possible to construct sequence networks and interconnect them according to the fault type. As a result, unbalanced faults can be treated using systematic circuit relationships rather than by solving the full coupled phase-domain system directly.

The first part of this chapter introduces the phasor representation of sinusoidal quantities, the symmetrical component transformation, the physical meaning of sequence components, sequence network interconnections for common shunt faults, and the calculation or estimation of sequence impedances. These concepts are not only theoretical; they are required by many protection and fault location algorithms because relays and disturbance recorders usually process voltage and current phasors during fault events.

The second part applies this foundation to conventional impedance-based fault-location methods. These methods relate the apparent impedance seen from one or more terminals to the known impedance of the protected line. One-ended techniques are simple but can be affected by fault resistance and remote infeed. Two-ended techniques can improve accuracy, although they may require synchronized measurements or angle correction. Three-terminal methods extend the same principles to tapped lines, where the faulted section must first be identified.

Thus, this chapter connects the mathematical treatment of unbalanced faults with practical conventional fault-location algorithms. The material provides the reference framework needed later when comparing conventional impedance-based methods with AI-based approaches.

2.2 Phasors

A phasor is a complex number that represents the magnitude and phase angle of a steady state sinusoidal signal. In AC circuits, voltages and currents during steady state conditions are sinusoidal.

A sinusoidal voltage can be expressed using a cosine function:

$$v(t) = V_{\max} \cos(\omega t + \theta), \tag{2.1}$$

where $v(t)$ is the time domain voltage, ω is the angular frequency (rad/s), $V_{\max}$ is the peak magnitude (V), and θ is the phase angle (rad). Since RMS values are typically preferred in AC analysis, and $V_{\text{rms}} = V_{\max}/\sqrt{2}$, Equation (2.1) becomes:

$$v(t) = \sqrt{2}\, V_{\text{rms}} \cos(\omega t + \theta) \tag{2.2}$$

Using Euler's formula ($e^{jx} = \cos x + j \sin x$), the time domain signal is transformed to the complex domain:

$$v(t) = \Re\left\{\sqrt{2}\,V_{\text{rms}}e^{j(\omega t+\theta)}\right\} = \Re\left\{V_{\text{rms}}e^{j\theta} \times \sqrt{2}e^{j\omega t}\right\}, \tag{2.3}$$

where $\Re$ denotes the real part of a complex number.

The time independent complex quantity $V_{\text{rms}}e^{j\theta}$ is the phasor. It has magnitude equal to the RMS value and angle θ. The term $\sqrt{2}e^{j\omega t}$ rotates at angular velocity ω, completing one revolution per cycle.

Phasors can be expressed in two forms:

$$V = V_{\text{rms}}\cos\theta + jV_{\text{rms}}\sin\theta \quad \text{(rectangular form)}, \tag{2.4}$$

$$V = V_{\text{rms}}\angle\theta \quad \text{(polar form)} \tag{2.5}$$

The rectangular form is convenient for addition and subtraction, while the polar form simplifies multiplication and division. Phasors are widely used because they greatly simplify steady state AC circuit analysis [1].

2.3 Theory of Symmetrical Components

The method of symmetrical components decomposes an unbalanced set of three phase voltage or current phasors into three balanced sets, known as sequence components: zero sequence, positive sequence, and negative sequence, as illustrated in Figure 7.

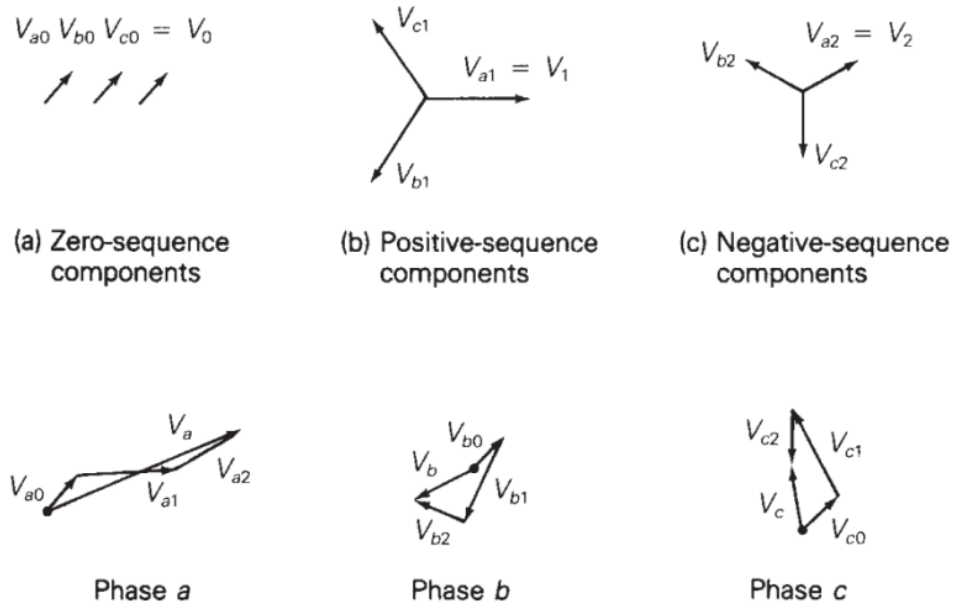


Figure 7: Symmetrical-component decomposition [39]

- **Zero sequence components**: Equal magnitude with zero phase shift between phases.
- **Positive sequence components**: Equal magnitude, same phase sequence as the system (ABC), with 120° phase shifts.
- **Negative sequence components**: Equal magnitude, opposite phase sequence (ACB), with 120° phase shifts.

Mathematically, the three phase currents are expressed as:

$$I_a = I_{a0} + I_{a1} + I_{a2}, \tag{2.6}$$
$$I_b = I_{b0} + I_{b1} + I_{b2}, \tag{2.7}$$
$$I_c = I_{c0} + I_{c1} + I_{c2} \tag{2.8}$$

These expressions can be simplified by referencing all sequence components to the A phase. Since zero sequence components are equal in magnitude and phase:

$$I_{b0} = I_{c0} = I_{a0} \tag{2.9}$$

To relate the B and C phases to A for positive and negative sequences, the operator a is introduced. This operator has unit magnitude and a phase angle of 120°:

$$a = 1\angle 120^\circ = e^{j2\pi/3} \tag{2.10}$$

Multiplying a phasor by a rotates it by 120° counterclockwise without changing its magnitude, while multiplying by a^2 rotates it by 240°. A useful identity is:

$$1 + a + a^2 = 0 \tag{2.11}$$

Using the a operator, the sequence components for B and C phases are expressed in terms of A phase:

$$I_{b1} = a^2 I_{a1}, \quad I_{c1} = a I_{a1}, \tag{2.12}$$
$$I_{b2} = a I_{a2}, \quad I_{c2} = a^2 I_{a2} \tag{2.13}$$

Substituting these relations yields:

$$I_a = I_{a0} + I_{a1} + I_{a2}, \tag{2.14}$$
$$I_b = I_{a0} + a^2 I_{a1} + a I_{a2}, \tag{2.15}$$
$$I_c = I_{a0} + a I_{a1} + a^2 I_{a2} \tag{2.16}$$

Dropping the a subscript (understanding that symmetrical components refer to A phase), the transformation from sequence components to phase quantities is:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \cdot \begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix} \tag{2.17}$$

The inverse transformation, from phase quantities to sequence components, is:

$$\begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \tag{2.18}$$

These transformations apply to voltages as well as currents. Two assumptions were made in deriving these equations:

1. The system has an ABC phase sequence. For ACB rotation, I_b and I_c must be swapped.

2. A phase is the reference phase. Under balanced conditions, any phase can serve as reference. During unbalanced faults, the reference phase should be selected based on the fault type.

2.4 Interconnecting Sequence Networks

A three phase power system can be represented by three sequence networks: positive sequence, negative sequence, and zero sequence. Only the positive sequence network contains sources, as generators produce only positive sequence voltages. Under balanced conditions, the three networks are isolated, meaning only positive sequence currents flow. When an unbalanced fault occurs, the negative sequence and (in some cases) zero sequence networks become connected to the positive sequence network at the fault point.

The following subsections describe the sequence-network interconnections for different shunt-fault types. The phase currents at the fault point are denoted

$$I_{Fa}^{f}, \quad I_{Fb}^{f}, \quad I_{Fc}^{f}, \tag{2.19}$$

and the corresponding phase voltages are

$$V_{Fa}^{f}, \quad V_{Fb}^{f}, \quad V_{Fc}^{f} \tag{2.20}$$

2.4.1 Three Phase Faults

A three-phase fault is balanced. Therefore, only the positive-sequence network carries fault current in the ideal symmetrical model. The fault point F_1 connects to the reference bus N_1 through fault resistance R_f, as shown in Figure 8(a).

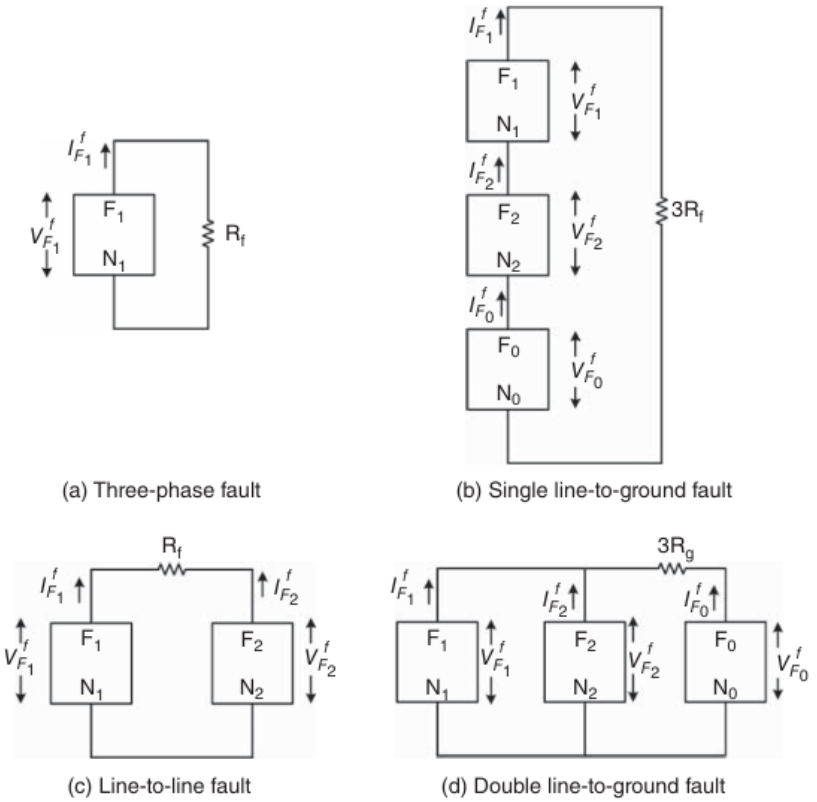


Figure 8: Sequence-network connections for shunt faults [1]

2.4.2 Single Line to Ground Fault

For an A-G fault with resistance R_f, the fault-point boundary conditions are:

$$I_{Fb}^{f} = I_{Fc}^{f} = 0, \tag{2.21}$$

$$V_{Fa}^{f} = R_f I_{Fa}^{f} \tag{2.22}$$

Using A phase as the reference (the faulted phase), the sequence currents become:

$$I_{F0}^f = I_{F1}^f = I_{F2}^f = \frac{1}{3} I_{Fa}^f \tag{2.23}$$

The sequence voltage condition is:

$$V_{F0}^f + V_{F1}^f + V_{F2}^f = 3R_f I_{F1}^f \tag{2.24}$$

These conditions require the three sequence networks to be connected in series, as shown in Figure 8(b). The fault resistance appears as $3R_f$.

Important: For single line to ground faults, always use the faulted phase as the reference.

2.4.3 Line to Line Fault

For a BC fault with resistance R_f, the fault-point boundary conditions are:

$$I_{Fa}^f = 0, \tag{2.25}$$
$$I_{Fb}^f = -I_{Fc}^f, \tag{2.26}$$
$$V_{Fb}^f - V_{Fc}^f = R_f I_{Fb}^f \tag{2.27}$$

Using A phase as the reference (the unfaulted phase), the sequence currents are:

$$I_{F0}^f = 0, \quad I_{F1}^f = -I_{F2}^f \tag{2.28}$$

The voltage condition yields:

$$V_{F1}^f - V_{F2}^f = R_f I_{F1}^f \tag{2.29}$$

Thus, only positive- and negative sequence networks are present, connected in parallel as shown in Figure 8(c).

Important: For line to line faults, always use the unfaulted phase as the reference.

2.4.4 Double Line to Ground Fault

For a BC-G fault with ground resistance R_g, the fault-point boundary conditions are:

$$I_{Fa}^f = 0, \tag{2.30}$$
$$V_{Fb}^f = V_{Fc}^f = R_g(I_{Fb}^f + I_{Fc}^f) \tag{2.31}$$

Using A phase as the reference (the unfaulted phase), the sequence conditions are:

$$I_{F0}^f + I_{F1}^f + I_{F2}^f = 0, \tag{2.32}$$
$$V_{F1}^f = V_{F2}^f, \tag{2.33}$$
$$V_{F0}^f = V_{F2}^f + 3R_g I_{F0}^f \tag{2.34}$$

All three sequence networks participate in the interconnection, with the zero-sequence branch connected through $3R_g$, as shown in Figure 8(d).

Important: For double line to ground faults, always use the unfaulted phase as the reference.

2.5 Sequence Impedances of Three Phase Lines

Three methods for obtaining sequence impedances are presented: calculation, measurement, and estimation.

2.5.1 Method 1: Calculating Sequence Impedances

Carson's equations calculate self and mutual impedances using conductor geometry and type [40, 41]:

$$z_{ii} = R_{Ci} + 0.00159f + j0.004657f \log_{10}\left(\frac{2160\sqrt{f}}{GMR_i}\right) \quad [\Omega/\text{mi}], \tag{2.35}$$

$$z_{ik} = 0.00159f + j0.004657f \log_{10}\left(\frac{2160\sqrt{f}}{d_{ik}}\right) \quad [\Omega/\text{mi}], \tag{2.36}$$

where R_{Ci} is conductor resistance, f is frequency, GMR_i is geometric mean radius, and d_{ik} is conductor spacing.

For a three phase line with neutral conductor n, the primitive impedance matrix is partitioned:

$$\begin{bmatrix}[V_{pg}]\\ [V_{ng}]\end{bmatrix} = \begin{bmatrix}[V^r_{pg}]\\ [V^r_{ng}]\end{bmatrix} + \begin{bmatrix}[Z_{pp}] & [Z_{pn}]\\ [Z_{np}] & [Z_{nn}]\end{bmatrix}\begin{bmatrix}[I_p]\\ [I_n]\end{bmatrix} \tag{2.37}$$

Assuming a perfectly grounded neutral ($V_{ng} = V^r_{ng} = 0$), Kron reduction eliminates the neutral:

$$[Z_{abc}] = [Z_{pp}] - [Z_{pn}][Z_{nn}]^{-1}[Z_{np}] \tag{2.38}$$

Transforming to sequence components:

$$[Z_{seq}] = [A]^{-1}[Z_{abc}][A] = \begin{bmatrix} Z_0 & 0 & 0\\ 0 & Z_1 & 0\\ 0 & 0 & Z_2\end{bmatrix} \quad [\Omega/\text{mi}] \tag{2.39}$$

For a transposed line with reciprocal phase impedances, $Z_1 = Z_2$. The zero-sequence impedance is generally larger than the positive-sequence impedance, but its value depends strongly on conductor geometry, grounding, earth resistivity, shield wires, and mutual coupling. The sequence matrix is diagonal for an ideally transposed or geometrically symmetrical line [40, 41].

2.5.2 Method 2: Measuring Sequence Impedances

With the line de-energized and its remote end connected according to the commissioning test arrangement, controlled currents can be injected to measure the line impedance [42, 43]. A balanced positive-sequence injection gives

$$Z_1 = \frac{V_1}{I_1}, \tag{2.40}$$

whereas an equal in-phase three-phase injection gives

$$Z_0 = \frac{V_0}{I_0} \tag{2.41}$$

Equivalent phase-loop tests can also be used, but their conversion to sequence impedance depends on the exact current-return and grounding arrangement. The test connections and conversion equations must therefore be stated together rather than applying one loop formula to every setup.

2.5.3 Method 3: Estimating Sequence Impedances

Sequence impedances can be estimated during unbalanced conditions (single pole open, ground faults) using recorded voltage and current phasors.

Single Pole Open Condition When one pole of a breaker is open, all three sequence networks connect in parallel at the open point, as shown in Figures 9 and 10. The positive sequence impedance is estimated from negative sequence quantities [44]:

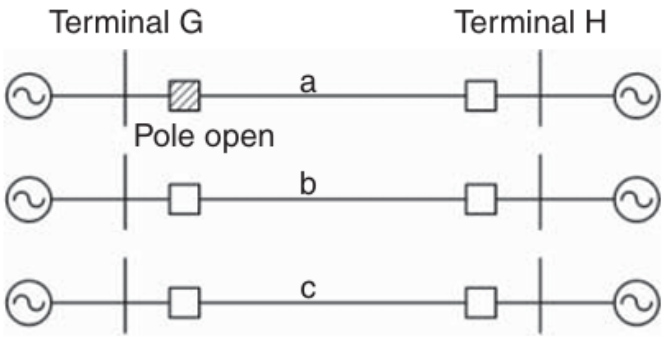


Figure 9: Single-pole-open two-terminal line [1]

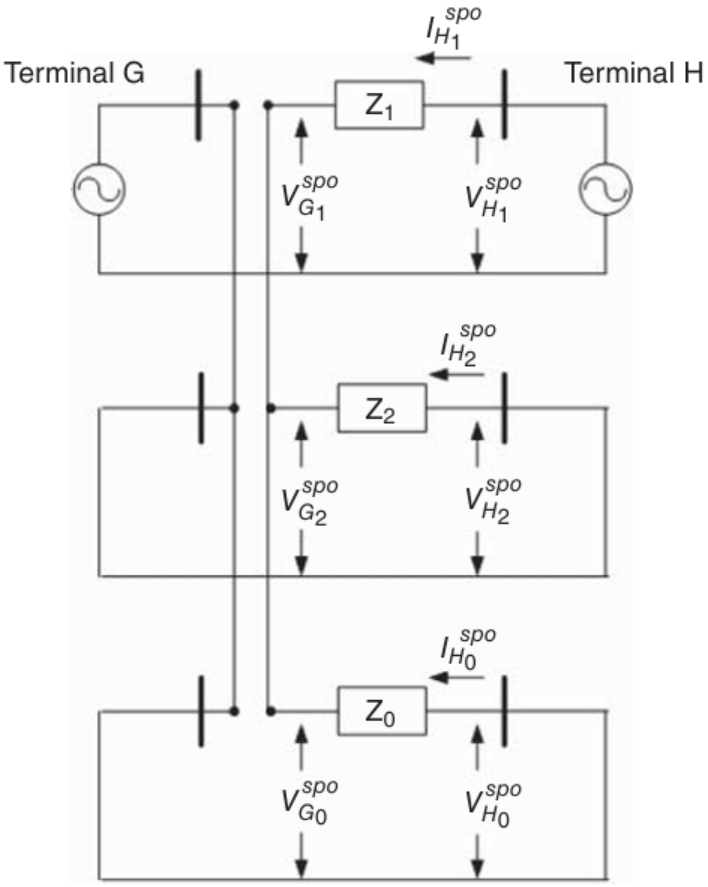


Figure 10: Sequence networks for a single-pole-open line [1]

$$Z_1 = \frac{V_{H2}^{spo} - V_{G2}^{spo}}{I_{H2}^{spo}} \tag{2.42}$$

The zero sequence impedance is:

$$Z_0 = \frac{V_{H0}^{spo} - V_{G0}^{spo}}{I_{H0}^{spo}} \tag{2.43}$$

Accurate estimation requires sufficient load current and angular separation, line-side voltage transformers, and synchronized measurements from both ends.

Ground Fault Condition Sequence impedances can also be estimated using voltage and current phasors recorded during single-line-to-ground or double-line-to-ground faults. Complex two-ended phasors must share a common reference, either through time synchronization or through an explicit angle-alignment procedure.

2.6 Conventional Impedance-Based Fault-Location Methods

Conventional fault-location techniques estimate the position of a fault by using the relationship between measured voltage and current phasors and the known impedance of the protected line. These methods are physically interpretable and remain widely used in protection and post-fault analysis. The following sections present one-ended, two-ended, and three-terminal impedance-based methods, starting from the basic apparent-impedance principle and then extending the formulation to account for remote infeed, synchronization, and multi-terminal network configurations.

2.6.1 One-Ended Impedance-Based Fault-Location Algorithms

One-ended impedance-based fault-location algorithms estimate the distance to a fault using measurements taken from a single terminal of a transmission line [45]. The voltage and current phasors recorded by a digital relay at the local end during the fault are used to compute the apparent impedance between the relay location and the fault point. By comparing this apparent impedance with the known positive-sequence line impedance per unit length, the per-unit or absolute distance to the fault is obtained.

The principal advantages of one-ended algorithms are their simplicity of implementation and reliance on data from only one line terminal. Their accuracy nevertheless depends on fault resistance, remote infeed, source impedance, and line-model assumptions.

2.6.1.1 Principle

The fundamental principle of one-ended impedance-based fault-location methods can be illustrated by analyzing a single-line-to-ground (A-G) fault on a two-terminal transmission line [1].

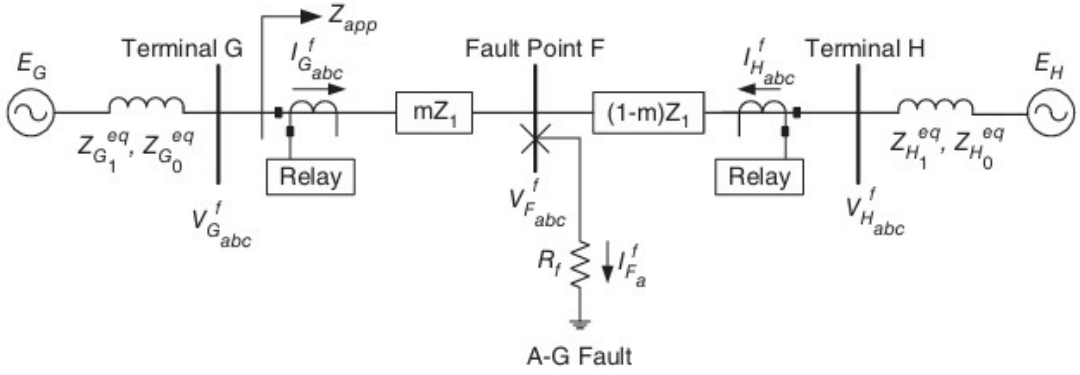


Figure 11: Two-terminal line for one-ended fault location [1]

Consider the homogeneous line shown in Figure 11, with total length L, connecting terminals G and H. Its total positive- and zero-sequence impedances are denoted by Z_1 and Z_0, respectively. The equivalent sources at terminals G and H are represented by ideal voltage sources E_G and E_H behind their respective sequence impedances $(Z_{eq}^{G1}, Z_{eq}^{G0}, Z_{eq}^{H1}, Z_{eq}^{H0})$.

For an internal A-G fault occurring at a fraction m of the line length from bus G, with a fault resistance R_f, the voltage and current phasors at terminal G (V^f_{Gabc}, I^f_{Gabc}) are recorded by a digital relay. The primary challenge in one-ended methods is that the total fault current I^f_{Fa} is unknown, as it includes a contribution from the remote terminal H.

To derive the apparent impedance measured from terminal G, the sequence networks

are interconnected as depicted in Figure 12. The sequence voltages at the fault point F can be expressed using local measurements and the unknown fault location m:

$$V_{F1}^f = V_{G1}^f - mZ_1 I_{G1}^f \quad [\text{V}], \tag{2.44}$$

$$V_{F2}^f = V_{G2}^f - mZ_1 I_{G2}^f \quad [\text{V}], \tag{2.45}$$

$$V_{F0}^f = V_{G0}^f - mZ_0 I_{G0}^f \quad [\text{V}] \tag{2.46}$$

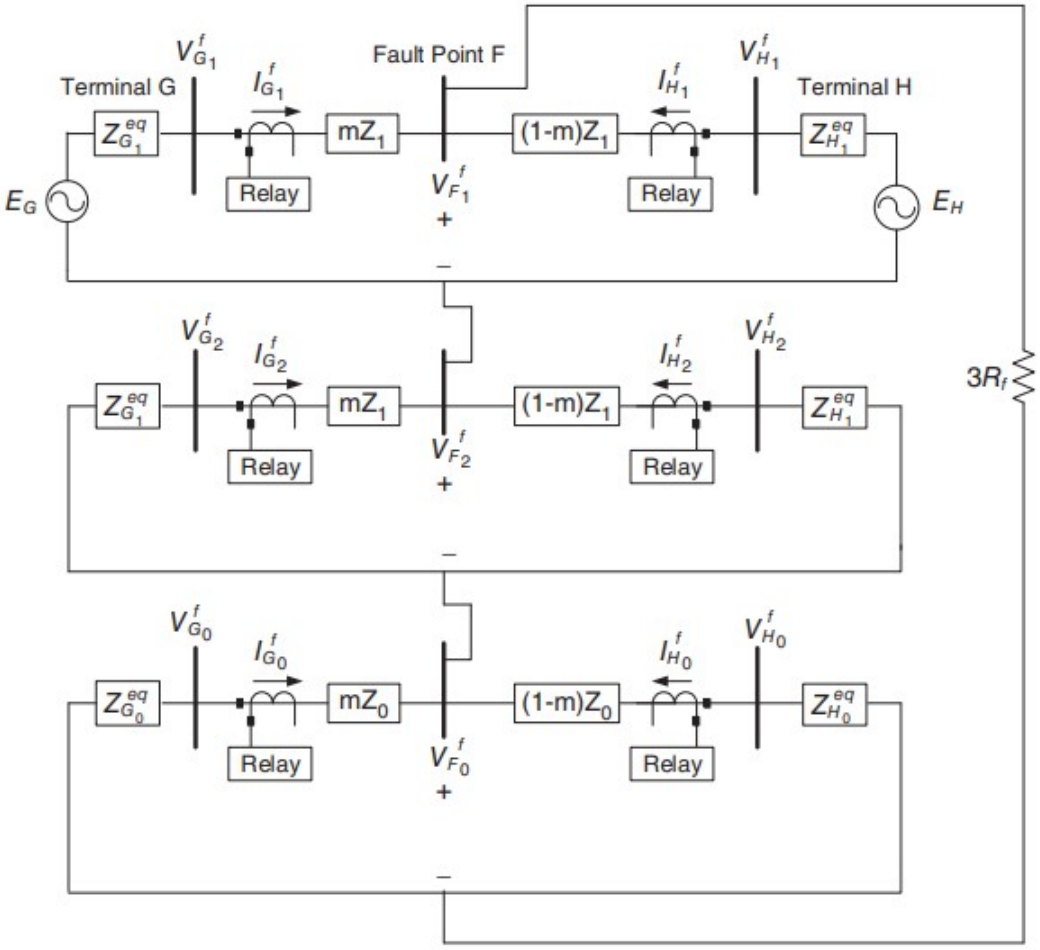


Figure 12: SLG sequence network for a two-terminal line [1]

Summing these sequence components yields the voltage of the faulted phase A at the fault point, V_{Fa}^f. By equating this voltage to the voltage drop across the fault resistance ($R_f I_{Fa}^f$) and performing algebraic manipulation including the introduction of the zero sequence current compensation factor, the following relationship is obtained:

$$V_{Ga}^f = mZ_1 \left(I_{Ga}^f + kI_{G0}^f \right) + R_f I_{Fa}^f \quad [\text{V}], \tag{2.47}$$

where k is the zero sequence compensation factor defined as:

$$k = \frac{Z_0 - Z_1}{Z_1} \tag{2.48}$$

From this, the apparent impedance (Z_{app}) measured by the relay at G is defined as:

$$Z_{app} = \frac{V_{Ga}^f}{I_{Ga}^f + kI_{G0}^f} = mZ_1 + R_f \left(\frac{I_{Fa}^f}{I_{Ga}^f + kI_{G0}^f} \right) \quad [\Omega] \tag{2.49}$$

This equation can be generalized for all fault types by selecting appropriate voltages and currents (e.g., faulted phase quantities with zero sequence compensation for phase to ground faults, or line to line quantities for phase faults):

$$Z_{app} = \frac{V_G}{I_G} = mZ_1 + R_f \left(\frac{I_F}{I_G} \right) \quad [\Omega]. \tag{2.50}$$

Equation (2.50) is the cornerstone of one-ended impedance-based fault-location algorithms. It highlights the core difficulty of the approach: the apparent impedance is a function of the desired fault location m, the fault resistance R_f, and the unknown magnitude and phase angle of the total fault current I_F. Specialized algorithms therefore seek to reduce or compensate for the influence of these unavailable quantities [1].

2.6.1.2 The Takagi Method

The Takagi method, introduced by Takagi et al. [1], improves one-ended impedance-based fault location for resistive faults in loaded systems. It uses the superposition principle to remove the pre-fault load-current component and to reduce the effect of fault resistance under its current-angle assumption.

Principle of Superposition The superposition principle states that a linear network operating under fault conditions can be decomposed into the sum of two distinct networks [1]:

1. **Pre-fault network**: also called the load network, it represents the system in its steady-state condition before the fault occurs.
2. **Pure-fault network**: also called the superimposed network, it contains only the components generated by the fault.

Figure 13 illustrates this decomposition for a single-line-to-ground fault on a two-terminal transmission line. In the pure-fault network, all ideal voltage sources are short-circuited, and the circuit is driven solely by the negative of the pre-fault voltage at the fault point ($-V_{F1}^{pre}$). This network contains only the fault-induced components and is therefore free from load current.

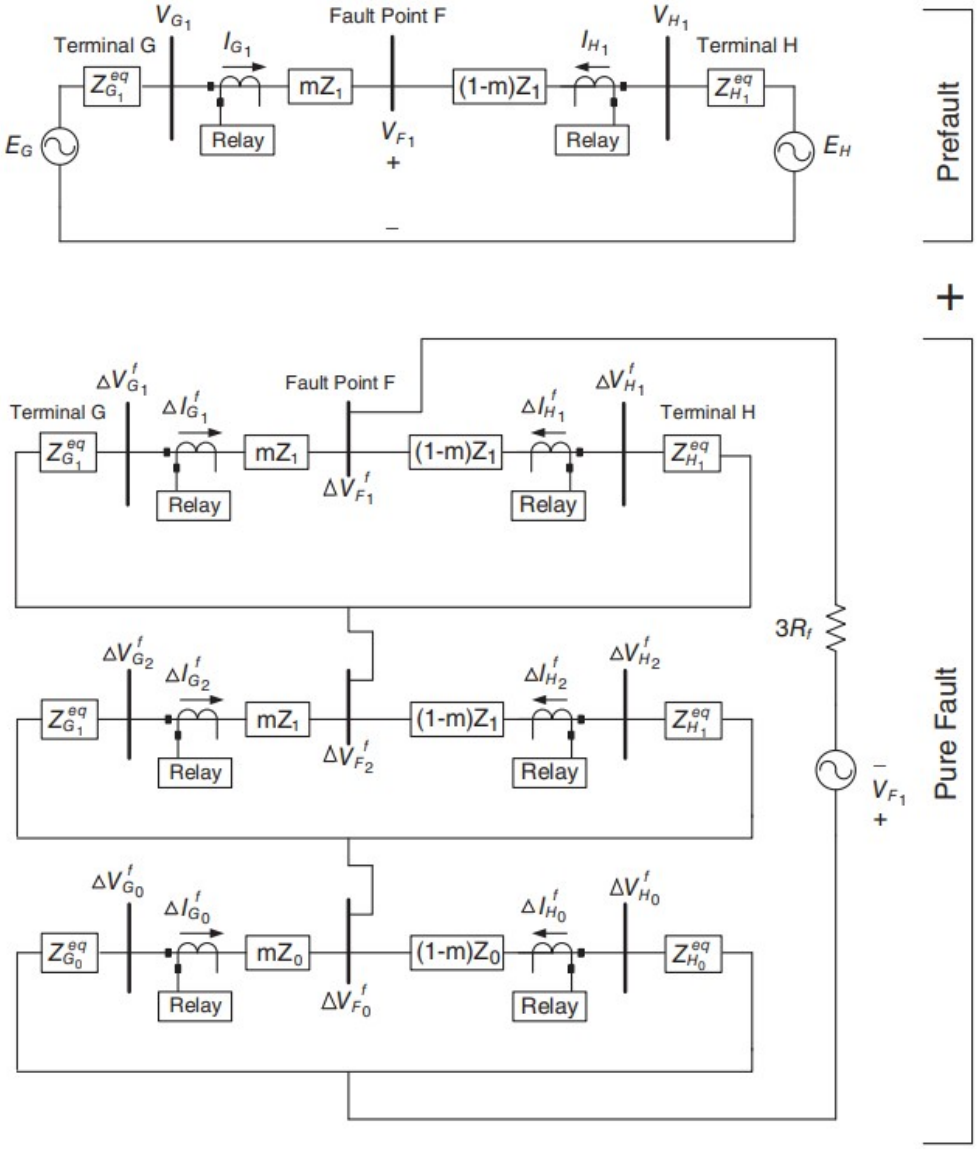


Figure 13: SLG superposition model [1]

Definition of the Pure Fault Current From the pure-fault network, a quantity called the *pure-fault current* (ΔI_G) is defined at the relay location (terminal G). This current phasor represents the change in current caused exclusively by the fault and is obtained by subtracting the pre-fault current phasor from the fault current phasor for the appropriate phase loop.

For a general fault, the pure fault current can be expressed in terms of its sequence components as [1]:

$$\Delta I_G = \Delta I_{G0}^f + \Delta I_{G1}^f + \Delta I_{G2}^f \tag{2.51}$$

where ΔI_{G0}^f, ΔI_{G1}^f, and ΔI_{G2}^f are the zero-, positive-, and negative-sequence pure-fault current phasors, respectively. Because the pre-fault system is balanced (containing no negative- or zero-sequence components), the following simplifications apply:

$$\Delta I_{G0}^f = I_{G0}^f \tag{2.52}$$

$$\Delta I_{G2}^f = I_{G2}^f \tag{2.53}$$

$$\Delta I_{G1}^f = I_{G1}^f - I_{G1}^{pre} \tag{2.54}$$

where I_{G1}^{pre} is the pre-fault positive-sequence current.

For an A-G fault, this simplifies to [1]:

$$\Delta I_G = I_{G0}^f + (I_{G1}^f - I_{G1}^{pre}) + I_{G2}^f \tag{2.55}$$

Table 1 provides the general definition of ΔI_G for various fault types, where the appropriate phase quantities are selected according to standard practice (faulted phase for single line to ground faults, or line to line quantities for phase faults).

Table 1: Pure-fault current ΔI_G [1]

Fault Type	ΔI_G
A-G	$I^f_{G_a} - I^{pre}_{G_a}$
B-G	$I^f_{G_b} - I^{pre}_{G_b}$
C-G	$I^f_{G_c} - I^{pre}_{G_c}$
AB, AB-G, ABC	$(I^f_{G_a} - I^{pre}_{G_a}) - (I^f_{G_b} - I^{pre}_{G_b})$
BC, BC-G, ABC	$(I^f_{G_b} - I^{pre}_{G_b}) - (I^f_{G_c} - I^{pre}_{G_c})$
CA, CA-G, ABC	$(I^f_{G_c} - I^{pre}_{G_c}) - (I^f_{G_a} - I^{pre}_{G_a})$

Fault Location Formulation The fundamental equation for one-ended fault location, derived from the interconnection of sequence networks (Figure 12), relates the apparent impedance to the fault parameters [1]:

$$Z_{app} = \frac{V_G}{I_G} = mZ_{1L} + R_f \left(\frac{I_F}{I_G}\right) \tag{2.56}$$

where V_G and I_G are the appropriate voltage and current quantities for the fault type, Z_{1L} is the total positive-sequence line impedance, m is the per-unit fault distance, R_f is the fault resistance, and I_F is the total fault current at the fault point [1]. The term I_F is unknown in one-ended methods because it includes contributions from the remote terminal.

The key insight of the Takagi method is that in a homogeneous system, the phase angle of the pure fault current ΔI_G is approximately equal to the phase angle of the total fault current I_F [1]. This property allows the elimination of the unknown fault resistance term.

Multiplying the one-ended apparent-impedance relation by the complex conjugate of ΔI_G (ΔI^*_G) yields:

$$V_G \Delta I^*_G = mZ_{1L} I_G \Delta I^*_G + R_f I_F \Delta I^*_G \tag{2.57}$$

Under the assumption that $\angle I_F \approx \angle \Delta I_G$, the term $R_f I_F \Delta I^*_G$ becomes purely real (resistive). Taking only the imaginary part eliminates the fault resistance term entirely:

$$\Im\left(V_G \Delta I^*_G\right) = m\Im\left(Z_{1L} I_G \Delta I^*_G\right) \tag{2.58}$$

Solving for the fault distance m gives the fundamental Takagi formula:

$$\boxed{m = \frac{\Im\left(V_G \times \Delta I^*_G\right)}{\Im\left(Z_{1L} \times I_G \times \Delta I^*_G\right)} \quad \text{[pu]}} \tag{2.59}$$

The actual distance to the fault is obtained by multiplying the per-unit value m by the total line length L.

Discussion and Limitations The Takagi method effectively compensates for load current and fault resistance by leveraging the pure fault current derived from the superposition principle. As demonstrated in the literature [1], this method significantly improves accuracy compared to conventional techniques when faults involve resistance in a loaded system.

However, the accuracy of the Takagi method relies on the assumption that the phase angles of ΔI_G and I_F are aligned. This holds most closely in *homogeneous systems*, where the source and line impedance angles are similar. In non-homogeneous systems, a phase mismatch produces a residual *reactance error* [1]. This limitation motivated subsequent algorithms, such as the Modified Takagi method and current-distribution-factor methods.

2.6.1.3 The Modified Takagi Method

The Modified Takagi method, also referred to as the zero-sequence-current-based method, addresses the original method's sensitivity to system non-homogeneity [1]. The original Takagi relation reduces load-current and fault-resistance effects but assumes that the pure fault current and total fault current are approximately aligned. In practical networks, differences between source- and line-impedance angles can therefore produce reactance error.

The Modified Takagi method mitigates this limitation through two key enhancements:

1. It uses the zero sequence current instead of the pure fault current to account for load during single line to ground faults.

2. It introduces an angle-correction factor that compensates for system nonhomogeneity when source-impedance data are available.

Modified Takagi Method Using Zero Sequence Current For single-line-to-ground faults, the Modified Takagi method substitutes the pure fault current ΔI_G with three times the zero-sequence current ($3I^f_{G0}$). Under the usual assumption of a balanced pre-fault system, the zero-sequence current is a fault-produced quantity and does not contain the positive-sequence load current. The fault distance from terminal G is computed as [1]:

$$m = \frac{\Im\left(V_G \times 3I^{f*}_{G0}\right)}{\Im\left(Z_1 \times I_G \times 3I^{f*}_{G0}\right)} \quad \text{[pu]} \tag{2.60}$$

where V_G and I_G are the appropriate voltage and current quantities for the fault type, Z_1 is the total positive-sequence line impedance, and I^f_{G0} is the zero-sequence current measured at terminal G during the fault [1]. The asterisk denotes the complex conjugate. Multiplying the per-unit value m by the total line length L yields the fault location in distance units.

Angle Correction for Non Homogeneous Systems To compensate for system non-homogeneity, the Modified Takagi method introduces an angle-correction factor derived from the zero-sequence network. Figure 14 shows the zero-sequence network during a ground fault, where the current-division ratio between the fault point and terminal G can be expressed as [1]:

$$\frac{I^f_{F0}}{I^f_{G0}} = \frac{Z^{eq}_{G0} + Z_0 + Z^{eq}_{H0}}{(1-m)Z_0 + Z^{eq}_{H0}} = d_0 \angle \beta_0 \tag{2.61}$$

where Z^{eq}_{G0} and Z^{eq}_{H0} are the zero-sequence source impedances behind terminals G and H, respectively, Z_0 is the total zero-sequence line impedance, d_0 is the current-distribution-factor magnitude, and β_0 represents the phase-angle mismatch due to system nonhomogeneity. In a homogeneous system, $\beta_0 = 0$, indicating that I^f_{F0} and I^f_{G0} are in phase.

Applying an angle correction of $e^{-j\beta_0}$ to the fault location computation effectively forces the system to appear homogeneous, improving accuracy. However, calculating β_0

requires knowledge of the fault distance m, creating a circular dependency. The Modified Takagi method resolves this through an iterative three step procedure [1]:

1. Calculate a preliminary fault location estimate m_{prelim} using Equation (2.60).
2. Use m_{prelim} in the current-division relation to compute the angle correction factor β_0.
3. Recalculate the fault location with the angle correction applied:

$$m = \frac{\Im\left(V_G \times 3I_{G0}^{f*} \times e^{-j\beta_0}\right)}{\Im\left(Z_1 \times I_G \times 3I_{G0}^{f*} \times e^{-j\beta_0}\right)} \quad [\text{pu}] \tag{2.62}$$

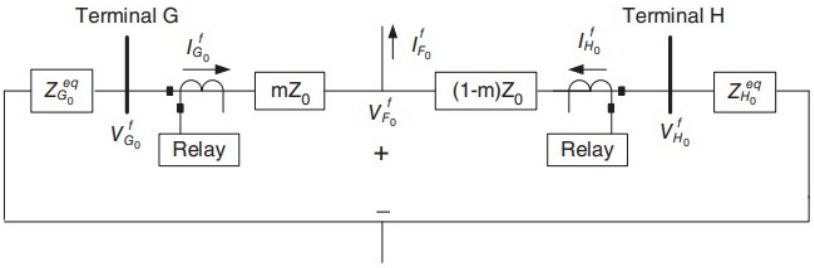


Figure 14: Zero-sequence angle correction [1]

The success of this angle correction depends on accurate knowledge of the zero sequence source impedances. If the local source impedance Z_{G0}^{eq} is unknown, it can be estimated from fault measurements using [1]:

$$Z_{G0}^{eq} = -\frac{V_{G0}^{f}}{I_{G0}^{f}} \quad [\Omega] \tag{2.63}$$

Angle correction also requires the remote source impedance Z_{H0}^{eq}, which is often unavailable. Therefore, the Modified Takagi method is commonly applied without angle correction, as given by Equation (2.60).

Modified Takagi Method Using Negative Sequence Current An alternative formulation of the Modified Takagi method uses negative sequence current instead of zero sequence current. This approach offers two significant advantages [1]:

1. Negative sequence current is present for all unbalanced faults (single line to ground, line to line, and double line to ground), making the method applicable to a wider range of fault types.
2. The negative-sequence network is typically more homogeneous than the zero-sequence network, as transformer connections often introduce additional nonhomogeneity only in the zero-sequence network.

The fault location using negative sequence current is computed as:

$$m = \frac{\Im\left(V_G \times I_{seq}^{*}\right)}{\Im\left(Z_1 \times I_G \times I_{seq}^{*}\right)} \quad [\text{pu}] \tag{2.64}$$

where I_{seq} depends on the fault type as defined in Table 2. For single line to ground faults, $I_{seq} = I_{G2}^f$ (the negative sequence current). For line to line and double line to ground faults, $I_{seq} = jI_{G2}^f$.

Table 2: I_{seq} by fault type [1]

Fault Type	I_{seq}
Single line to ground	I_{G2}^f
Line to line, double line to ground	jI_{G2}^f

Summary and Limitations The Modified Takagi method enhances the original Takagi method by:

- Using zero sequence current to simplify implementation for ground faults.
- Providing an angle-correction mechanism to compensate for system nonhomogeneity when source-impedance data are available.
- Offering a negative sequence based formulation that applies to all unbalanced faults and benefits from greater network homogeneity.

The primary limitation of the angle corrected version is its dependence on accurate remote source impedance data, which is often unavailable in practice. Consequently, the uncorrected version (Equation (2.60)) remains widely used, while the negative sequence formulation (Equation (2.64)) provides a robust alternative for unbalanced faults.

2.6.2 Two-Ended Impedance-Based Fault-Location Methods

Two-ended impedance-based fault-location algorithms use voltage and current measurements captured at both ends of a two-terminal transmission line. Compared with one-ended methods, the additional terminal measurements can reduce sensitivity to fault resistance, load current, remote infeed, and source-impedance assumptions. The methods below differ mainly in whether the terminal phasors are synchronized.

Synchronized Method The synchronized method assumes that measurements from both line terminals share a common time reference, typically supplied by the Global Positioning System (GPS) or another global navigation satellite system (GNSS).

Any of the three symmetrical components may be used for fault-location computation. Under a balanced pre-fault condition, negative-sequence components offer distinct advantages because they exclude positive-sequence load current and are unaffected by zero-sequence mutual coupling, uncertainty in zero-sequence line impedance, and infeed from zero-sequence tapped loads [46, 47]. To illustrate the fault-location principle, consider the negative-sequence network during an unbalanced fault, as shown in Figure 15.

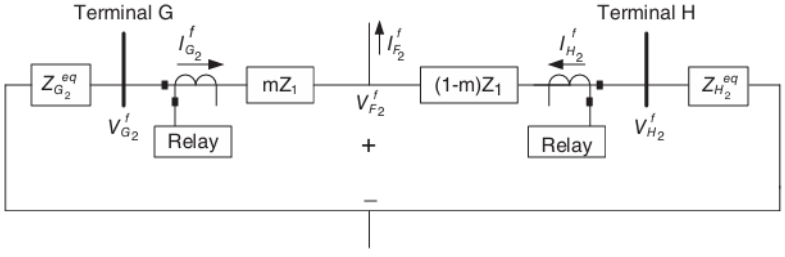


Figure 15: Negative-sequence network for an unbalanced fault

Let I_{G2}^f and I_{H2}^f be directed from their respective terminals toward the line and the fault. The negative-sequence voltage at the fault point F, denoted V_{F2}^f, can then be expressed from both terminals as follows:

From terminal G:

$$V_{F2}^f = V_{G2}^f - mZ_1 I_{G2}^f \quad [\text{V}], \tag{2.65}$$

From terminal H:

$$V_{F2}^f = V_{H2}^f - (1-m)Z_1 I_{H2}^f \quad [\text{V}], \tag{2.66}$$

where:

- V_{G2}^f and I_{G2}^f are the negative sequence voltage and current phasors measured at terminal G during the fault,
- V_{H2}^f and I_{H2}^f are the negative sequence voltage and current phasors measured at terminal H during the fault,
- Z_1 is the total positive-sequence line impedance,
- m is the per unit distance from terminal G to the fault point,
- $(1-m)$ is the per unit distance from terminal H to the fault point.

Since the fault point voltage V_{F2}^f must be identical regardless of the terminal from which it is calculated, equating the two terminal voltage expressions yields the fault distance m:

$$\boxed{m = \frac{V_{G2}^f - V_{H2}^f + Z_1 I_{H2}^f}{\left(I_{G2}^f + I_{H2}^f\right) Z_1}} \quad [\text{pu}] \tag{2.67}$$

Equation (2.67) is applicable for locating any unbalanced fault, including single line to ground, line to line, and double line to ground faults.

Extension to Balanced Faults During a three phase balanced fault, negative sequence components are absent. In such cases, the same fault location principle is applied to the positive sequence network. The distance to the fault is then computed as [1, 45]:

$$\boxed{m = \frac{V_{G1}^f - V_{H1}^f + Z_1 I_{H1}^f}{\left(I_{G1}^f + I_{H1}^f\right) Z_1}} \quad [\text{pu}], \tag{2.68}$$

where V_{G1}^f and I_{G1}^f are the positive sequence voltage and current phasors measured at terminal G during the fault, and V_{H1}^f and I_{H1}^f are the corresponding quantities measured at terminal H.

This approach does not require the individual unbalanced fault class to be known in advance. A threshold on negative-sequence magnitude can be used to select the unbalanced-fault formulation, whereas balanced three-phase faults require the positive-sequence formulation. The per-unit fault distance m obtained from Equation (2.67) or Equation (2.68) is converted to an absolute distance by multiplying by the total line length L.

Unsynchronized Method When Intelligent Electronic Devices (IEDs) are installed at both ends of a transmission line, the captured voltage and current waveforms may not be time aligned. This lack of synchronization can arise from different sampling rates between the IEDs, the absence of GPS equipment, or a malfunctioning time synchronization device.

To overcome this challenge, a synchronization operator $e^{j\delta}$ is introduced to align the measurements from terminal G relative to those from terminal H [48].

The voltage at the fault point can be expressed using either positive sequence or negative sequence components. For a given sequence component i (where $i = 1$ denotes positive sequence and $i = 2$ denotes negative sequence), the fault point voltage is written as:

From terminal G, incorporating the synchronization correction:

$$V_{Fi}^{f} = V_{Gi}^{f} e^{j\delta} - mZ_1 I_{Gi}^{f} e^{j\delta} \quad \text{[V]}, \tag{2.69}$$

From terminal H, which serves as the reference:

$$V_{Fi}^{f} = V_{Hi}^{f} - (1-m)Z_1 I_{Hi}^{f} \quad \text{[V]} \tag{2.70}$$

In these expressions:

- V_{Gi}^{f} and I_{Gi}^{f} are the sequence voltage and current phasors recorded at terminal G during the fault,
- V_{Hi}^{f} and I_{Hi}^{f} are the corresponding quantities recorded at terminal H,
- Z_1 represents the total positive-sequence line impedance,
- m is the per unit distance from terminal G to the fault point,
- δ is the unknown synchronization angle between the two terminal measurements.

Negative sequence components are typically employed for locating unbalanced faults (single line to ground, line to line, and double line to ground), whereas positive sequence components are used for balanced three phase faults.

Since both expressions represent the same fault point voltage, equating them yields the synchronization operator:

$$e^{j\delta} = \frac{V_{Hi}^{f} - (1-m)Z_1 I_{Hi}^{f}}{V_{Gi}^{f} - mZ_1 I_{Gi}^{f}} \tag{2.71}$$

The synchronization operator can be eliminated without explicitly determining the angle δ by taking the absolute value of both sides. Noting that $|e^{j\delta}| = 1$, the following relationship is obtained:

$$1 = \left| \frac{V_{Hi}^{f} - (1-m)Z_1 I_{Hi}^{f}}{V_{Gi}^{f} - mZ_1 I_{Gi}^{f}} \right| \tag{2.72}$$

This equation contains only the unknown fault distance m. By squaring both sides, expanding, and collecting terms, a standard quadratic equation emerges:

$$Am^2 + Bm + C = 0, \tag{2.73}$$

where the coefficients are defined as:

$$A = \left| Z_1 I_{Gi}^{f} \right|^2 - \left| Z_1 I_{Hi}^{f} \right|^2 \tag{2.74}$$

$$B = -2 \times \mathrm{Re}\left[V_{Gi}^{f} \left(Z_1 I_{Gi}^{f} \right)^* + \left(V_{Hi}^{f} - Z_1 I_{Hi}^{f} \right) \left(Z_1 I_{Hi}^{f} \right)^* \right] \tag{2.75}$$

$$C = \left| V_{Gi}^{f} \right|^2 - \left| V_{Hi}^{f} - Z_1 I_{Hi}^{f} \right|^2 \tag{2.76}$$

The per unit distance to the fault is then obtained by solving the quadratic formula:

$$\boxed{m = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}} \quad \text{[pu]} \tag{2.77}$$

Disambiguating the Two Solutions The quadratic equation generally produces two numerical roots. A root outside the physical interval $[0, 1]$ is rejected. If both roots lie inside the interval, the ambiguity must be resolved using an additional criterion, such as another sequence component, a consistency check on the recovered synchronization angle, or information from the protection system. A second sequence calculation can help, but it is not guaranteed to produce a unique root in every case.

The final per-unit fault distance m is converted to absolute distance by multiplying by the total line length L. In ideal data, a significant negative-sequence quantity indicates an unbalanced fault, whereas a balanced three-phase fault is treated with positive-sequence quantities. In practice, a threshold is required because measurement errors and system asymmetry can produce small negative-sequence components.

2.6.3 Three-Terminal Impedance-Based Fault-Location Methods

Three-terminal transmission lines, as shown in Figure 16, are used in utility networks as cost-effective solutions for system expansion and voltage support. However, the presence of three sources and a tap point increases the complexity of fault location. Three-terminal methods address this challenge by using measurements from all three terminals in a systematic two-step approach [1].

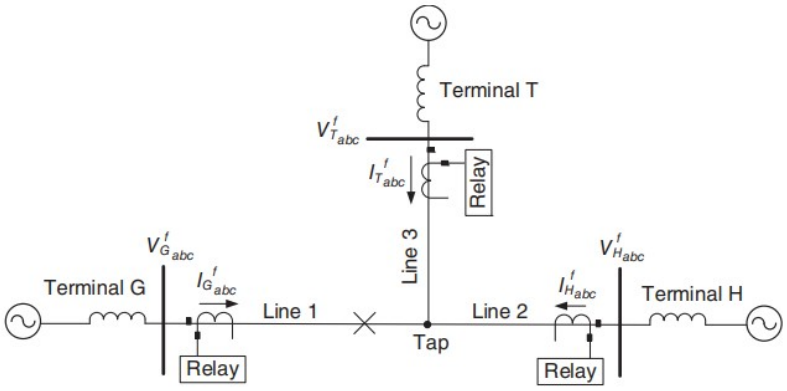


Figure 16: General three-terminal transmission-line system

2.6.3.1 General Two-Step Approach

All three-terminal methods follow the same fundamental procedure:

1. **Identification of the faulted line section**: Determine which of the three line sections contains the fault.

2. **Fault location within the identified section**: Reduce the three-terminal system to an equivalent two-terminal system and apply two-ended fault-location principles.

The specific implementation depends on whether the measurements are synchronized and on which sequence quantities are available. The following formulations apply this common procedure without repeating the general derivation.

Synchronized Method This method requires all three relays to be synchronized to a common GPS time reference.

Faulted Section Identification When an unbalanced fault occurs, the negative sequence tap voltage is calculated from each terminal:

$$V_{\text{Tap}}^{f} = V_{G2}^{f} - Z_{1,L1} I_{G2}^{f}, \tag{2.78}$$
$$V_{\text{Tap}}^{f} = V_{H2}^{f} - Z_{1,L2} I_{H2}^{f}, \tag{2.79}$$
$$V_{\text{Tap}}^{f} = V_{T2}^{f} - Z_{1,L3} I_{T2}^{f} \tag{2.80}$$

The tap voltage calculated using measurements from the faulted line differs from those obtained from the healthy lines. For a fault on Line 1, the values from terminals H and T should agree within measurement tolerance, while the value from terminal G differs. Similar logic applies for faults on Line 2 or Line 3.

Reduction to Two Terminal System Once the faulted section is identified, the two remote terminals (operating in parallel) are combined into an equivalent source. The system is reduced as follows:

$$V_{\text{loc}}^{f} = V_{G2}^{f}, \quad I_{\text{loc}}^{f} = I_{G2}^{f}, \quad Z_{1,\text{line}} = Z_{1,L1}, \tag{2.81}$$
$$I_{\text{Tap}}^{f} = I_{H2}^{f} + I_{T2}^{f}, \tag{2.82}$$
$$Z_{\text{Tap}}^{\text{eq}} = \frac{(Z_{1,L3} + Z_{T2}^{\text{eq}}) \times (Z_{1,L2} + Z_{H2}^{\text{eq}})}{Z_{1,L2} + Z_{H2}^{\text{eq}} + Z_{1,L3} + Z_{T2}^{\text{eq}}} \tag{2.83}$$

Fault Location After selecting sequence $i = 2$ for an unbalanced fault or $i = 1$ for a balanced three-phase fault, the distance is calculated as

$$m = \frac{V_{\text{loc},i}^{f} - V_{\text{Tap},i}^{f}}{(I_{\text{loc},i}^{f} + I_{\text{Tap},i}^{f}) Z_{1,\text{line}}} \quad \text{[pu]} \tag{2.84}$$

Unsynchronized Method This method uses fault voltage and current phasors from all terminals. The relays need not be time synchronized.

Faulted Section Identification The negative-sequence tap-voltage magnitude is calculated from each terminal using the tap-voltage expressions above. The magnitude computed from the faulted line differs from those of the healthy lines. For a fault on Line 1, the values of $|V_{\text{Tap}}^{f}|$ from terminals H and T should agree within tolerance, while the value from terminal G differs.

Angle Correction for Unsynchronized Measurements Since the remote relays are not synchronized, their currents cannot be added directly. The alignment angle of Remote Terminal 2 with respect to Remote Terminal 1 is the phase-angle difference:

$$\delta = \angle V_{\text{Tap},R1}^{f} - \angle V_{\text{Tap},R2}^{f} \tag{2.85}$$

The negative sequence current at the tap point is then:

$$I_{\text{Tap}}^{f} = I_{H2}^{f} + I_{T2}^{f} e^{j\delta} \tag{2.86}$$

Fault Location The fault distance is found from

$$m = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}, \tag{2.87}$$

with the following coefficients:

$$A = \left| Z_{1,\text{line}} I^f_{\text{loc}} \right|^2 - \left| Z_{1,\text{line}} I^f_{\text{Tap}} \right|^2, \tag{2.88}$$

$$B = -2 \times \text{Re}\left[V^f_{\text{loc}} \left(Z_{1,\text{line}} I^f_{\text{loc}} \right)^* + \left(V^f_{\text{Tap},R1} - Z_{1,\text{line}} I^f_{\text{Tap}} \right) \left(Z_{1,\text{line}} I^f_{\text{Tap}} \right)^* \right], \tag{2.89}$$

$$C = \left| V^f_{\text{loc}} \right|^2 - \left| V^f_{\text{Tap},R1} - Z_{1,\text{line}} I^f_{\text{Tap}} \right|^2 \tag{2.90}$$

A root in the physical interval $0 \leq m \leq 1$ is retained. If both roots lie in this interval, an additional consistency criterion is required, as in the two-terminal unsynchronized formulation.

Unsynchronized Negative Sequence Method This method [49] uses only fault current phasors and source impedance data. The three relays need not be synchronized.

Faulted Section Identification The negative sequence tap voltage is calculated from each terminal using source impedances:

$$V^f_{\text{Tap}} = -\left(Z^{\text{eq}}_{G2} + Z_{1,L1} \right) I^f_{G2}, \tag{2.91}$$

$$V^f_{\text{Tap}} = -\left(Z^{\text{eq}}_{H2} + Z_{1,L2} \right) I^f_{H2}, \tag{2.92}$$

$$V^f_{\text{Tap}} = -\left(Z^{\text{eq}}_{T2} + Z_{1,L3} \right) I^f_{T2} \tag{2.93}$$

The magnitude from the faulted line differs from those of the healthy lines. For a fault on Line 1, the values of $|V^f_{\text{Tap}}|$ from terminals H and T should agree within tolerance, while the value from G differs.

Angle Correction and Fault Location After identifying the faulted section, the system is reduced to an equivalent two terminal configuration. An angle correction is applied to align the remote currents. The tap current is obtained as $I^f_{\text{Tap}} = I^f_{H2} + I^f_{T2} e^{j\delta}$.

The fault distance is found using the quadratic formula $m = (-B \pm \sqrt{B^2 - 4AC})/(2A)$, with intermediate complex quantities:

$$a + jb = I^f_{\text{loc}} \times Z^{\text{eq}}_{\text{loc}}, \tag{2.94}$$

$$c + jd = Z_{1,\text{line}} \times I^f_{\text{loc}}, \tag{2.95}$$

$$e + jf = Z^{\text{eq}}_{\text{Tap}} + Z_{1,\text{line}}, \tag{2.96}$$

$$g + jh = Z_{1,\text{line}} \tag{2.97}$$

The coefficients are then:

$$A = \left| I^f_{\text{Tap}} \right|^2 (g^2 + h^2) - (c^2 + d^2), \tag{2.98}$$

$$B = -2 \left| I^f_{\text{Tap}} \right|^2 (eg + fh) - 2(ac + bd), \tag{2.99}$$

$$C = \left| I^f_{\text{Tap}} \right|^2 (e^2 + f^2) - (a^2 + b^2) \tag{2.100}$$

Synchronized Line Current Differential Method This method [50] uses local fault voltage and current phasors together with synchronized fault currents from the two remote terminals.

Fault Location Estimates The fault distance is computed under three separate assumptions:

$$m_G = \frac{\mathrm{Im}\left(V_G I_{\mathrm{pol}}^*\right)}{\mathrm{Im}\left(Z_{1,L1} I_G I_{\mathrm{pol}}^*\right)} \quad \text{(assuming fault on Line 1)}, \tag{2.101}$$

$$m_H = \frac{\mathrm{Im}\left(V_H I_{\mathrm{pol}}^*\right)}{\mathrm{Im}\left(Z_{1,L2} I_H I_{\mathrm{pol}}^*\right)} \quad \text{(assuming fault on Line 2)}, \tag{2.102}$$

$$m_T = \frac{\mathrm{Im}\left(V_T I_{\mathrm{pol}}^*\right)}{\mathrm{Im}\left(Z_{1,L3} I_T I_{\mathrm{pol}}^*\right)} \quad \text{(assuming fault on Line 3)}, \tag{2.103}$$

where I_{pol} is the polarization current derived from local and remote measurements.

Identifying the Faulted Section The line that yields a value of m between 0 and 1 per unit is the faulted section, and that m is taken as the fault location. For a fault on Line 1, m_G will be less than 1 per unit while m_H and m_T will exceed 1 per unit. If more than one m satisfies the criterion, the fault is likely very close to the tap point.

2.7 General Conclusion

This chapter established the analytical basis for conventional fault-location methods. The first part showed that symmetrical components provide a systematic way to analyze unbalanced faults by decomposing phase-domain voltages and currents into positive-, negative-, and zero-sequence networks. This decomposition is essential because most practical faults are unsymmetrical and cannot be represented accurately using only a balanced single-phase equivalent.

The second part applied these sequence-domain concepts to impedance-based fault-location algorithms. One-ended methods are attractive because they require measurements from only one terminal, but they are affected by fault resistance, remote infeed, and source-impedance assumptions. Two-ended methods improve accuracy by using measurements from both line terminals, although they may require synchronization or additional angle-correction procedures. Three-terminal methods extend the same principles to tapped or multiterminal lines, where the algorithm must first identify the faulted section before estimating the distance to the fault.

Overall, conventional impedance-based techniques remain important because they are physically interpretable and directly linked to transmission-line parameters. However, their accuracy depends on reliable phasor measurements, correct sequence-impedance values, accurate line models, and appropriate handling of synchronization and remote-source effects. These strengths and limitations provide the reference point for evaluating the AI-based fault-location methods introduced in the following chapter.

Chapter 3 AI-Based Methods

Summary

This chapter presents the artificial-intelligence-based methods used in the thesis for fault classification and fault-location estimation. It begins with supervised learning, validation, and cross-validation, since a high training score is not enough to demonstrate that a model will perform well on unseen fault cases. It then explains artificial neural networks, support vector machines, and random forests from theoretical and practical perspectives. Finally, it introduces a distance-only hybrid structure in which ANN or Random Forest regression corrects the remaining error of a conventional impedance estimate.

3.1 General Introduction

The fault-location problem in power systems is strongly influenced by nonlinear relationships among voltage, current, fault resistance, source impedance, network topology, and fault type. Conventional impedance-based algorithms provide physically interpretable estimates, but their performance may degrade under complex operating conditions, distributed generation, measurement noise, or uncertain system parameters. Artificial-intelligence-based methods offer an alternative data-driven approach by learning the mapping between measured electrical quantities and fault descriptors directly from simulated or measured examples.

In this chapter, three supervised-learning families are considered: artificial neural networks (ANNs), support vector machines (SVMs), and random forests (RFs). These models were selected because they represent different learning principles. ANN models learn nonlinear mappings through trainable weights and activation functions. SVM models rely on margin maximization and kernel-based learning, with support vector classification (SVC) used for fault-type prediction and support vector regression (SVR) used for continuous distance estimation. Random forest models combine many decision trees through ensemble learning, using averaging for regression and majority voting for classification.

The objective of this chapter is not only to introduce these algorithms but also to explain how they are adapted to the fault-location task. Attention is therefore given to the input feature vector, the distinction between regression and classification targets, the role of preprocessing, the mathematical loss or splitting criteria, and the performance metrics used to evaluate each method. Since a high score on the training data alone is insufficient to demonstrate that a model is reliable, the chapter also introduces validation and cross-validation. This provides the theoretical basis required before comparing AI-based methods with conventional impedance-based fault-location techniques.

3.2 Model Validation and Cross-Validation

In supervised learning, a model is trained from a set of known input-output examples. For a fault-location problem, each example may contain measured or simulated electrical features as input and a target, such as the fault type or fault distance, as output. The objective is not simply to reproduce the examples used during training; it is to generalize by producing accurate predictions for fault cases not seen during the learning process.

This distinction is important because a flexible model can sometimes memorize the training data. In that case, the training error becomes small, but the model may perform poorly when the operating condition, fault resistance, fault location, or fault type changes. This phenomenon is known as overfitting. The opposite case is underfitting, where the model is too simple to capture the real relationship between the electrical measurements and the target. Validation is therefore used to estimate how well the trained model is expected to behave on unseen data [51, 52].

A basic approach is to divide the available dataset into three parts. The training set is used to adjust the model parameters. The validation set is used to compare model choices, such as network structure, kernel selection, regularization strength, or tree complexity. The test set is kept aside until the end and is used only to report the final generalization performance. This separation avoids choosing a model only because it fits one particular set of examples.

Cross-validation gives a more stable estimate while using the available dataset efficiently. In k-fold cross-validation, the dataset is divided into k approximately equal subsets called folds. The model is trained k times. In each run, one fold is used for validation and the remaining $k-1$ folds are used for training. If E_j is the validation error

obtained on fold j, the mean cross-validation error is:

$$E_{\mathrm{CV}} = \frac{1}{k} \sum_{j=1}^{k} E_j \tag{3.1}$$

For regression, E_j can be an error measure such as MAE, MSE, or RMSE. For classification, it can be the error rate or another loss for which lower values are better. Scores for which higher values are better, such as accuracy or F1, are averaged directly rather than described as errors. The same validation principle can therefore be used for ANN, SVM, and RF models, even though their internal learning mechanisms differ.

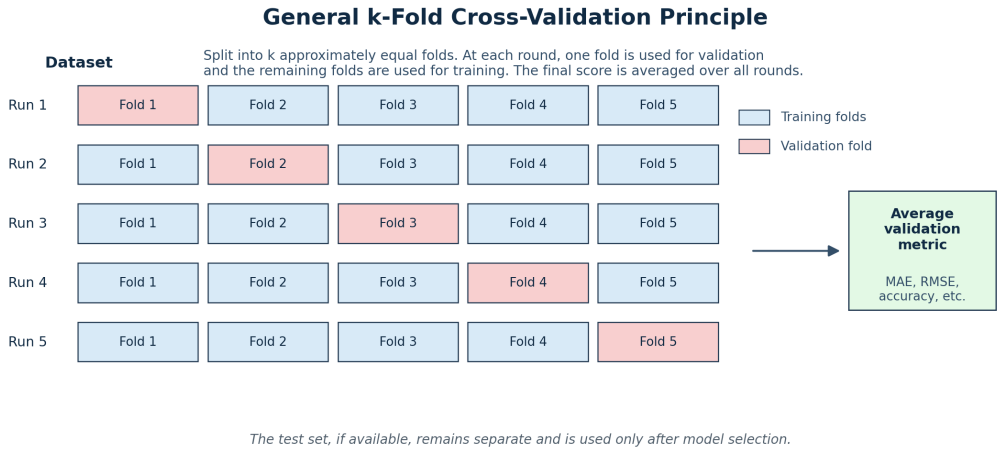


Figure 17: k-fold cross-validation principle

The value of k controls the balance between computational cost and the stability of the validation estimate. A small value of k trains fewer models and is faster, but each score may depend more strongly on the particular split. A large value uses more data for training in each run but requires more repeated training. Cross-validation may be used to compare candidate models and select hyperparameters. When no separate test set is reserved, out-of-fold predictions can instead be assembled so that every case is evaluated exactly once by a model that was not trained on that case [53].

All data-dependent preprocessing must be fitted separately within each training fold. For example, missing-value imputation statistics, scaling parameters, and feature-selection rules are calculated from the training folds only and then applied unchanged to the held-out fold. Fitting these operations before the split would leak information from the validation data and produce an optimistic estimate of performance.

For power-system fault diagnosis, cross-validation is especially useful because the generated cases may contain many combinations of fault type, resistance, location, and operating point. A random split can accidentally make the validation set too similar to the training set, which may give an optimistic impression of performance. Careful validation helps reveal whether the model is learning a general relationship from the electrical features or merely exploiting repeated patterns in the simulated cases. Cross-validation therefore supports the credibility of the final fault-location results independently of the specific AI algorithm used.

3.3 Artificial-Neural-Network-Based Fault Location

3.3.1 Theoretical Background of Artificial Neural Networks

Artificial neural networks (ANNs) are data-driven computational models inspired by the way biological nervous systems process information. In engineering applications, an ANN learns a mapping between an input vector and an output target by adjusting weights and biases from examples. This property is useful in transmission-line fault diagnosis because the relationship among measured voltages, currents, fault resistance, fault type, operating condition, and fault distance is nonlinear and difficult to represent with a single closed-form equation. Neural networks are therefore widely used for fault detection, classification, and location regression in power systems [54, 55, 56].

The elementary processing unit of an ANN is the artificial neuron. For an input vector $\mathbf{x} = [x_1, x_2, \ldots, x_n]^T$, the neuron computes a weighted sum and then applies a nonlinear activation function:

$$z = \sum_{i=1}^{n} w_i x_i + b = \mathbf{w}^T \mathbf{x} + b \tag{3.2}$$

$$a = \phi(z) \tag{3.3}$$

where w_i are trainable weights, b is a bias term, z is the induced local field, $\phi(\cdot)$ is the activation function, and a is the neuron output. Without the nonlinear function $\phi(\cdot)$, stacking many layers would still be equivalent to one linear transformation. Nonlinear activations such as the sigmoid, hyperbolic tangent, and rectified linear unit allow the network to approximate nonlinear functions [54, 55].

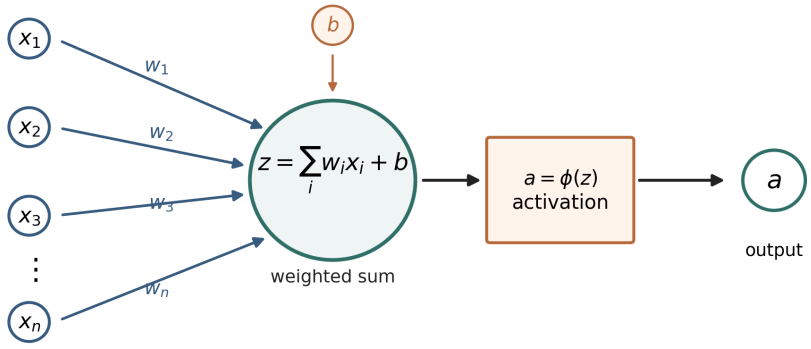


Figure 18: Artificial neuron model

Common activation functions are:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{3.4}$$

$$\tanh(z) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}} \tag{3.5}$$

$$\mathrm{ReLU}(z) = \max(0, z) \tag{3.6}$$

The sigmoid function is useful when an output must remain between 0 and 1, whereas a linear output is commonly used for unrestricted regression. Consequently, a sigmoid output is appropriate when fault distance is represented explicitly as a normalized line fraction, while a linear output is more suitable when the predicted quantity is not bounded.

3.3.2 Multilayer Feedforward ANN

A multilayer perceptron (MLP) consists of an input layer, one or more hidden layers, and an output layer. In a feedforward ANN, information moves from the input layer to the output layer without feedback loops. For layer l, the forward propagation equations are:

$$\mathbf{z}^{(l)} = \mathbf{W}^{(l)}\mathbf{a}^{(l-1)} + \mathbf{b}^{(l)} \tag{3.7}$$

$$\mathbf{a}^{(l)} = \phi^{(l)}\left(\mathbf{z}^{(l)}\right) \tag{3.8}$$

where $\mathbf{a}^{(0)} = \mathbf{x}$, $\mathbf{W}^{(l)}$ is the weight matrix of layer l, $\mathbf{b}^{(l)}$ is its bias vector, and $\mathbf{a}^{(l)}$ is the layer output. The hidden layers progressively transform the original measurements into internal features that are more suitable for classification or regression. The ability of multilayer networks to learn such internal representations is one of the main reasons for their success [55, 57].

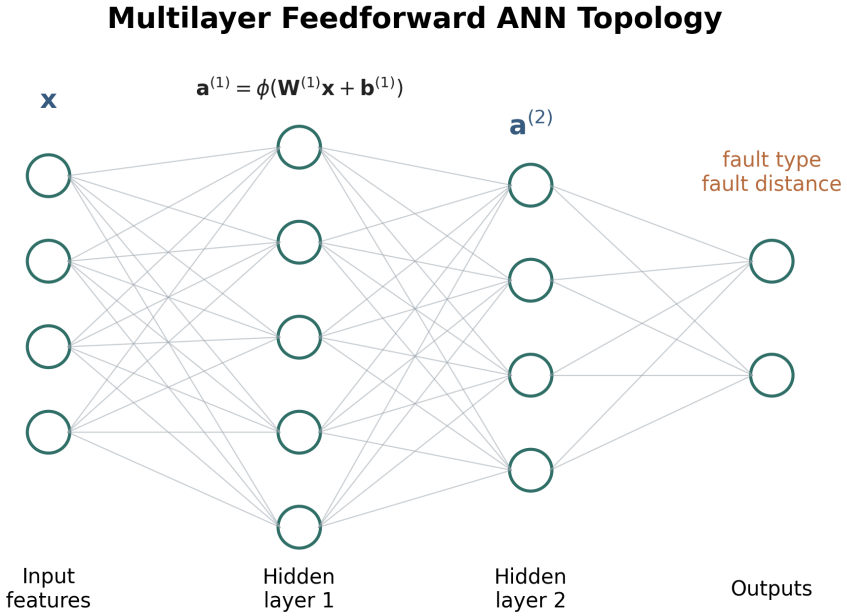


Figure 19: Feedforward ANN topology

In transmission-line fault applications, the input vector may contain phase-domain or sequence-domain measurements such as voltage magnitudes, voltage angles, current magnitudes, current angles, pre-fault quantities, post-fault quantities, and changes between pre-fault and post-fault states. Figure 19 illustrates the general multilayer structure used to map electrical features to fault-classification and location outputs.

3.3.3 Preprocessing of Electrical Measurements

Before training, the electrical measurements must be arranged into a numerical feature matrix. For a dataset of N simulated or measured cases, the training set can be written as:

$$\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{N} \tag{3.9}$$

where $\mathbf{x}_i$ is the feature vector of case i, and $\mathbf{y}_i$ is the corresponding target. In fault classification, $\mathbf{y}_i$ may be a one-hot-encoded vector representing the fault type. In fault location, y_i is usually a continuous value such as the fault distance in kilometres or the normalized distance fraction along the line.

Preprocessing is important because voltage, current, impedance, and resistance features can have different units and numerical ranges. If one feature has a much larger

numerical scale than the others, it can dominate the gradient during training. A common approach is standardization:

$$x' = \frac{x - \mu}{\sigma} \tag{3.10}$$

where μ and σ are the mean and standard deviation computed from the training data. Another common approach is min-max normalization:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{3.11}$$

or, for a symmetric range,

$$x' = 2\frac{x - x_{\min}}{x_{\max} - x_{\min}} - 1 \tag{3.12}$$

Normalization improves numerical stability and usually accelerates convergence, especially when gradient-based optimization is used [54, 55].

3.3.4 Training and Backpropagation

Training an ANN means finding the weights and biases that minimize a loss function. For fault-distance regression, a common loss is the mean squared error:

$$J(\Theta) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{3.13}$$

where Θ represents all trainable parameters, y_i is the true distance, and $\hat{y}_i$ is the network estimate. For fault classification with C classes, the softmax function can be used at the output layer:

$$\hat{p}_{i,c} = \frac{e^{z_{i,c}}}{\sum_{k=1}^{C} e^{z_{i,k}}} \tag{3.14}$$

and the cross entropy loss is:

$$J(\Theta) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log(\hat{p}_{i,c}) \tag{3.15}$$

Mean absolute error is another valid regression loss and is the one retained for the distance output of the ANN used in this thesis. The fault-type output uses sparse categorical cross-entropy because its class targets are integer encoded. In the executed multitask model, the two losses are combined through task-specific weights.

The backpropagation algorithm computes the gradient of the loss with respect to every weight and bias by applying the chain rule from the output layer back to the earlier layers [57]. For a parameter θ, gradient descent updates it as:

$$\theta \leftarrow \theta - \eta\frac{\partial J}{\partial \theta} \tag{3.16}$$

where η is the learning rate. In practice, variants such as stochastic gradient descent, adaptive optimizers, or Levenberg–Marquardt training may be used depending on the network size and the application [54, 55].

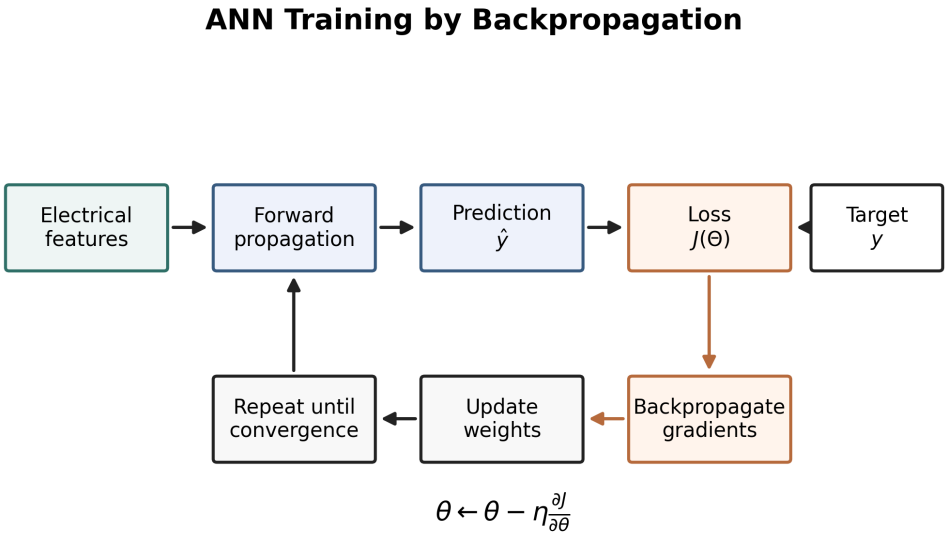


Figure 20: ANN training cycle

3.3.5 ANN Outputs for Fault Classification and Location

In a fault-diagnosis framework, classification and regression can be handled by separate networks or by a multi-output architecture. The implementation used in this thesis is a multitask network: shared hidden layers learn a common electrical representation, after which separate output branches perform fault-type classification and distance regression. The classification branch identifies the fault type, for example AG, BG, CG, AB, AC, BC, ABG, ACG, BCG, ABC, or $ABCG$, while the regression branch estimates the distance to the fault. If the output is normalized as a line fraction $\hat{m}$, the estimated physical distance is:

$$\hat{d} = \hat{m}L \tag{3.17}$$

where L is the total line length. Figure 19 shows an example multilayer ANN architecture used for fault type classification and fault location estimation.

3.3.6 Main ANN Parameters and Their Meaning

The practical behavior of an ANN is controlled by both trainable parameters and design parameters. The trainable parameters are the weights and biases. They are adjusted during learning and define the final mapping between the electrical input features and the predicted outputs. The design parameters, often called hyperparameters, are chosen before training and determine the shape and training behavior of the network [54, 55].

The number of hidden layers controls the depth of the network. A shallow network may not have enough flexibility to represent the nonlinear relationship between voltage and current measurements and fault location. A very deep network, however, can become harder to train and may overfit if the dataset is not sufficiently representative. The number of neurons in each hidden layer controls the width of the representation. More neurons allow the network to learn richer combinations of the input features, but they also increase the number of weights and the risk of memorizing noise.

The activation function determines how each layer transforms its input. ReLU is commonly used in hidden layers because it is simple, computationally efficient, and less affected by vanishing gradients than saturating functions such as sigmoid or hyperbolic tangent. Sigmoid remains useful at an output when the desired value is bounded between 0 and 1. For this reason, if the fault distance is represented as a normalized line fraction,

a sigmoid output is physically meaningful. A linear output is more suitable when the regression target is unbounded or when values outside the training range are acceptable.

The learning rate η controls the size of each weight update during training. If it is too large, the loss may oscillate or diverge; if it is too small, training becomes slow and may stop before reaching a useful solution. The optimizer defines how gradients are used to update the weights. Basic gradient descent uses the direct gradient, while adaptive optimizers modify the effective step size during training.

The batch size is the number of samples processed before one weight update. Small batches introduce more randomness, which can sometimes improve generalization, while large batches give smoother gradient estimates but require more memory. The number of epochs indicates how many times the training algorithm passes through the full dataset. Too few epochs may underfit the data, while too many epochs may overfit.

Regularization parameters are also important. Dropout randomly disables a fraction of hidden neurons during training, forcing the network not to depend too strongly on one path through the model. Batch normalization normalizes intermediate activations inside the network and can make training more stable. Together, these choices help the ANN learn a useful electrical representation instead of only memorizing the training cases [55].

3.3.7 Strengths and Practical Considerations

The ANN approach is useful for fault diagnosis because it can approximate complex nonlinear mappings between electrical measurements and fault descriptors. This is important in transmission line applications, where the measured voltage and current patterns are affected by fault resistance, fault location, operating point, source contribution, and fault type. A multilayer ANN can combine these effects through its hidden layers and can therefore learn relationships that are difficult to express with one explicit analytical equation.

Another advantage of ANN models is their flexibility. The same general architecture can be adapted for classification, regression, or multitask learning. The executed experiment uses a shared multitask structure so that both outputs can benefit from common voltage, current, and sequence-component representations, while task-specific branches retain the flexibility required by the continuous and categorical targets.

However, ANN models also require careful use. Their inputs should usually be scaled because gradient-based training is sensitive to numerical range. The number of layers, number of neurons, activation functions, learning rate, regularization, and training duration must be selected carefully. If the network is too simple, it may underfit the fault behavior. If it is too flexible, it may overfit the simulated cases. For this reason, validation is essential when using ANN models for fault-location studies.

3.4 Support-Vector-Machine-Based Fault Location

3.4.1 Theoretical Background of Support Vector Machines

Support Vector Machines (SVMs) are supervised learning models developed within the framework of statistical learning theory. In this work, the SVM family is used in two separate forms: Support Vector Classification (SVC) for categorical fault decisions and Support Vector Regression (SVR) for continuous fault location estimation. The classification form constructs a decision boundary that separates classes while maximizing the margin between the boundary and the nearest training samples. These nearest samples are called support vectors because they determine the final position of the separating hyperplane [58, 59]. This maximum margin principle is important because it attempts to improve generalization, not only fit the training data.

For a binary classification problem, the training set is written as:

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}, \qquad y_i \in \{-1, +1\} \tag{3.18}$$

where $\mathbf{x}_i$ is the feature vector and y_i is the class label. In the linearly separable case, the separating hyperplane is:

$$f(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b = 0 \tag{3.19}$$

where $\mathbf{w}$ is the normal vector to the hyperplane and b is the bias. The two margin boundaries are:

$$\mathbf{w}^T\mathbf{x} + b = +1, \qquad \mathbf{w}^T\mathbf{x} + b = -1 \tag{3.20}$$

The geometric margin is equal to $2/\|\mathbf{w}\|$. Therefore, maximizing the margin is equivalent to minimizing $\|\mathbf{w}\|^2$, subject to correct classification of the training samples:

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 \tag{3.21}$$

subject to:

$$y_i(\mathbf{w}^T\mathbf{x}_i + b) \geq 1, \qquad i = 1, 2, \ldots, N \tag{3.22}$$

Figure 21 illustrates the separating hyperplane, the margin, and the support vectors.

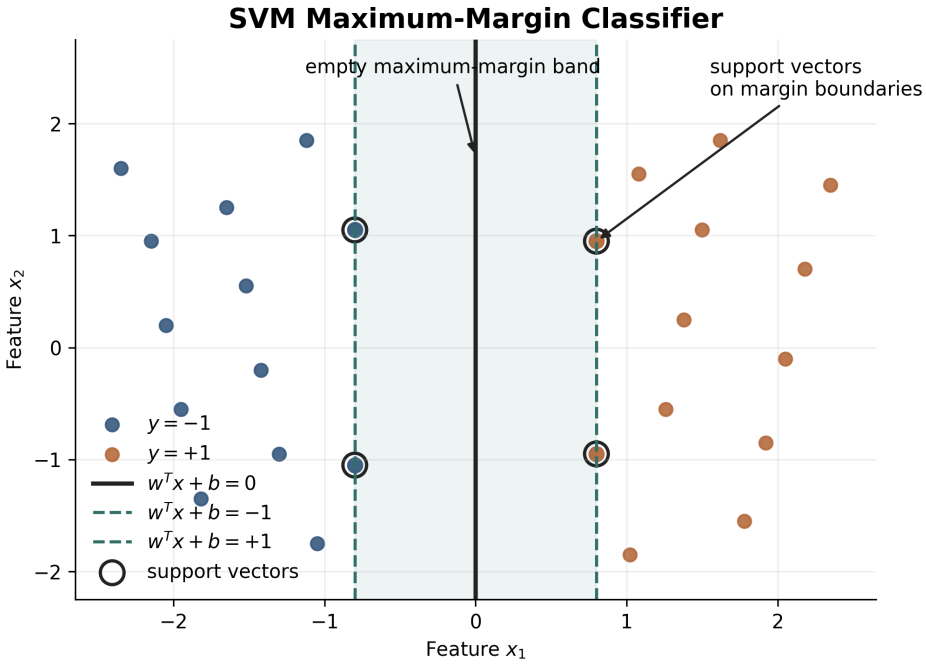


Figure 21: Maximum-margin SVM classifier

3.4.2 Soft Margin SVM

In practical fault diagnosis datasets, perfect linear separation is rarely possible because of noise, measurement uncertainty, different fault resistances, and overlapping operating conditions. The soft margin SVM introduces slack variables ξ_i to allow controlled violations of the margin:

$$\min_{\mathbf{w},b,\xi} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{N}\xi_i \tag{3.23}$$

subject to:

$$y_i(\mathbf{w}^T\mathbf{x}_i + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0 \tag{3.24}$$

The parameter C controls the trade-off between a large margin and training errors. A large C penalizes misclassification strongly, producing a narrower margin that fits the training data more closely. A smaller C allows more violations, often giving a smoother decision boundary and better generalization when the data contain noise.

3.4.3 Kernel Trick for Nonlinear SVC and SVR

Fault behavior in power systems is usually nonlinear. The measured voltage and current features may not be separable by a straight line or plane in the original feature space. To address this, SVMs use a mapping function $\phi(\cdot)$ that transforms the original features into a higher dimensional space:

$$\mathbf{x} \rightarrow \phi(\mathbf{x}) \tag{3.25}$$

Instead of computing $\phi(\mathbf{x})$ explicitly, the kernel trick evaluates inner products in the transformed space using:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j) \tag{3.26}$$

This makes nonlinear classification and nonlinear regression possible while keeping the computation efficient [60]. Figure 22 shows the idea: data that are not linearly separable in the original space can become separable after an implicit transformation.

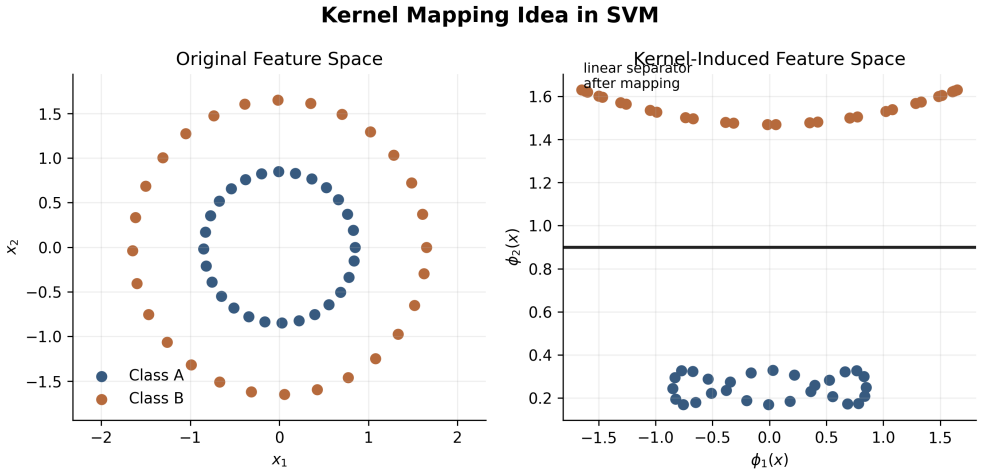


Figure 22: Kernel mapping for nonlinear separation

For both SVC and SVR, a common choice is the radial basis function (RBF) kernel:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2\right) \tag{3.27}$$

where γ controls the width of the kernel. A small γ gives a smoother decision boundary because each support vector has a broader influence. A large γ gives a more localized influence, which can fit complex patterns but may increase the risk of overfitting.

After training, the SVC decision function is:

$$f(\mathbf{x}) = \text{sign}\left(\sum_{i=1}^{N} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b\right) \tag{3.28}$$

where α_i are the Lagrange multipliers. Only samples with nonzero α_i contribute to the decision function; these are the support vectors.

3.4.4 Support Vector Regression

While SVC is used to assign a sample to a discrete class, SVR is used when the target variable is continuous. In fault-location studies, this continuous target may be the estimated distance to the fault, expressed, for example, in kilometres or as a percentage of the line length. For a regression dataset:

$$\mathcal{D}_r = \{(\mathbf{x}_i, t_i)\}_{i=1}^{N}, \qquad -\infty < t_i < \infty, \tag{3.29}$$

SVR searches for a function:

$$\hat{t} = f(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) + b \tag{3.30}$$

that is as flat as possible while keeping the prediction error within an acceptable tolerance. Instead of penalizing every error directly, SVR uses an ε-insensitive loss function:

$$L_\varepsilon(t_i, \hat{t}_i) = \max\left(0, \left|t_i - \hat{t}_i\right| - \varepsilon\right) \tag{3.31}$$

Errors smaller than ε are ignored, which forms an ε-tube around the regression function. Samples outside this tube become support vectors because they influence the fitted regression curve. The soft margin SVR optimization problem can be written as:

$$\min_{\mathbf{w},b,\xi,\xi^*} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{N}(\xi_i + \xi_i^*) \tag{3.32}$$

subject to:

$$t_i - \mathbf{w}^T\phi(\mathbf{x}_i) - b \le \varepsilon + \xi_i, \tag{3.33}$$

$$\mathbf{w}^T\phi(\mathbf{x}_i) + b - t_i \le \varepsilon + \xi_i^*, \tag{3.34}$$

$$\xi_i, \xi_i^* \ge 0 \tag{3.35}$$

The parameter C controls the penalty applied to samples outside the ε-tube, while ε controls how much error is considered acceptable. A larger ε produces a wider tolerance tube and a smoother model, whereas a smaller ε forces the model to follow the training data more closely. With a kernel function, the final SVR prediction has the form:

$$\hat{t} = f(\mathbf{x}) = \sum_{i=1}^{N}(\alpha_i - \alpha_i^*)\,K(\mathbf{x}_i, \mathbf{x}) + b \tag{3.36}$$

Thus, in this thesis, SVC and SVR are not merged into one model. They are trained and evaluated as separate tasks: SVC handles the classification objective, while SVR handles the continuous fault location regression objective.

3.4.5 SVM Workflow for Fault Diagnosis

In a transmission-line or distribution-network fault-location problem, the SVM input vector may contain voltage magnitudes, current magnitudes, phase angles, sequence components, or extracted time-domain features. These features are usually scaled before training because SVMs are sensitive to feature magnitude. Hyperparameters such as C, γ, and, for SVR, ε, are selected using validation data or cross-validation. The trained SVC can classify the fault type, distinguish between faulted and non-faulted conditions, or identify the most probable faulted section, while the trained SVR estimates the continuous fault distance. Figure 23 summarizes this workflow.

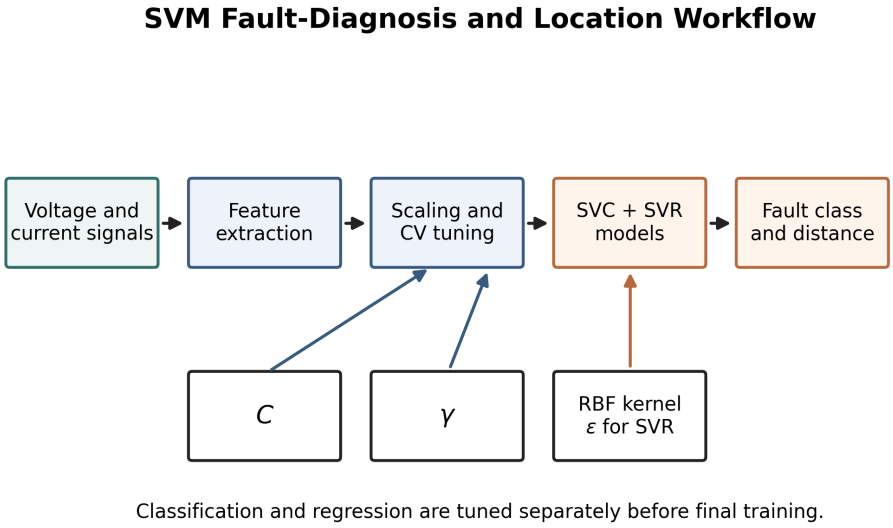


Figure 23: SVM fault-diagnosis workflow

3.4.6 Main SVM Parameters and Their Meaning

The first important SVM choice is the kernel. A linear kernel searches for a linear decision boundary or regression function in the original feature space. This can be appropriate when the classes are nearly linearly separable. In fault location problems, however, the measured voltage and current patterns are usually affected by nonlinear interactions among fault resistance, phase coupling, sequence components, and terminal contributions. A nonlinear kernel, especially the RBF kernel, is therefore often more suitable because it allows the model to build curved boundaries without explicitly creating all nonlinear feature combinations [59, 60].

The parameter $\mathbf{C}$ controls how strongly the model penalizes training errors. A high C gives less tolerance to errors and tends to fit the training data more closely. This may improve training accuracy, but it can also make the model sensitive to noise. A lower C allows more violations of the margin or the regression tube, which often gives a smoother model.

For the RBF kernel, $\boldsymbol{\gamma}$ controls how far the influence of each support vector extends. A small γ gives a broad influence and a smoother decision surface. A large γ makes the influence more local, allowing more complex boundaries but increasing the risk of overfitting.

For SVR, the parameter $\boldsymbol{\varepsilon}$ defines the width of the insensitive tube around the regression function. Errors smaller than ε are ignored during training. A larger ε produces a smoother regression function because the model accepts a wider tolerance, while a smaller ε forces the model to follow the training targets more closely.

Other numerical parameters also affect training. The tolerance controls the stopping criterion of the optimization algorithm, and maximum iterations limits the amount of computation allowed. These parameters do not define the electrical meaning of the model, but they help ensure that the optimization process is stable and reproducible.

3.4.7 Strengths and Practical Considerations

The main strength of the SVM family is its margin-based learning principle. In classification, SVC does not only try to separate the fault classes; it tries to separate them with the largest possible margin. This makes the method less dependent on individual training points that are far from the decision boundary and gives it a strong theoretical

basis for generalization [59].

For nonlinear fault location data, the kernel formulation is especially useful. The RBF kernel allows the model to represent curved decision boundaries and nonlinear regression functions without explicitly constructing a large number of nonlinear features. This is suitable for electrical fault data because voltage magnitudes, current contributions, phase angles, sequence components, and fault resistance interact in a nonlinear way.

SVM models also have practical limitations. They are sensitive to feature scaling because the kernel depends on distances in feature space. Their behavior also depends strongly on hyperparameters such as C, γ, and ε. A very large C or γ can make the model too local and sensitive to noise, while values that are too small can make the model too smooth. In addition, SVC and SVR are naturally separated formulations, so classification and distance regression are usually trained as two models rather than one shared model.

3.5 Random-Forest-Based Fault Location

A random forest (RF) is an ensemble-learning method that combines many decision trees to obtain robust classification and regression models. The method was introduced by Breiman as a way of reducing the high variance of individual decision trees through bootstrap sampling and random feature selection [61]. In this work, the RF family is used in two separate forms: a random forest regressor (RFR) for continuous fault-distance estimation and a random forest classifier (RFC) for categorical fault-type identification. This separation is appropriate because the distance target is numerical, whereas the fault-type target is discrete.

3.5.1 Theoretical Background of Random Forests

3.5.1.1 Decision Tree Splitting

A decision tree recursively partitions the input space into smaller regions. For a training set:

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}, \tag{3.37}$$

where $\mathbf{x}_i$ is the electrical feature vector and y_i is either a continuous target or a class label, each internal node selects a feature and a threshold that best separates the samples. For a regression tree, a common splitting objective is to reduce the target variance or squared error. If a node S is split into left and right child nodes S_L and S_R, the impurity reduction can be written as:

$$\Delta I = I(S) - \frac{|S_L|}{|S|} I(S_L) - \frac{|S_R|}{|S|} I(S_R), \tag{3.38}$$

where, for regression,

$$I(S) = \frac{1}{|S|} \sum_{i \in S} (y_i - \bar{y}_S)^2 \tag{3.39}$$

For classification, the impurity is commonly measured using the Gini index:

$$G(S) = 1 - \sum_{k=1}^{K} p_k^2, \tag{3.40}$$

where p_k is the proportion of samples in node S that belong to class k. A smaller impurity means that the node contains samples that are more homogeneous with respect to the target.

3.5.1.2 Bootstrap Sampling and Random Feature Selection

A single decision tree can fit complex nonlinear patterns, but it is often sensitive to noise and small changes in the training data. A random forest reduces this instability

by training T trees on different bootstrap samples. A bootstrap sample is created by drawing N samples with replacement from the original training set. In addition, at each split, only a randomly selected subset of input features is considered. These two sources of randomness make the trees diverse, so their errors are less correlated. Figure 24 illustrates this training principle.

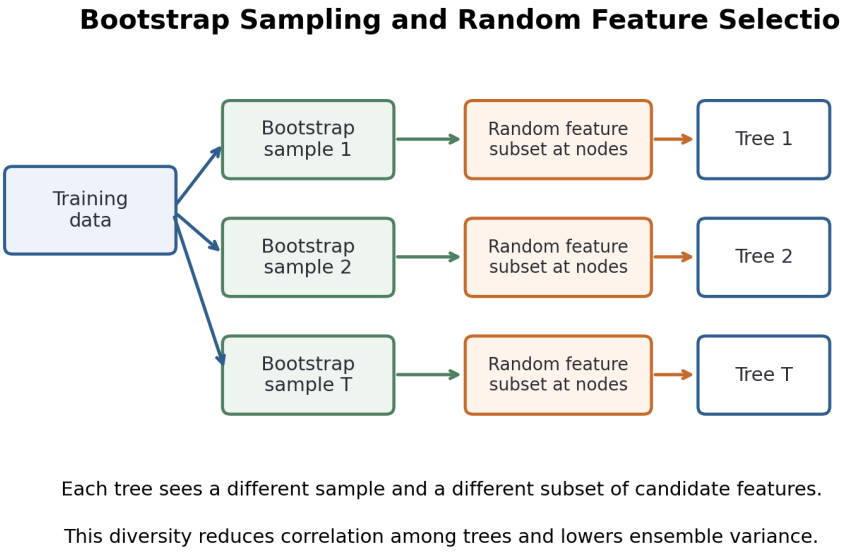


Figure 24: Bootstrap sampling and random feature selection

3.5.1.3 Random Forest Ensemble Structure

The complete RF model is therefore an ensemble:

$$\mathcal{F} = \{h_1(\mathbf{x}), h_2(\mathbf{x}), \ldots, h_T(\mathbf{x})\}, \tag{3.41}$$

where $h_t(\mathbf{x})$ is the prediction of tree t. Figure 25 summarizes the architecture of the forest.

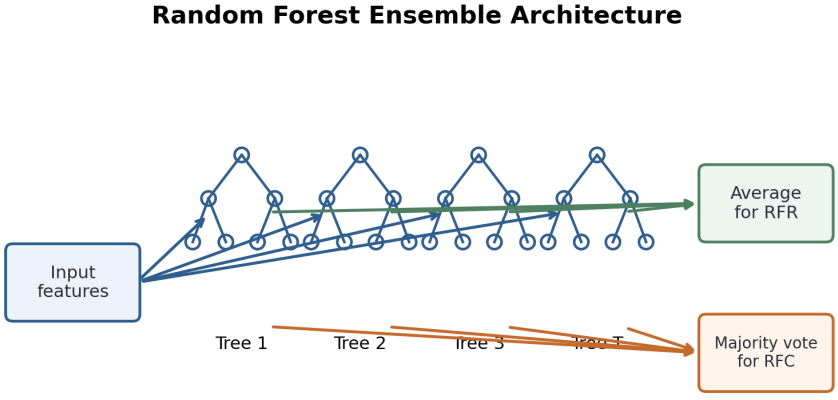


Figure 25: Random Forest ensemble architecture

3.5.2 Random Forest Regression and Classification

3.5.2.1 Random Forest Regressor

For fault location regression, each tree outputs a numerical estimate of the fault distance. The forest prediction is the average of the individual tree predictions:

$$\hat{d}(\mathbf{x}) = \frac{1}{T}\sum_{t=1}^{T} h_t(\mathbf{x}), \tag{3.42}$$

where $\hat{d}$ is the estimated normalized or physical fault distance. Averaging reduces variance because unusually high or low predictions from individual trees are smoothed by the ensemble.

3.5.2.2 Random Forest Classifier

For fault-type classification, each tree produces a class-probability vector. In the implementation used here, the forest averages these probabilities and selects the class with the largest mean probability:

$$\hat{P}(c = k \mid \mathbf{x}) = \frac{1}{T}\sum_{t=1}^{T} \hat{P}_t(c = k \mid \mathbf{x}), \tag{3.43}$$

$$\hat{c}(\mathbf{x}) = \arg\max_k \hat{P}(c = k \mid \mathbf{x}), \tag{3.44}$$

where $\hat{P}_t(c = k \mid \mathbf{x})$ is the probability assigned by tree t, based on the class proportions in its reached leaf. Thus, RF naturally supports both forms required in this thesis: RFR estimates the continuous fault location, while RFC identifies the categorical fault type.

3.5.3 Main Random Forest Parameters and Their Meaning

The most visible random-forest parameter is the number of trees, usually denoted T or implemented as `n_estimators`. Increasing the number of trees generally makes the ensemble more stable because more individual predictions are averaged or voted together. After a certain point, however, adding trees mainly increases computation time while giving only a small improvement [61, 62].

The maximum depth of a tree controls how far each tree is allowed to split. Deep trees can represent detailed nonlinear patterns, but they can also fit noise or very specific training cases. Shallower trees are smoother and less sensitive to individual samples, but they may underfit if the electrical relationship is complex.

The parameter maximum features, often written as `max_features`, controls how many input variables are considered at each split. Using all features can make the trees too similar to one another. Considering only a random subset makes the trees more diverse, which is one of the main reasons random forests reduce variance compared with a single decision tree.

The parameters minimum samples split and minimum samples leaf control how easily the tree can create small branches. A small value allows the tree to follow fine details in the training data, while a larger value forces each split or terminal leaf to contain more samples. Larger values can reduce overfitting and make the learned structure less sensitive to isolated cases.

The bootstrap option determines whether each tree is trained on a sample drawn with replacement from the training set. Bootstrap sampling is central to the random-forest method because it gives each tree a different view of the data. The random state does not

change the mathematical method, but it fixes the random sampling and feature selection so that the same experiment can be reproduced.

For classification, the splitting criterion is often the Gini impurity or entropy. For regression, the criterion is usually based on squared error or variance reduction. These criteria decide which threshold split is most useful at each node. In fault location applications, this means the forest searches for voltage, current, resistance, or sequence component thresholds that separate cases into more homogeneous fault classes or distance ranges.

3.5.4 Out-of-Bag Estimation

Because each tree is trained on a bootstrap sample, some training samples are not selected for that tree. These unused samples are called out-of-bag (OOB) samples. They can provide an internal estimate of generalization error by evaluating each sample only with trees for which it was OOB [61, 62]. OOB scoring is a general RF capability; the results in this thesis use the common five-fold out-of-fold protocol instead.

3.5.5 Feature Importance

Random forests can also provide feature-importance information. In impurity-based importance, a feature is considered important if splits using that feature produce large impurity reductions across the forest. For feature x_j, a simplified importance score can be written as:

$$\mathrm{VI}(x_j) = \frac{1}{T} \sum_{t=1}^{T} \sum_{v \in \mathcal{V}_{t,j}} \Delta I(v), \tag{3.45}$$

where $\mathcal{V}_{t,j}$ is the set of nodes in tree t that split on feature x_j, and $\Delta I(v)$ is the impurity reduction at node v. In the context of fault location, this can indicate which electrical measurements contribute strongly to the prediction, such as voltage magnitudes, current magnitudes, phase angles, sequence components, or impedance-related features. However, impurity-based importance does not establish physical causality and can be biased when predictors differ greatly in cardinality or are strongly correlated.

3.5.6 RF Fault Location Workflow

The RF-based fault-location workflow used in this thesis is summarized in Figure 26. For each simulated fault event, electrical measurements are collected and converted into input features:

- real and imaginary parts, magnitudes, and phase angles of terminal voltages and currents;
- pre-fault, faulted-state, and incremental quantities;
- positive-, negative-, and zero-sequence components; and
- additional known descriptors used by the study, such as fault resistance.

These features serve as inputs to the RF models. The RFR output is the fault distance estimate, while the RFC output is the fault type label.

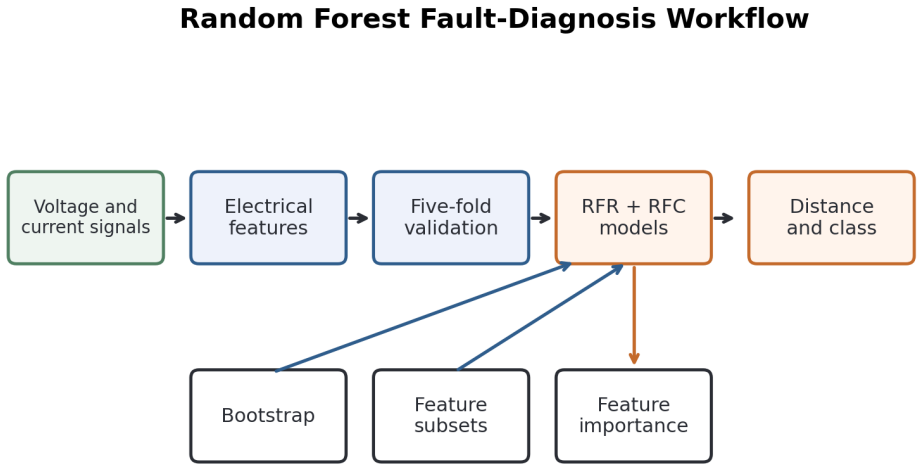


Figure 26: Random Forest fault-diagnosis workflow

3.5.7 Strengths and Practical Considerations

The RF approach offers several benefits for power system fault diagnosis:

- **Nonlinear mapping**: RF can model nonlinear relationships between electrical measurements and fault descriptors without requiring an explicit analytical line model.
- **Separated outputs**: RFR handles fault distance regression, while RFC handles fault type classification.
- **Limited scaling requirements**: Unlike ANN and SVM models, RF does not require feature standardization because tree splits depend on ordered thresholds rather than gradient-based optimization or distances.
- **Interpretability**: Feature-importance values can indicate which voltage, current, sequence, or impedance-related quantities influence the model most strongly, subject to the limitations noted above.
- **Variance reduction**: Ensemble averaging generally makes the model less sensitive than a single tree to fluctuations in the training sample.

The main practical advantage of RF is that it is insensitive to monotonic rescaling of individual features. This is because decision trees use threshold-based splits rather than gradient updates or distance-based kernels. Missing values must still be handled consistently if the selected implementation does not support them directly. RF can also provide feature-importance values, which offer limited insight into the electrical quantities that influence the predictions.

RF also has limitations. It can require a large number of trees for stable results, which increases memory usage and computation time. Very deep trees may still overfit local patterns if the dataset is not representative. In addition, feature importance should be interpreted carefully because correlated input variables can share importance or make one feature appear less important than it physically is. Therefore, RF is robust and practical, but it still requires validation and careful interpretation.

3.6 Hybrid Fault-Distance Estimation

The preceding ANN, SVM, and random-forest approaches learn fault information directly from electrical features. A complementary strategy is to combine an analytical line model with a learning algorithm. In this hybrid approach, the conventional two-ended impedance procedure is not used only to produce one final distance value. Its intermediate location factors and distance estimates are retained as a compact representation of the fault. An ANN or Random Forest regressor then learns how to correct the systematic error that remains in a selected analytical estimate.

This formulation is dedicated to fault-distance estimation. It does not require a fault-type classifier, and the learning stage does not receive the complete voltage and current feature vector used by the purely data-driven models. Instead, it receives quantities already obtained from the conventional equations, such as real and imaginary parts, magnitudes, and angles of sequence-based location factors, together with raw and physically bounded distance candidates. The model therefore learns from information that has already been organized according to the transmission-line equations.

3.6.1 Generation of Physics-Based Candidates

Let V_s, I_s, V_r, and I_r denote synchronized voltage and current phasors measured at the sending and receiving terminals, and let Z_L be the total series impedance of the line. After symmetrical-component transformation, two-ended location factors can be calculated for the positive- and negative-sequence networks. A representative positive-sequence factor is

$$m_1 = \frac{V_{s1} - V_{r1} - Z_L I_{r1}}{Z_L \left(I_{s1} - I_{r1}\right)} \tag{3.46}$$

The same principle can be applied to incremental quantities, where the pre-fault component is removed from the faulted-state phasor:

$$\Delta V_1 = V_{1,\text{fault}} - V_{1,\text{pre}}, \qquad \Delta I_1 = I_{1,\text{fault}} - I_{1,\text{pre}} \tag{3.47}$$

The corresponding incremental two-ended factor is

$$m_{\Delta 1} = \frac{\Delta V_{s1} - \Delta V_{r1} - Z_L \Delta I_{r1}}{Z_L \left(\Delta I_{s1} - \Delta I_{r1}\right)} \tag{3.48}$$

Because terminal orientation determines the direction in which the location factor is measured, a normalized distance candidate can be written in the form

$$\hat{d}_{\text{phys}} = \text{clip}(1 - \Re\{m_{\Delta 1}\},\, 0,\, 1)\,, \tag{3.49}$$

where $\text{clip}(\cdot, 0, 1)$ limits the estimate to the physical line interval. The dimensional distance is then $\hat{D}_{\text{phys}} = L\hat{d}_{\text{phys}}$, where L is the line length in kilometres.

Rather than discarding the remaining analytical information, the hybrid input vector $\mathbf{p}$ collects a bank of conventional physics descriptors. It may include

- raw and clipped conventional distance estimates;
- positive- and negative-sequence two-ended location factors;
- incremental sequence location factors;
- remote-end analytical candidates; and
- the real part, imaginary part, magnitude, and angle of each complex factor.

These quantities are much more compact than a complete electrical feature bank, while preserving several alternative views of the same physical fault-location problem.

3.6.2 Residual Learning and Final Estimate

The analytical candidate already provides an approximate location. Consequently, the ANN or random forest is trained to estimate only its remaining error rather than reconstructing the complete distance mapping from the beginning. For a training case with true normalized distance d, the residual target is

$$r = d - \hat{d}_{\text{phys}} \tag{3.50}$$

The learning model receives the physics vector $\mathbf{p}$ and predicts

$$\widehat{r} = f_\theta(\mathbf{p}), \tag{3.51}$$

where f_θ denotes either the ANN or the random forest regressor. The corrected estimate is

$$\hat{d} = \text{clip}\Big(\hat{d}_{\text{phys}} + \widehat{r},\, 0,\, 1\Big), \qquad \hat{D} = L\hat{d} \tag{3.52}$$

For both hybrid implementations, the physics descriptors are imputed and standardized using parameters calculated from the training fold only. Standardization is essential for ANN optimization and was retained for the RF pipeline to provide both regressors with the same transformed candidate matrix; it does not alter the ordering used by tree thresholds. The ANN learns a nonlinear continuous correction, while the random forest averages the corrections produced by many regression trees. In both cases, preprocessing and model fitting remain inside each training fold, and the validation fold is transformed with the unchanged training parameters.

The complete applied pipeline is shown later in Figure 34. The analytical stage first converts synchronized terminal measurements into sequence and incremental quantities. Several impedance-based candidates are then retained. One candidate acts as the reference estimate, while the complete compact candidate bank is supplied to the residual learner. Finally, the predicted correction is added to the reference and constrained to the physical line interval.

3.6.3 Methodological Interest of the Hybrid Structure

The hybrid structure preserves an explicit connection with transmission-line physics while allowing a data-driven model to represent effects that are difficult to capture with one closed-form expression. These effects may include fault resistance, non-ideal terminal contributions, measurement deviations, operating-point changes, and inverter-based generation. Since the learning stage receives physically meaningful candidates instead of the entire raw feature vector, its role is clearly defined: it refines an analytical estimate rather than replacing the electrical model.

This arrangement also provides a useful hierarchy of estimates. The conventional candidate remains available as an interpretable analytical output, and the ANN or RF correction can be examined separately. If the learning stage is unavailable, the analytical branch can still provide a distance estimate. If it is available, the final result combines the general physical structure of the line equations with the nonlinear correction capability of machine learning.

3.7 Evaluation Metrics for Classification and Regression

The same evaluation principles can be applied to ANN, SVM, and random forest models because the metrics assess the predictions rather than the internal learning mechanism. For fault-distance regression, the principal metrics are mean absolute error (MAE), mean

squared error (MSE), root mean squared error (RMSE), and the coefficient of determination R^2:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{3.53}$$

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{3.54}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{3.55}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{3.56}$$

MAE gives the average error magnitude, MSE gives greater weight to large errors through squaring, and RMSE returns the error measure to the original target unit. The coefficient R^2 compares the residual error with the error obtained by predicting the target mean. A value of 1 indicates perfect predictions, 0 indicates performance equal to that mean baseline, and a negative value indicates poorer performance. When distance is normalized by line length, MAE and RMSE may also be reported as percentages; multiplying the estimated fraction by the line length gives the corresponding physical distance in kilometres.

For classification, a confusion matrix records the true and predicted classes. For one class, its entries are interpreted as true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN). A row-normalized confusion matrix divides each row by the number of true samples in that class, so every row sums to one when the class is present. Accuracy measures the overall proportion of correct decisions:

$$\mathrm{Accuracy} = \frac{N_{\mathrm{correct}}}{N_{\mathrm{total}}} \tag{3.57}$$

Precision measures the reliability of the samples assigned to a class, while recall measures the proportion of its real samples that are recovered:

$$\mathrm{Precision} = \frac{TP}{TP + FP}, \qquad \mathrm{Recall} = \frac{TP}{TP + FN} \tag{3.58}$$

Their harmonic mean is the F1 score:

$$\mathrm{F1} = 2\frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{3.59}$$

In a multiclass problem, these quantities can be calculated separately for each fault type. Macro averaging gives every class equal importance, whereas weighted averaging accounts for the number of samples belonging to each class. These complementary measures are useful because a high global accuracy can otherwise conceal poor behavior for an individual fault category. All reported metrics must be calculated from validation or test cases that were not used to fit the evaluated model [51, 52].

3.8 Comparison of the Three Learning Methods

Although ANN, SVM, and RF can all be applied to fault classification and distance estimation, they do not learn in the same way. Table 3 summarizes their main similarities and differences from the viewpoint of transmission-line fault analysis.

Table 3: Comparison of ANN, SVM, and RF methods

Criterion	**ANN**	**SVM**	**RF**
Learning principle	Learns weights and biases in layered nonlinear transformations.	Uses support vectors to define a maximum-margin classifier or an ε-insensitive regressor.	Averages or votes across many trees trained with bootstrap sampling and random feature selection.
Classification and regression	Can use one multitask network or separate networks.	Uses separate SVC and SVR formulations.	Uses separate classifier and regressor forests.
Nonlinear behaviour	Learns complex voltage-current interactions from representative data.	Represents nonlinear separation through kernels, especially RBF.	Captures nonlinear threshold rules through tree splits.
Preprocessing sensitivity	Needs scaling and imputation for stable gradient-based training.	Needs scaling because kernels depend on feature-space distances.	Less affected by scaling, but missing values still require treatment.
Main strengths	Expressive and suitable for shared representations between related tasks.	Margin-based formulation with strong theoretical grounding.	Practical, robust, and able to report feature-importance indicators.
Main limitations	Sensitive to architecture, training control, and data representativeness.	Sensitive to C, γ, and ε; large datasets can increase computation.	Many trees increase time, and correlated inputs can complicate feature importance.
Interest for fault location	Useful when the mapping from measurements to fault descriptors is highly nonlinear.	Useful when classes or distances are well described by kernel-based boundaries.	Useful as a stable baseline for mixed nonlinear electrical features.

3.9 General Conclusion

The AI-based approaches presented in this chapter provide flexible alternatives to purely analytical fault-location methods. Instead of deriving the fault distance only from impedance equations, these models learn the relationship between electrical measurements and fault descriptors from data. This makes them suitable for situations where the mapping is nonlinear or affected by several interacting variables, such as fault resistance, loading condition, fault type, and measurement location.

Artificial neural networks are powerful nonlinear approximators and can represent complex relationships between input measurements and output targets. Their performance, however, depends strongly on appropriate preprocessing, suitable architecture selection, and representative training data. Support vector machines provide a different learning principle based on margin maximization and kernel functions. In this work, the SVM family is divided into SVC for classification and SVR for regression, which matches the two main fault-analysis objectives. Random forests provide an ensemble-based approach with limited scaling requirements and feature-importance measures that must be interpreted carefully.

Overall, the three methods complement each other. ANN models are flexible and expressive, SVM models are theoretically well grounded through margin-based learning, and RF models reduce the variance of individual decision trees through ensemble averaging. Their common purpose in this thesis is to estimate the fault location and identify the fault type from electrical features, allowing their performance to be compared with conventional impedance-based techniques in later analysis. The hybrid extension serves the distance-estimation task specifically by learning a correction to a conventional analytical

estimate.

Chapter 4 Methodology and Simulation Framework

Summary

This chapter explains how the study was carried out, from the simulated power system to the final evaluation protocol. It first presents the software tools used to build the simulation model, automate the cases, process the data, train the models, and generate the figures. It then describes the IEEE 39-bus New England system, its MATLAB/Simulink implementation, and the validation step used to confirm that the dynamic response of the model remains coherent after a cleared fault. The chapter also presents the three selected operating points, the automated fault simulation campaign, the integration of renewable inverter-based generation, and the transformation of voltage and current signals into usable electrical features. Finally, it explains the preprocessing choices, the five-fold validation protocol, and the practical application of the ANN, SVM, RF, hybrid, and conventional fault-location methods.

4.1 General Introduction

The objective of this chapter is to describe the complete methodological framework used in this work. The study is based on a simulated transmission network, from which voltage and current measurements are extracted under different fault conditions. These measurements are then processed and used by conventional impedance-based methods, data-driven methods, and hybrid analytical-learning methods.

The methodology is organized as a sequential process. First, the IEEE 39-bus New England test system is modeled and validated. Second, three steady-state operating points are obtained using Newton–Raphson power-flow calculations. Third, fault scenarios covering these operating points are generated automatically in MATLAB/Simulink. Fourth, the exported voltage and current signals are converted into numerical features suitable for learning algorithms and conventional equations. Finally, the machine-learning and hybrid models are evaluated using a five-fold validation procedure, while the conventional impedance-based method is used as an analytical reference.

4.2 Software Tools Used

Several software tools were used together to move from the physical simulation of faults to the final numerical evaluation of the methods. MATLAB and Simulink formed the main simulation environment. MATLAB was used to prepare the operating cases, automate the repeated simulations, export the measured signals, and organize the generated data. Simulink was used to represent the dynamic electrical network itself, including nominal-π transmission lines, synchronous machines, control blocks, measurement blocks, and fault-insertion blocks.

Python was used after the simulation stage for data handling, preprocessing, model training, evaluation, and result visualization. The machine-learning part was implemented with scikit-learn, which provides standard and reproducible implementations of classification, regression, preprocessing, and cross-validation tools [63]. Matplotlib was used to generate the figures and performance plots presented in this work [64]. Spyder was used as the main Python development environment because it provides an interactive workflow for inspecting variables, running scripts, and checking intermediate results. Anaconda was used to manage the Python environment and the scientific computing packages required by the study.



(a) MATLAB



(b) Simulink



(c) Python



(d) Matplotlib



(e) scikit-learn



(f) Spyder



(g) Anaconda

Figure 27: Main software tools

4.3 Studied Power System

The studied network is the IEEE 39-bus New England test system, which is widely used for transmission-system analysis [65]. It represents a simplified model of a high-voltage transmission network located in the northeastern United States. The system

comprises 39 buses, 10 synchronous generators, 19 loads, 34 transmission lines, and 12 transformers. The nominal frequency is 60 Hz, and the main transmission voltage level is 345 kV. For buses operating at different voltage levels, the following nominal voltages are assumed: 138 kV for Bus 12, 230 kV for Bus 20, and 16.5 kV for Buses 30 through 38.

The single-line diagram of the studied system is provided in Annexe 1.

The use of this test system is appropriate for the present study because it contains multiple transmission corridors, several synchronous generators, transformer connections, and different line lengths. These characteristics allow the methods to be evaluated across diverse electrical paths within the modeled network.

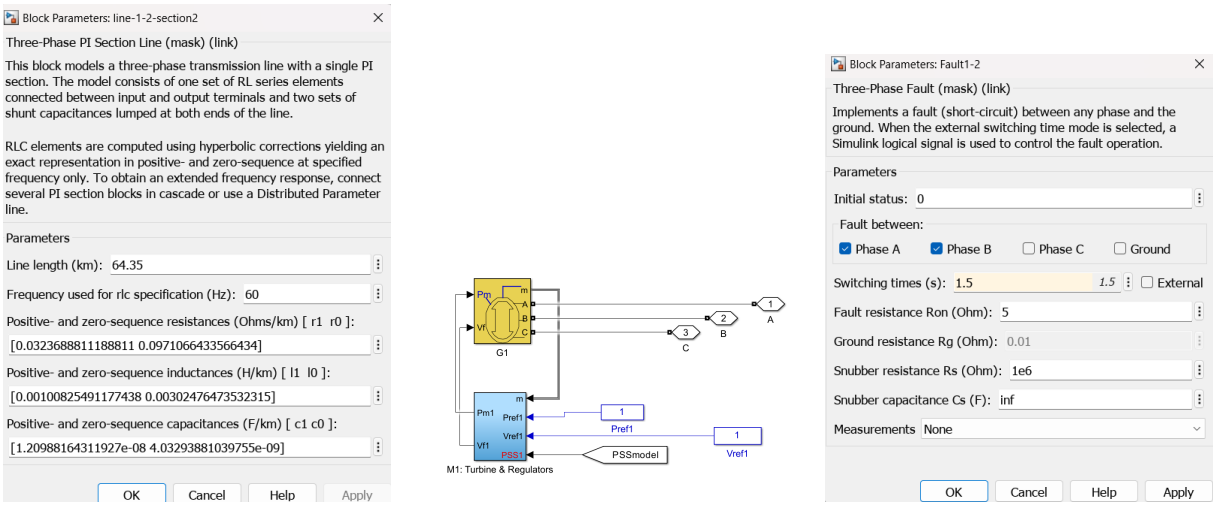
4.3.1 Network Components

The transmission lines are represented by nominal-π section models. This representation accounts for the longitudinal series impedance and shunt capacitance of each line, which is sufficient for the phasor-based transmission-line fault-location analysis considered in this work. The line data are expressed in per unit on a 100 MVA base and converted into the parameters required by the simulation model. Since the original reference data do not provide all physical line lengths directly, the lengths are estimated consistently from the per-unit impedance data and the assumed line reactance per unit length.

The generating units are modeled using synchronous machine blocks. Each synchronous machine is connected to turbine governor and voltage regulation blocks in order to reproduce the electromechanical response of the generators. Power System Stabilizer (PSS) blocks are included in the system to improve damping and to represent the dynamic control behavior of the generating units during and after disturbances.

The loads are modeled as constant active and reactive power demands. The active and reactive load values are based on the IEEE 39-bus data reported in [66]. The transformer data are also represented using per-unit parameters on the 100 MVA base, with tap settings included where required.

The complete MATLAB/Simulink model view is provided in Annexe 2.



(a) Nominal-π line block (b) Generator and control block (c) Fault block

Figure 28: Main Simulink blocks

4.4 Simulation Environment

The network was implemented in MATLAB/Simulink. Simulink was selected because it allows the electrical network, synchronous machines, control systems, fault blocks, and measurement blocks to be represented in a single dynamic simulation environment. This

is important for fault-location studies because the measured voltage and current signals depend not only on the static network impedance but also on the transient response of generators, controllers, and inverter-based sources.

The general simulation structure consists of the following elements:

- PI section blocks for transmission lines;
- synchronous machine blocks for conventional generators;
- turbine governor and excitation/regulator blocks connected to the generators;
- PSS blocks for damping of electromechanical oscillations;
- fault blocks inserted at selected locations on the transmission lines;
- voltage and current measurement blocks at the sending and receiving ends of the faulted line;
- MATLAB scripts used to configure and execute the simulation cases automatically.

The role of MATLAB was not limited to post-processing. It was also used to automate the simulation campaign. This was necessary because the retained database contains 21,317 simulated cases. Performing this number of simulations manually would be impractical and would introduce a high risk of configuration errors. The automation script modifies the case parameters, launches the Simulink model, waits for the simulation to finish, and exports the required measurements.

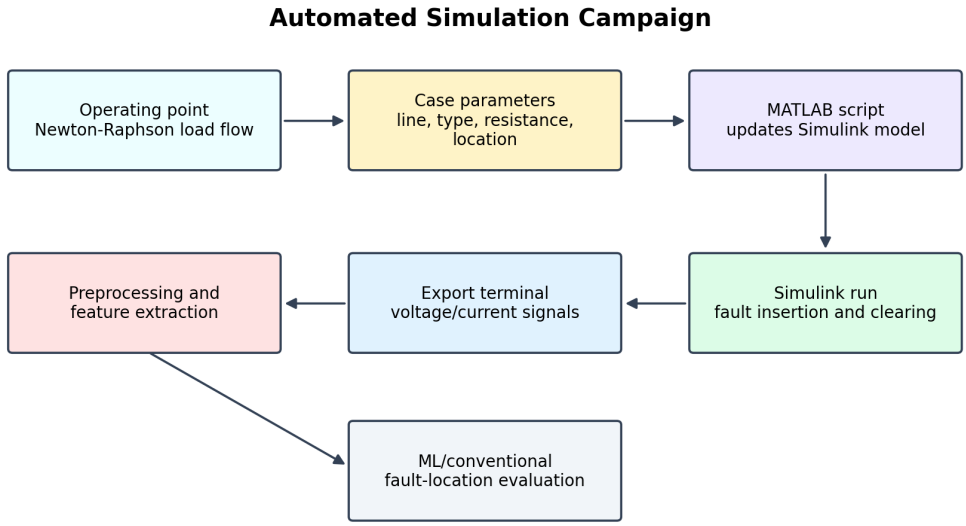


Figure 29: Automated fault-simulation loop

4.5 Operating Point Selection

The voltage profile, line loading, generator dispatch, and power-flow direction can influence the measured terminal voltages and currents during a fault. Three operating points were therefore selected from actual hourly ISO New England demand data reported through the U.S. Energy Information Administration Form EIA-930 [67, 68]. The nominal, light-load, and peak-load conditions were all used in the simulation campaign and are represented within the single database of 21,317 cases.

The steady-state operating points were obtained using Newton–Raphson power-flow calculations. The objective was not to study the power-flow algorithm itself, but to obtain coherent pre-fault voltage magnitudes, phase angles, generator injections, and line flows. The three points were selected as follows:

- **Nominal operating point:** the reference condition, corresponding to the base production and consumption values.
- **Light load operating point:** a low consumption condition selected from the lower part of a representative daily load curve.
- **Peak load operating point:** a high consumption condition selected from the maximum region of the load curve.

This selection represents three levels of network stress: low loading, normal loading, and peak demand. Cases from these three conditions form one combined simulation campaign containing 21,317 valid cases. Thus, 21,317 is the total number of cases across the operating points; it is not the number of cases generated separately at each operating point.

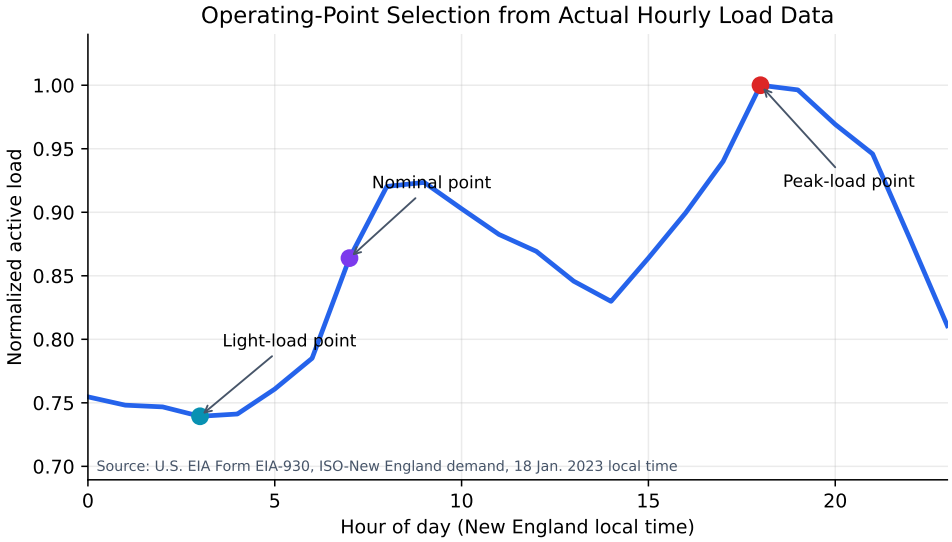


Figure 30: Operating-point selection

4.6 Dynamic Validation of the Simulink Model

Before generating the complete fault-location database, the dynamic behavior of the Simulink model was verified. A temporary fault was applied and then cleared after a specified clearing time. The rotor speeds of the synchronous generators were observed during the disturbance and after fault clearing.

During the fault, the generator speeds are expected to deviate from their pre-fault values because the electrical power output changes suddenly while the mechanical input power cannot change instantaneously. After clearing the fault, the system should not diverge. Instead, the speeds should settle toward stable values, with oscillations damped by the generator control systems and PSS blocks. This behavior confirms that the model is dynamically coherent and suitable for producing fault-location data.

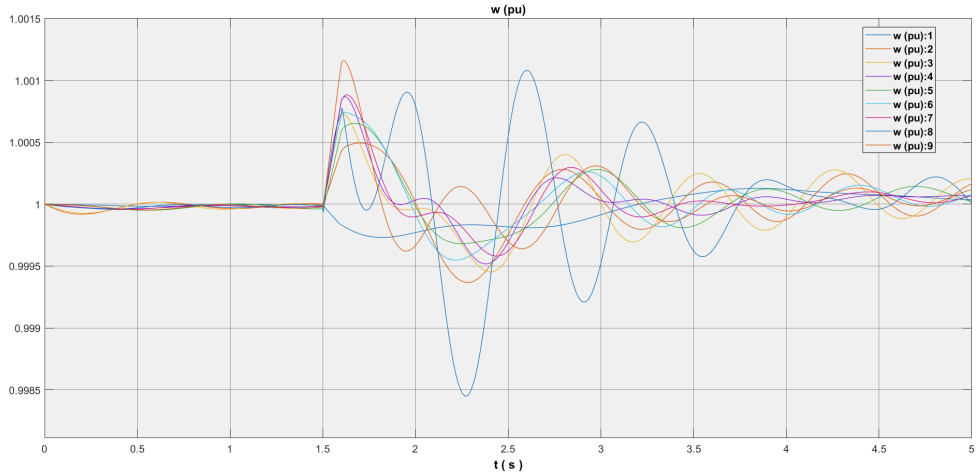


Figure 31: Generator speed during a cleared three-phase fault

The figure shows that the transient speed deviation during the fault is followed by a stable post-fault response. This numerical validation supports the use of the Simulink model for the subsequent automated simulation campaign.

4.7 Fault Simulation Campaign

For each operating point, faults were applied at different locations along the selected transmission lines. The simulation parameters varied across the following dimensions:

- faulted line;
- fault location along the line;
- fault type;
- fault resistance;
- operating point.

The considered fault types include single-line-to-ground faults, phase-to-phase faults, double-line-to-ground faults, and three-phase faults:

$$AG,\ BG,\ CG,\ AB,\ AC,\ BC,\ ABG,\ ACG,\ BCG,\ ABC,\ ABCG \tag{4.1}$$

For every case, three-phase voltages and currents are measured at both ends of the faulted line. These measurements are exported from Simulink and used for conventional fault location and machine-learning feature extraction. After feature extraction and data-quality checks, the retained database contains 21,317 valid cases.

The $ABCG$ class represents a three-phase-to-ground fault and is treated as distinct from the ABC three-phase fault.

Table 4: Composition of the simulated fault database

Simulation dimension	Considered values or scope
Operating conditions	Nominal, light-load, and peak-load operating points
Transmission lines	34 lines of the studied network
Fault locations	19 positions, from 5% to 95% of the line length in 5% steps
Fault resistances	0.01 Ω, 2 Ω, and 5 Ω
Fault classes	11 classes: AG, BG, CG, AB, AC, BC, ABG, ACG, BCG, ABC, and ABCG
Measurements	Three-phase voltages and currents recorded at both line terminals
Total retained size	21,317 valid simulated fault cases across the three operating points

4.8 Renewable Inverter Integration

In addition to the base synchronous-generator system, renewable inverter-based generation was included to study its effect on fault-location performance. This step is important because inverter-interfaced resources do not contribute fault current in the same way as synchronous machines. Instead, their injected current is shaped by the inverter control strategy, current limits, and voltage-support requirements, which can modify the voltage-current relationship used by line-protection and fault-location functions [69, 70].

The renewable source is represented through a controlled current-injection model. The controller computes balanced three-phase current references from the measured voltage and the active- and reactive-power references. A phase-locked loop (PLL) is used to estimate the grid-voltage angle. The current reference is then expressed in the dq frame, where the direct-axis component is associated with active power and the quadrature-axis component is associated with reactive power.

The controller first reconstructs the three-phase voltage phasors from their real and imaginary components. It then extracts the positive-sequence voltage:

$$V_1 = \frac{1}{3}\left(V_a + aV_b + a^2V_c\right), \qquad a = e^{j2\pi/3} \tag{4.2}$$

Using the positive sequence prevents the PLL from following the negative- and zero-sequence components produced by an unbalanced fault. The implemented PI PLL uses a sampling interval of 1 ms, proportional gain $K_p = 20$, and integral gain $K_i = 200$. Its angle estimate is retained between simulation steps through persistent controller states.

$$I_d^* = \frac{P_{ref}}{V_d}, \qquad I_q^* = -\frac{Q_{ref}}{V_d} \tag{4.3}$$

During a voltage dip, the inverter provides additional reactive current according to a voltage support law:

$$I_q^* \leftarrow I_q^* + K_{qv}\max(0, V_{dip} - |V|) \tag{4.4}$$

The total current is limited by the inverter current capacity:

$$\sqrt{(I_d^*)^2 + (I_q^*)^2} \le I_{max} \tag{4.5}$$

The implemented limit is $I_{max} = 1.2$ p.u. When the requested reactive current alone exceeds this value, it is limited to I_{max} and the active current reference is set to zero. Otherwise, the maximum admissible active current is

$$I_{d,\max} = \sqrt{I_{\max}^2 - (I_q^*)^2} \tag{4.6}$$

This gives reactive-current priority during severe voltage dips. The voltage-support action begins below $V_{dip} = 0.9$ p.u. with $K_{qv} = 2$. Finally, the positive-sequence current command is rotated back to the network reference frame and shifted by $\pm 120°$ to obtain balanced phase-current commands. For the three-phase base, the current is calculated as $I_{base} = S_{base}/(\sqrt{3}V_{base})$, with $S_{base} = 100\,\text{MVA}$ and $V_{base} = 22\,\text{kV}$.

Listing 4.1 presents the core of the implemented controller. Initialization checks and small numerical safeguards are omitted from the listing for clarity, but remain present in the simulation function.

Listing 4.1: Core MATLAB implementation of the renewable inverter current controller.

```
% Positive-sequence voltage
va = complex(Vre(1), Vim(1));
vb = complex(Vre(2), Vim(2));
vc = complex(Vre(3), Vim(3));
a  = complex(-0.5, sqrt(3)/2);
V  = (va + a*vb + a^2*vc)/3;

% PI phase-locked loop
phase_error  = wrap_to_pi(angle(V) - theta_state);
pll_int_state = pll_int_state + dt*phase_error;
omega_dev = pll_kp*phase_error + pll_ki*pll_int_state;
theta_state = wrap_to_pi(theta_state + dt*omega_dev);

% Active- and reactive-current references
Vdq = V*exp(-1j*theta_state);
Vd  = real(Vdq);
Id_ref = Pref/Vd_safe;
Iq_ref = -Qref/Vd_safe;

% Reactive support during voltage dips
if abs(V) < Vdip
    Iq_ref = Iq_ref + Kqv*(Vdip - abs(V));
end

% Current limit with reactive-current priority
if abs(Iq_ref) > Imax
    Iq_ref = sign_with_zero(Iq_ref)*Imax;
    Id_ref = 0;
else
    Id_max = sqrt(max(Imax^2 - Iq_ref^2, 0));
    Id_ref = sign_with_zero(Id_ref)*min(abs(Id_ref), Id_max);
end

% Balanced three-phase current commands
Ipos = complex(Id_ref, Iq_ref)*exp(1j*theta_state);
Ia = Ipos*Ibase;
Ib = Ipos*exp(-1j*2*pi/3)*Ibase;
Ic = Ipos*exp( 1j*2*pi/3)*Ibase;
```

4.8.1 Inverter Validation and Study Scope

Before generating the fault cases, several preliminary tests were conducted to verify that replacing the synchronous generator with the inverter-based source did not disturb the initial operating condition of the network. Under steady-state conditions, the active and reactive powers injected by the inverter were checked against the corresponding powers supplied by the synchronous generator before its replacement. The values were found to match, showing that the inverter preserved the scheduled power injection at the selected operating point. The voltage and frequency behavior was also observed after the replacement, and the system remained stable. These checks were necessary to ensure that subsequent differences in fault behavior were caused by the inverter control and current-limiting characteristics rather than by an incorrectly initialized operating condition.

The fault simulations including renewable generation were conducted only at the nominal operating point. Extending the complete campaign to all operating points would have required repeating a very large number of dynamic Simulink runs. Since each run involves initialization, fault application, transient simulation, signal export, and post-processing, the total computation time becomes considerable when thousands of fault combinations are considered. Owing to the available time and computational resources, the nominal point was therefore retained as a representative case. This reduced study was intended to provide an initial assessment of how inverter-based generation affects conventional and data-driven fault-location methods, rather than an exhaustive evaluation over every loading condition.

This behavior changes the voltage–current relationship seen by impedance-based fault-location methods. Renewable integration is therefore a useful stress case for comparing conventional impedance-based estimation with data-driven models trained on voltage and current patterns affected by inverter control. The objective is not to reproduce every possible manufacturer-specific inverter response, but to test whether the studied methods remain reliable when the fault current is limited and control-dependent, as reported in protection studies involving inverter-based resources [69, 70].

4.9 Data Acquisition and Feature Extraction

The raw outputs of each simulation are the three-phase voltage and current measurements recorded at the two ends of the faulted line. The two terminals are denoted as the sending end and the receiving end, and the corresponding measured quantities are V_s, I_s, V_r, and I_r. Rather than being used only as raw time samples, these signals are converted into phasor-based descriptors so that each simulated case is represented by a fixed numerical vector.

For each terminal quantity, three signal states are extracted. The first is the pre-fault value, taken as the last available phasor before the fault-inception time. The second is the post-fault value, obtained by averaging the samples in a selected post-fault window. The third is the superimposed, or delta, value:

$$\Delta X = X_{\text{post}} - X_{\text{pre}} \tag{4.7}$$

This organization was used because the three states carry different information. The pre-fault value describes the loading condition before the disturbance, the post-fault value describes the disturbed network response, and the delta value emphasizes the change produced by the fault itself.

Each complex phasor is written as:

$$X = X_{re} + jX_{im} \tag{4.8}$$

Since the learning models operate on real-valued feature vectors, every complex phasor is represented through its real part, imaginary part, magnitude, and phase angle:

$$|X| = \sqrt{X_{re}^2 + X_{im}^2}, \qquad \angle X = \text{atan2}(X_{im}, X_{re}) \tag{4.9}$$

In addition to the phase-domain values, symmetrical components are computed for every three-phase voltage and current phasor:

$$X_0 = \frac{1}{3}(X_a + X_b + X_c), \tag{4.10}$$

$$X_1 = \frac{1}{3}(X_a + aX_b + a^2X_c), \tag{4.11}$$

$$X_2 = \frac{1}{3}(X_a + a^2X_b + aX_c), \qquad a = e^{j2\pi/3} \tag{4.12}$$

The zero-, positive-, and negative-sequence components are useful because they make the unbalance created by different fault types more visible. Ground and line-to-line faults tend to produce negative- and zero-sequence signatures, while balanced three-phase faults are mainly expressed through positive-sequence quantities.

For the machine-learning part, the retained input set is the electrical-core feature profile. It contains the fault resistance and the voltage/current descriptors from both line terminals:

`fault_resistance_ohm`, `send_v_*`, `send_i_*`, `recv_v_*`, `recv_i_*` (4.13)

This profile produced 289 input features. Columns that directly identify the case or contain the answer, such as the run index, line metadata, validation fold, true distance, and fault type, were excluded from the input matrix. This was necessary to prevent target leakage and to ensure that the models were learning from electrical behavior rather than from labels or identifiers. Fault resistance was retained because it is a controlled and known simulation parameter in this study. In a field implementation, it would need to be estimated independently or omitted when unavailable; consequently, the reported models should not be interpreted as directly deployable without addressing this input requirement.

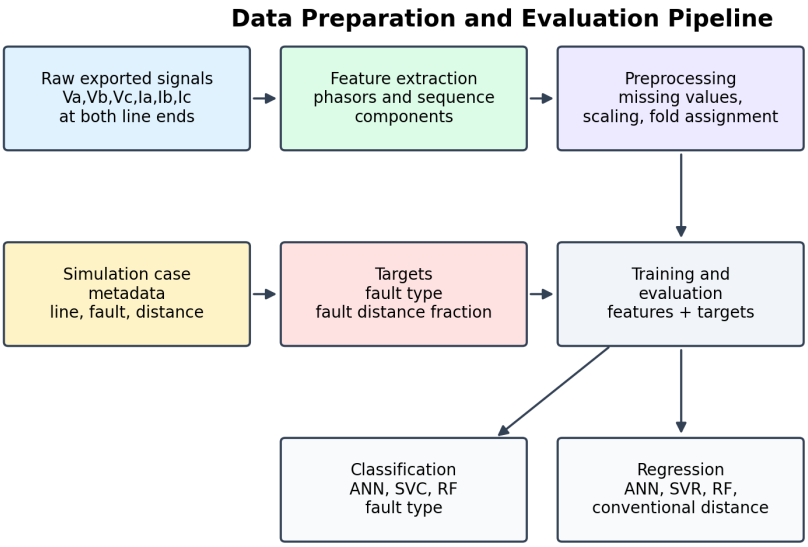


Figure 32: Preprocessing and evaluation pipeline

4.10 Data Preprocessing

The extracted case table was prepared using the principles introduced in the preceding chapter. The distance target was represented as the normalized fault distance fraction:

$$y = \frac{d_{\text{true}}}{L}, \tag{4.14}$$

where d_{true} is the simulated fault distance from the sending end and L is the total length of the faulted line. The normalized target is used because the studied network contains lines with different lengths. A fixed error in kilometres does not have the same meaning on a short line and on a long line. After prediction, the kilometre error is recovered as:

$$e_{\text{km}} = |\hat{y} - y|L \tag{4.15}$$

Missing numerical values were replaced by the median calculated from the training fold only. The same median was then applied unchanged to the validation fold; an undefined median was replaced by zero.

The ANN and SVM inputs were standardized using:

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}, \tag{4.16}$$

where μ_j and σ_j were calculated from the training fold only. This choice was retained because the electrical core vector combines quantities with different numerical ranges and because both gradient based ANN training and the distance based RBF kernel benefit from comparable feature scales.

The Random Forest pipeline used the median imputed values without standardization because its threshold decisions depend on feature ordering rather than numerical scale.

4.11 Five-Fold Validation Protocol

The machine-learning models were evaluated using k-fold cross-validation with $k = 5$. This value was chosen as a practical compromise between reliability and computational

cost. A smaller number of folds would make each validation subset larger, but the final estimate would depend more strongly on a small number of splits. A much larger number of folds would use slightly more data for training at each run, but it would also require more repeated training, which is costly for ANN, SVM, and RF models. With five folds, each run uses approximately 80% of the available cases for training and 20% for validation, and every case is validated once.

The fold assignment was generated before model training and stored in the dataset as the column `cv_fold`. The objective was to evaluate the models on cases that were not used during training while keeping the validation subsets representative of the studied fault scenarios. Therefore, the fold generation was designed so that the different line cases and fault categories were distributed across the five validation runs.

To avoid an unbalanced split, the cases were first separated according to the simulated line and fault type. Inside each subset, the cases were sorted using the fault distance, fault resistance, run index, and folder name. They were then assigned to the five folds in a round-robin order:

$$\text{fold}(i) = i \bmod 5 \tag{4.17}$$

This procedure distributes nearby fault cases across the folds instead of placing one continuous block of distances or resistances entirely in the same fold. As a result, each fold contains a representative mixture of line cases, fault types, fault locations, and fault resistances.

During each validation run, one fold is held out and the remaining four folds are used for training:

$$\mathcal{D}_{\text{val}}^{(j)} = \{(\mathbf{x}_i, y_i) : \texttt{cv_fold}_i = j\}, \tag{4.18}$$

$$\mathcal{D}_{\text{train}}^{(j)} = \{(\mathbf{x}_i, y_i) : \texttt{cv_fold}_i \neq j\} \tag{4.19}$$

This condition guarantees that the same simulated case cannot appear in both the training set and the validation set during the same run. The validation fold is also excluded from all preprocessing calculations. Numerical medians, standardization means, and standardization deviations are computed only from $\mathcal{D}_{\text{train}}^{(j)}$, then applied unchanged to $\mathcal{D}_{\text{val}}^{(j)}$. In addition, the `cv_fold` column itself, together with true distance, fault type, line identifiers, and folder identifiers, is removed from the input feature matrix. This prevents the model from using the fold number or target information as a shortcut.

The validation predictions from the five runs are assembled into one out-of-fold prediction vector, so every retained case is evaluated exactly once by a model that did not train on that case. The principal reported metrics are calculated from this complete out-of-fold vector. Fold-specific metrics are retained as diagnostics for variation among the five validation subsets. Because every physical line is represented in both the training and validation portions, this protocol measures interpolation to new fault cases on known modeled lines; it does not demonstrate generalization to an entirely unseen transmission line.

4.12 Application of Machine Learning Methods

The standalone machine-learning study was carried out using the electrical-core dataset containing 21,317 retained fault cases. The objective was to estimate two outputs from the same family of electrical measurements: the fault type and the fault distance. The classification target is the fault class, while the regression target is the normalized distance fraction y. The ANN handles both tasks jointly through a multitask architecture,

whereas SVM and Random Forest use separate classification and regression models. The hybrid methods described later perform distance regression only.

Table 5: Retained fault-location configurations

Method	Preprocessing	Retained configuration
ANN	Median imputation and standardization	Joint multitask network with shared layers $512-256-128$, distance branch $64-32$, and fault branch 64; batch normalization, ReLU activations, dropout 0.03, Adam, sigmoid distance output, and 11-unit softmax fault output
SVM	Median imputation and standardization	Separate RBF models: SVR with $C = 10$, $\gamma = 0.01$, and $\epsilon = 0.005$; SVC with $C = 30$ and $\gamma = \text{scale}$
Random Forest	Median imputation without scaling	Separate regressor and classifier; 100 trees, unrestricted depth, square root feature selection, bootstrap sampling, and `random_state` = 42
Hybrid ANN	Training-fold median imputation and standardization	Distance-only residual regressor using 27 conventional physics-derived inputs; reference from the incremental positive-sequence two-ended candidate; hidden layers $64-32-16$, linear correction output, Adam optimization, early stopping, and deterministic fold seeds
Hybrid RF	Training-fold median imputation and standardization	Distance-only residual regressor using the same 27 conventional physics-derived inputs and reference candidate; 500 trees, all candidate features available per split, bootstrap sampling, and `random_state` = 42
Conventional	Phasor preparation and sequence component calculation	Two-ended impedance-based distance calculation using synchronized terminal quantities and known line parameters

4.12.1 Artificial Neural Network

The ANN implementation used one joint multitask feedforward network receiving the 289 standardized electrical-core features. Its first layers are shared by both tasks so that the model can learn a common representation of the terminal voltages, currents, sequence components, and fault resistance. The shared representation then divides into a distance-regression branch and a fault-type-classification branch.

The shared trunk contains fully connected layers with 512, 256, and 128 neurons. The distance branch then contains 64 and 32 neurons, while the classification branch contains 64 neurons. Batch normalization and ReLU activation are applied in the hidden layers, followed by dropout with a rate of 0.03. The distance branch ends with one sigmoid unit because the normalized target is constrained to $y \in [0, 1]$, and the classification branch ends with an 11-unit softmax layer. Adam is used with a learning rate of 10^{-3}. Training is allowed to run for at most 120 epochs with a batch size of 256 and an early-stopping patience of 25 epochs.

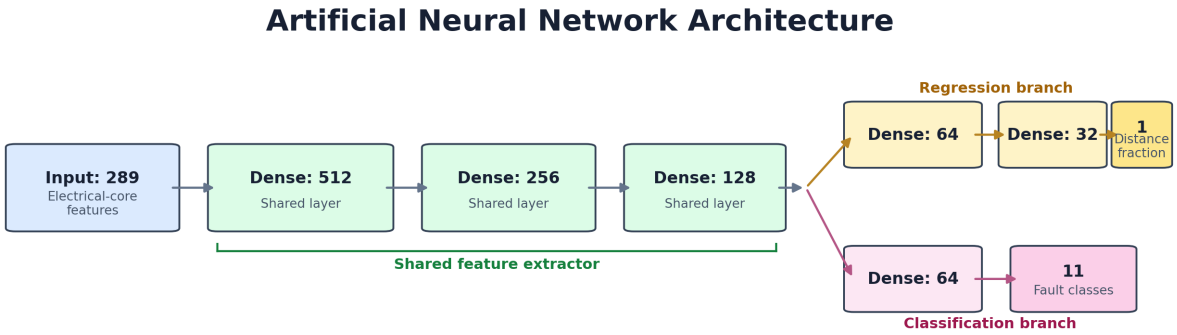


Figure 33: Joint ANN architecture

The distance output minimizes mean absolute error and the classification output minimizes sparse categorical cross-entropy. They are optimized together using the weighted multitask objective

$$\mathcal{L}_{\text{total}} = 8.0\,\mathcal{L}_{\text{distance}} + 0.3\,\mathcal{L}_{\text{fault}} \tag{4.20}$$

The larger distance-loss weight reflects the principal fault-location objective, while the classification task supplies a related learning signal through the shared representation. Early stopping monitors the validation distance MAE, and the learning rate is reduced when validation improvement stalls.

4.12.1.1 Choice of Network Depth and Number of Neurons

There is no universal formula that gives the ideal number of hidden layers or neurons for a fault-location problem. The architecture must provide enough capacity to represent nonlinear relations between the electrical measurements and each target without introducing unnecessary complexity. The selected tapered structure was therefore retained empirically by considering validation behavior and computational cost.

The first shared hidden layer receives the complete 289-dimensional electrical-core vector. The shared reduction,

$$289 \longrightarrow 512 \longrightarrow 256 \longrightarrow 128, \tag{4.21}$$

produces a common representation while preserving enough capacity to model interactions among terminal voltages, currents, sequence components, phase angles, and fault resistance. This representation is then processed by the $64-32$ distance branch and the 64-neuron fault branch. The task-specific layers allow the two outputs to specialize after benefiting from the same learned electrical representation. Batch normalization, dropout, and early stopping limit overfitting.

4.12.1.2 Weight Initialization and Reproducibility

Before training begins, the ANN weights must be assigned initial values. Initializing every weight to the same value would be unsuitable because neurons belonging to the same layer would produce identical outputs and receive identical gradient updates. They would therefore continue to learn the same function, preventing the layer from developing distinct feature representations. To break this symmetry, the dense layer weights were initialized with the default Keras Glorot uniform, also called Xavier uniform, initialization. For a layer with n_{in} input connections and n_{out} output connections, each initial weight is sampled from:

$$w_{ij} \sim \mathcal{U}\left(-\sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}}, \sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}}\right) \tag{4.22}$$

The interval therefore depends on the size of the layer rather than using one fixed range for the entire network. Its purpose is to keep the scale of the activations and gradients reasonably controlled as information moves forward and backward through the network at the beginning of training. This reduces the risk of starting with signals that are excessively large or extremely small.

The bias terms of the dense layers were initialized to zero. This does not create the symmetry problem described above because the corresponding weights are initialized differently. Dropout layers do not contain trainable weights; they randomly deactivate a fraction of the hidden activations only during training.

The initialization is only the starting point of the optimization. During training, backpropagation computes the gradient of the weighted multitask loss with respect to every trainable parameter. The shared layers therefore receive gradient contributions from both outputs, while each task-specific branch is updated by its corresponding loss. Adam then updates the weights and biases iteratively.

To support reproducibility, the TensorFlow and NumPy random seeds were set to 42 before the five-fold loop. A new multitask network is created for each fold and trained in a fixed order. Repeating the experiment with the same data order, software environment, and hardware settings therefore uses the same pseudorandom sequence; minor numerical differences can nevertheless occur on nondeterministic hardware kernels.

4.12.2 Support Vector Machine

The SVM approach used separate models for the two targets: SVR for the normalized distance fraction and SVC for the fault type. Both models received the same 289 median imputed and standardized electrical core features.

The retained SVR used the RBF kernel with $C = 10$, $\gamma = 0.01$, and $\epsilon = 0.005$. The retained SVC also used the RBF kernel, with $C = 30$ and $\gamma = \text{scale}$. The nonlinear kernel was selected because the measured response depends jointly on terminal quantities, fault resistance, phase unbalance, and sequence components; the selected ϵ prevents the regressor from reacting to every very small distance deviation. The general roles of C, γ, ϵ, and the RBF kernel are given in Chapter 3.

4.12.2.1 Model Initialization and Reproducibility

Unlike the ANN, the SVR and SVC do not begin with randomly initialized weights that are progressively updated through backpropagation. Their support vectors, intercepts, and dual coefficients are obtained by solving a convex optimization problem. Consequently, the user does not need to specify an initial weight matrix, and repeating the training with the same samples, preprocessing, and hyperparameters produces the same fitted solution apart from negligible numerical differences.

Before each fold, the SVM models were therefore configured through their hyperparameters rather than through trainable weight initialization. The kernel type, C, γ, ϵ for SVR, convergence tolerance, and maximum number of solver iterations were fixed before fitting. The standardization parameters were calculated only from the training portion of the fold and then applied unchanged to its validation portion. This keeps the preprocessing reproducible and prevents information from the validation cases from influencing the fitted SVM.

4.12.3 Random Forest

Separate Random Forest models were fitted for normalized distance regression and fault type classification. Both received the 289 median imputed electrical core features without standard scaling. Each model used 100 trees, unrestricted depth, the square root of the available features as candidates at each split, a minimum of two samples for splitting, and a minimum of one sample per leaf.

Square-root feature selection and bootstrap sampling were retained to diversify the trees, while the 100-tree ensemble provided a practical balance between prediction stability and computation. This method also supports feature-importance analysis and exposes threshold-based relations between the electrical quantities and the targets. The general ensemble, voting, averaging, and splitting principles are described in Chapter 3.

4.12.3.1 Random Initialization and Reproducibility

Random Forest does not initialize numerical weights. Its randomness enters through the construction of the ensemble. Each tree is trained from a bootstrap sample drawn from the training cases, and only a randomly selected subset of features is examined when a split is considered. These two mechanisms allow the individual trees to learn different structures even though they are fitted from the same training fold.

In this work, the pseudorandom generator was initialized with `random_state` $= 42$ for both the regressor and the classifier. The seed does not determine the tree parameters directly; it fixes the sequence of bootstrap samples and candidate feature subsets used during forest construction. Therefore, rerunning the method with the same fold, data order, software configuration, and seed recreates the same ensemble. The split thresholds and leaf predictions are then learned from the selected training samples. This initialization choice supports reproducible comparison while preserving the diversity required by the forest.

4.13 Application of the Hybrid Method

After the standalone ANN, SVM, and Random Forest models, a distance-only hybrid method was defined to combine the conventional two-ended formulation with ANN and Random Forest regression. This hybrid experiment was applied to the renewable-generation subset, which contains 1,140 valid cases from two studied lines, ten fault classes, three fault resistances, and 19 fault locations. The objective of this structure is not to send the complete 289-feature electrical vector to another model. Instead, the conventional calculation is first used to generate a compact set of analytical location descriptors. These intermediate impedance-location factors and distance estimates are then used to correct a reference analytical estimate.

No fault-type classifier is required in this hybrid distance-estimation stage. The retained inputs are produced directly from synchronized measurements at the two terminals and from the known line impedance. They contain:

- the raw conventional distance fraction;
- the conventional distance fraction clipped to the physical interval $[0, 1]$;
- positive- and negative-sequence two-ended location factors;
- incremental positive- and negative-sequence location factors;
- remote-end analytical location factors; and
- the real part, imaginary part, magnitude, angle, and converted distance candidate associated with each complex factor.

This selection produced a compact set of 27 analytical inputs. Raw phase and sequence voltage/current descriptors from the 289-feature machine-learning profile were not supplied directly to the hybrid regressors. They were used only by the analytical preprocessing stage to calculate the conventional location descriptors.

4.13.1 Reference Candidate and Residual Target

The reference estimate was obtained from the incremental positive-sequence two-ended factor:

$$m_{\Delta 1} = \frac{\Delta V_{s1} - \Delta V_{r1} - Z_L \Delta I_{r1}}{Z_L \left(\Delta I_{s1} - \Delta I_{r1}\right)} \tag{4.23}$$

Its normalized physical distance candidate is

$$\hat{d}_{\text{phys}} = \text{clip}(1 - \Re\{m_{\Delta 1}\},\, 0,\, 1) \tag{4.24}$$

Rather than predicting the complete fault distance independently, the ANN and Random Forest models learn the remaining analytical error. For each training case, the residual target is

$$r = d_{\text{true}} - \hat{d}_{\text{phys}} \tag{4.25}$$

If $\mathbf{p}$ denotes the vector of conventional analytical descriptors, the learned correction is

$$\widehat{r} = f_\theta(\mathbf{p}), \tag{4.26}$$

and the final normalized estimate is

$$\hat{d} = \text{clip}\Big(\hat{d}_{\text{phys}} + \widehat{r},\, 0,\, 1\Big) \tag{4.27}$$

The distance in kilometres is obtained only after prediction:

$$\hat{D} = L\hat{d}, \tag{4.28}$$

where L is the total line length.

4.13.2 Hybrid ANN and Random Forest Pipelines

For the hybrid ANN, missing candidate values were replaced using medians calculated from the training fold. The physics inputs were then standardized with the training-fold mean and standard deviation. A feedforward regression network received the standardized candidate vector and produced one continuous residual correction.

For the hybrid Random Forest, the same fold-specific median imputation and standardization were applied. Scaling is not mathematically required by tree thresholds, but it was retained in the executed hybrid pipeline so that both regressors received the same transformed candidate matrix. Standardization preserves the ordering of each feature and therefore does not change the set of threshold partitions available to a tree.

Both implementations used the existing five-fold assignments of the 1,140-case renewable subset. The fold sizes were 240, 240, 220, 220, and 220 cases. For fold j, the imputer, scaler, and residual regressor were fitted using only the four training folds. The unchanged preprocessing parameters and fitted model were then applied to the held-out fold. The five validation predictions were finally assembled as out-of-fold estimates so that every case was evaluated exactly once.

Figure 34 illustrates the complete application pipeline. The conventional branch converts synchronized measurements into a bank of analytical location descriptors. One descriptor provides the reference distance, while the complete descriptor vector is used by

ANN or Random Forest to predict a residual correction. The analytical estimate therefore remains visible and interpretable, while the learning stage is restricted to refining its remaining error.

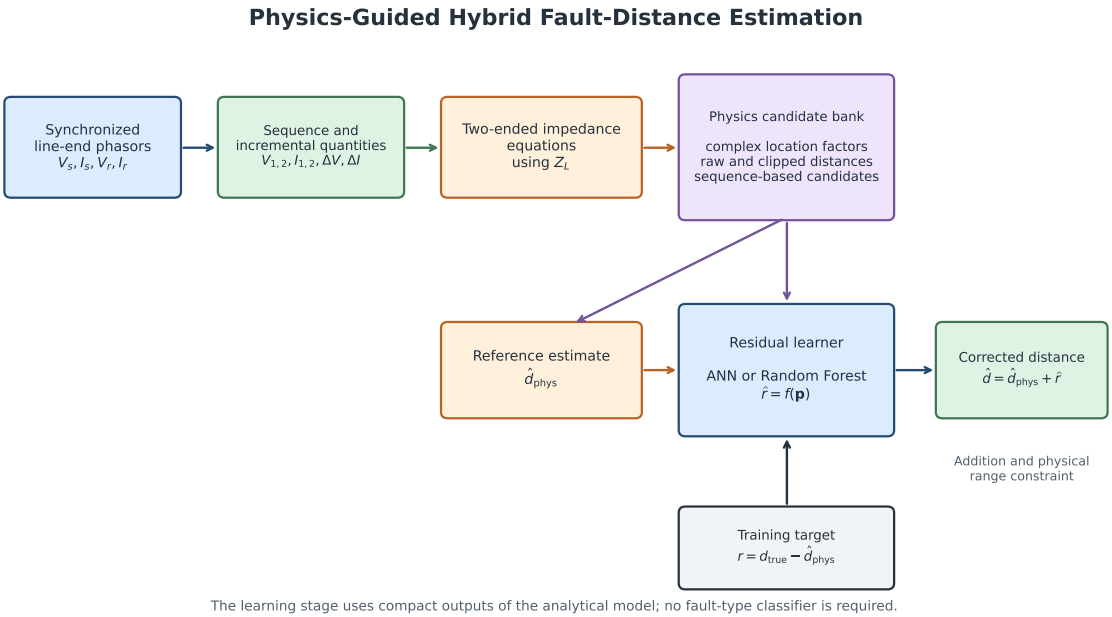


Figure 34: Hybrid fault-distance estimation pipeline

4.14 Application of the Conventional Method

The conventional method used in this work is a two-ended impedance-based fault-location procedure. It is a physics-based analytical method and does not fit numerical weights in the same way as ANN, SVM, or RF. Instead, it uses synchronized voltage and current phasors from both line terminals together with the known line impedance.

The basic idea is that if the fault is located at a fraction m from the sending end, the impedance from the sending terminal to the fault is mZ_L, while the impedance from the receiving terminal to the fault is $(1-m)Z_L$. In the exported Simulink measurements, the receiving-end current reference is opposite to the inward current convention used in the general derivation of Chapter 2. With the exported reference directions, the fault-point voltage is expressed from both sides as:

$$V_f = V_s - mZ_L I_s, \tag{4.29}$$

$$V_f = V_r + (1-m)Z_L I_r \tag{4.30}$$

Equating both expressions gives a distance estimate. One candidate form used in the implementation is:

$$m = \frac{V_s - V_r - Z_L I_r}{Z_L (I_s - I_r)} \tag{4.31}$$

The retained expression and current directions were fixed according to the terminal orientation and sign conventions of the Simulink measurements. The same analytical formulation was then applied without model training. Positive-sequence and superimposed quantities were used where appropriate to reduce the influence of pre-fault loading and emphasize the electrical changes produced by the fault.

4.15 Evaluation Metrics

The metric definitions and their interpretation are presented in Chapter 3. In this study, fault classification was evaluated using accuracy, normalized confusion matrices, class-wise precision, recall, and F1 score, together with macro and weighted averages where appropriate. These complementary measures were retained so that a high global score would not conceal errors affecting an individual fault class.

Fault-distance estimation was evaluated using MAE, MSE, RMSE, and R^2. Errors were calculated from the normalized line-distance fraction to permit comparison across lines of different lengths, and selected results were also converted to kilometres for physical interpretation. All metrics were calculated from the held-out fold in each cross-validation run and then aggregated across the five folds. Training cases were never included in the score reported for the same run.

4.16 General Conclusion

The methodology developed in this chapter provides a complete link between the power-system simulation and the final fault-location evaluation. The IEEE 39-bus New England system was modeled in MATLAB/Simulink, checked through dynamic validation, and used to generate a controlled set of fault cases over the selected operating points. The same measurement chain then supplied both the conventional impedance calculation and the machine-learning database, which makes the later comparison meaningful.

The chapter also defined the data rules that protect the validity of the results. Feature extraction was separated from target variables, preprocessing was fitted only on training folds, and the existing five-fold assignments were used to obtain out-of-fold predictions. The renewable inverter cases and the hybrid correction procedure were introduced as extensions of the same framework rather than as separate experiments with unrelated assumptions.

Overall, this methodology ensures that the reported results are traceable from the simulated voltage and current signals to the final performance metrics. It also clarifies the role of each method: the conventional algorithm provides an interpretable analytical estimate, the standalone learning models use the extracted electrical features, and the hybrid models learn only the residual correction of the analytical distance estimate.

Chapter 5 Results and Discussion

Summary

This chapter compares conventional impedance-based fault location with ANN, SVM, and Random Forest models under a common five-fold out-of-fold evaluation. Classification performance, distance-estimation accuracy, computation time, and sensitivity to measurement noise are examined together rather than as isolated indicators. Additional experiments assess PCA-based feature reduction and Bayesian SVM optimization. The analysis is then extended to cases with renewable inverter integration, where conventional estimates are also corrected by hybrid ANN and Random Forest regressors. The comparison highlights the practical trade-offs among nominal accuracy, robustness, computational cost, and interpretability.

5.1 General Introduction

The evaluation begins by examining ANN, SVM, Random Forest, and conventional methods separately, so that the strengths and weaknesses of each method are visible before the global comparison. Their distance-estimation and classification results are then compared using the same out-of-fold protocol. The remaining experiments examine feature reduction, parameter optimization, measurement noise, renewable inverter integration, and hybrid correction of the conventional estimate.

5.2 Artificial Neural Network Results

The ANN performs fault classification and distance estimation through its two output branches. The results below examine both tasks and the corresponding training behaviour.

5.2.1 Distance-Estimation Results

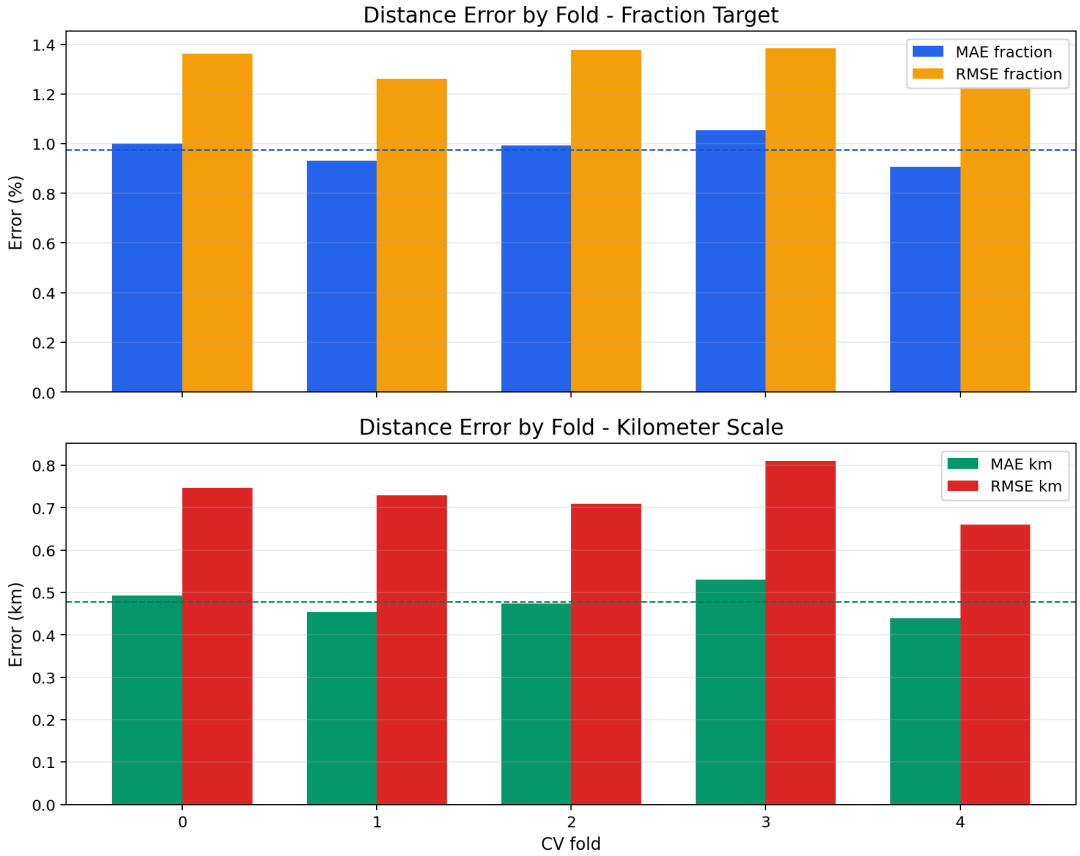


Figure 35: ANN distance-error boxplots by fold

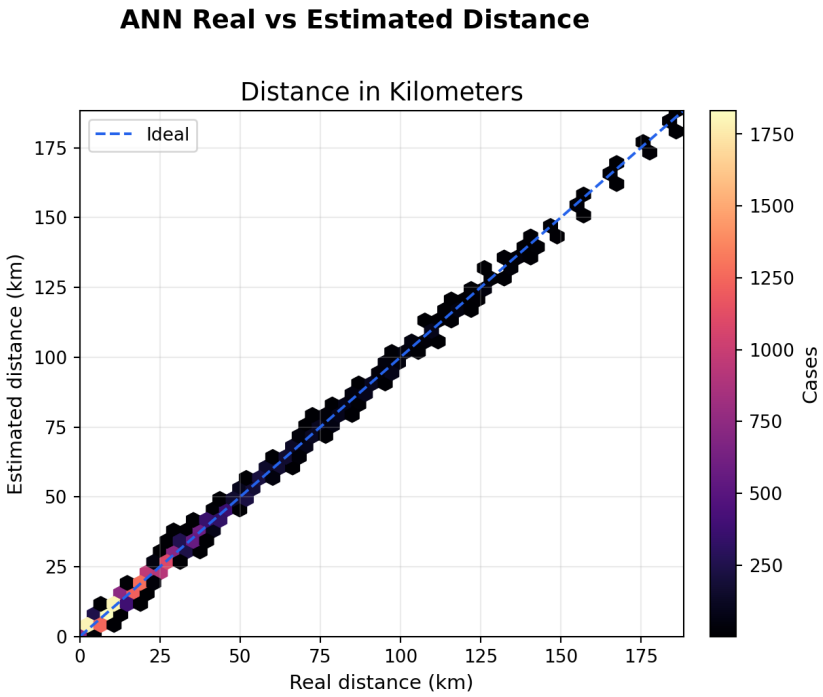


Figure 36: ANN real versus estimated distance

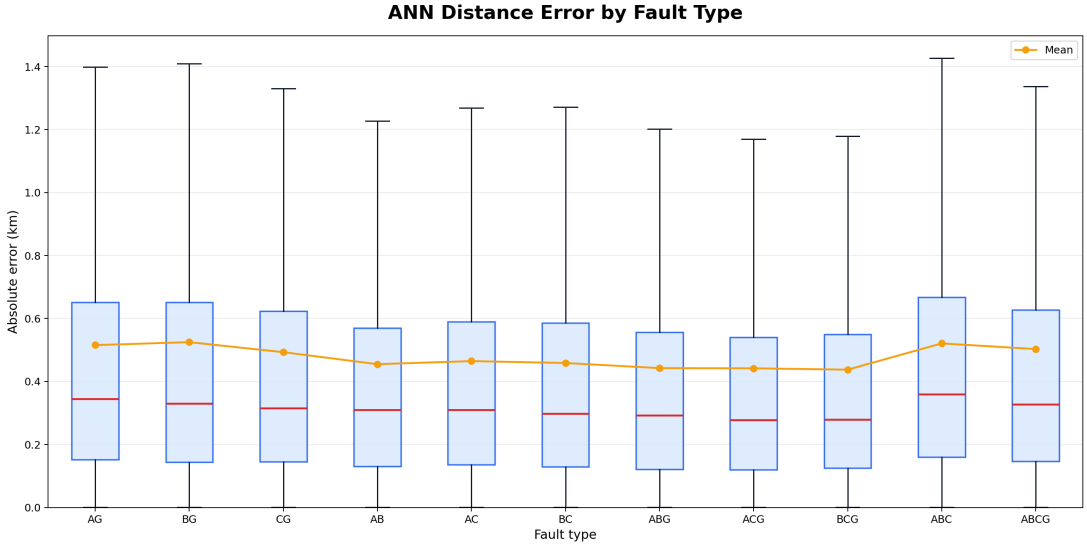


Figure 37: ANN error boxplots by fault type

The estimated distances remain close to the ideal diagonal over the full range. An MAE of 0.976% confirms that the typical error is small, whereas the higher RMSE of 1.324% reveals a limited number of larger deviations. Similar errors across the five folds rule out dependence on one favourable validation subset. Fault type still affects the error distribution, so some electrical conditions are harder to locate even when the physical distance is similar.

5.2.2 Fault-Classification Results

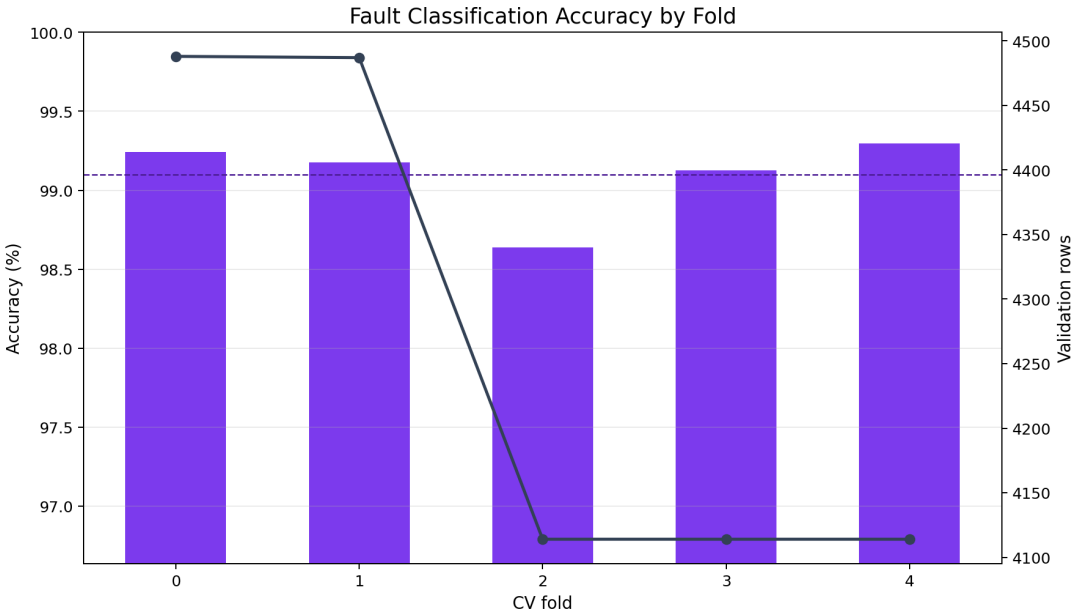


Figure 38: ANN accuracy by fold

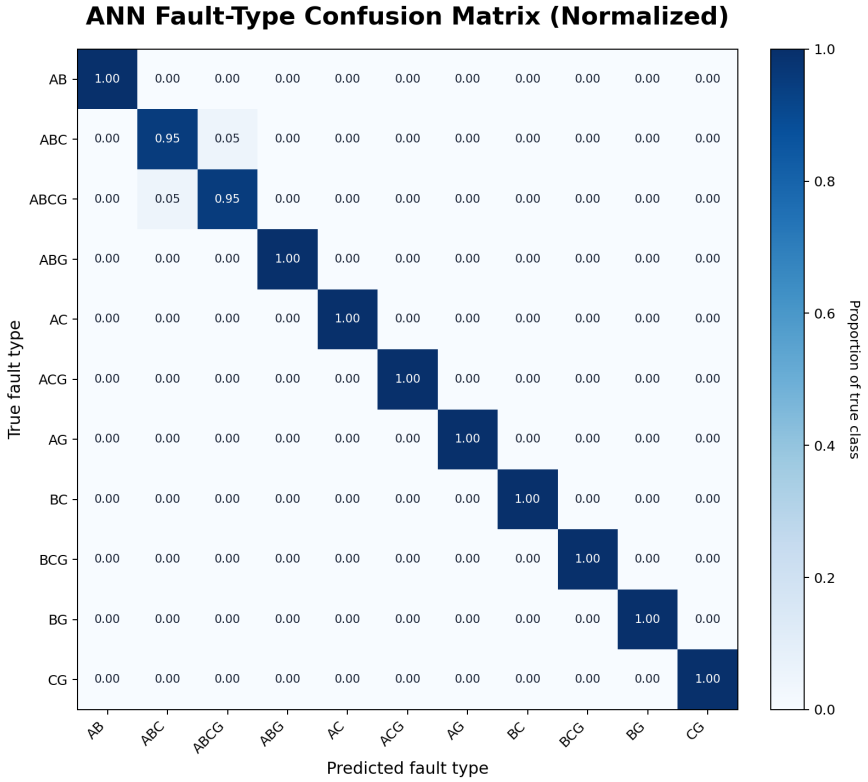


Figure 39: ANN normalized confusion matrix

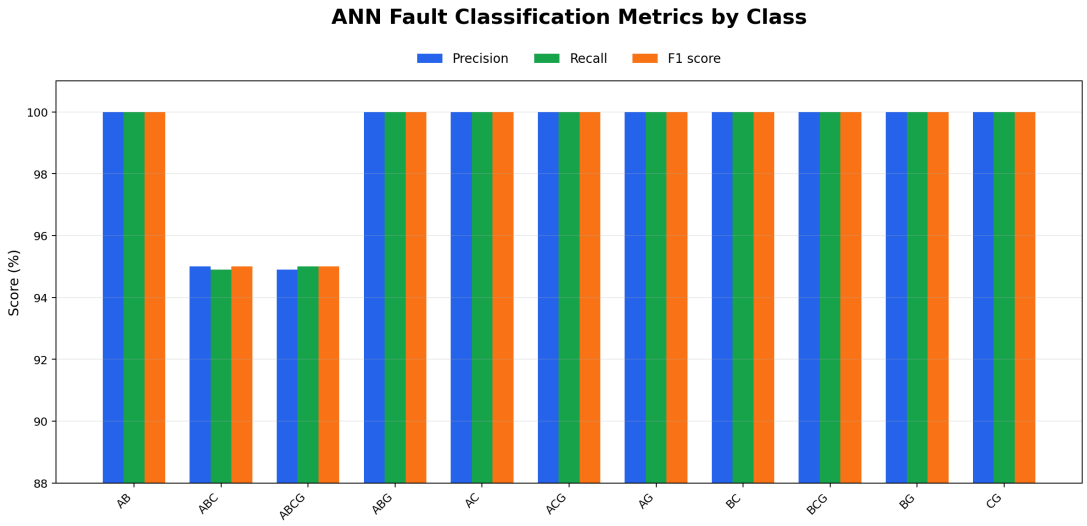


Figure 40: ANN metrics by fault type

The ANN classifies 99.099% of the cases correctly. Nearly all remaining confusion occurs between ABC and ABCG faults. Both involve all three phases, and the additional ground component in ABCG is not always sufficiently distinct to separate the two classes in every case. Performance remains very high for the other fault types, so the global accuracy does not hide a weak minority class.

5.2.3 Training Behaviour

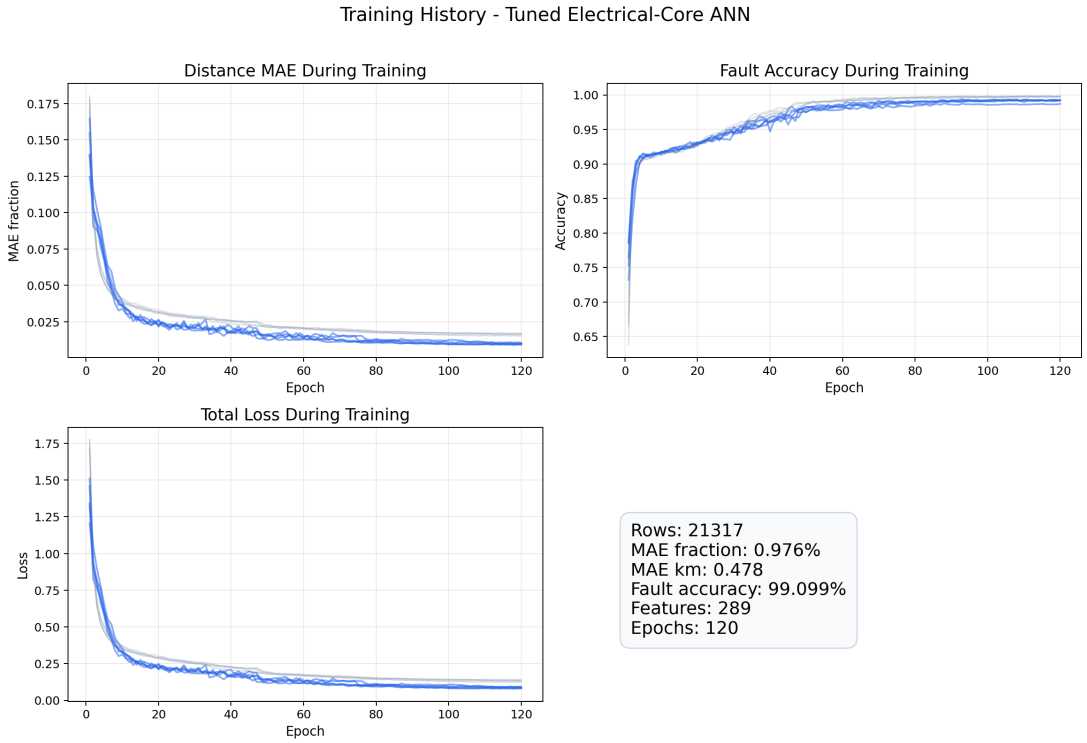


Figure 41: ANN training history

Training and validation losses decrease together and then stabilize. Their limited separation suggests that early stopping selected the model before serious overfitting developed. The shared layers therefore learned information that remained useful for both distance estimation and classification.

5.3 Support Vector Machine Results

SVM uses separate models for the two tasks: SVR estimates distance and SVC identifies the fault type.

5.3.1 Distance-Estimation Results

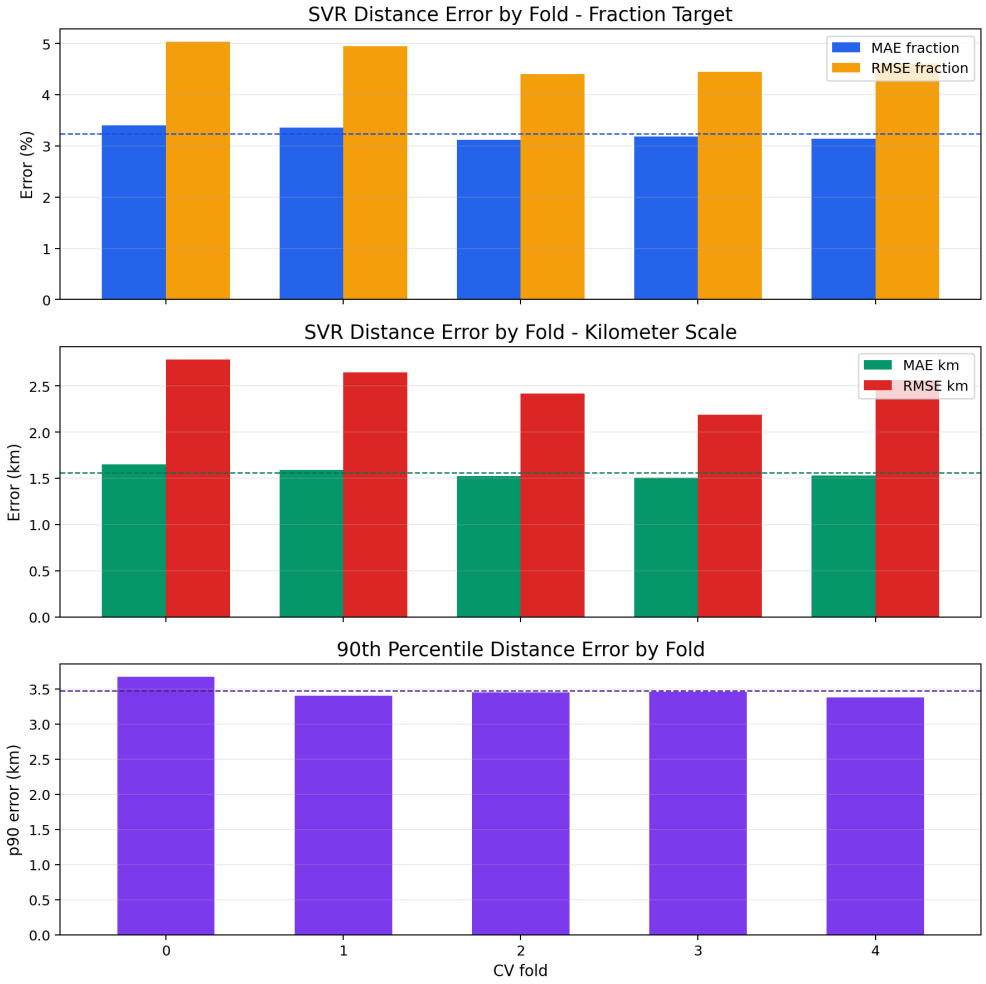


Figure 42: SVR distance-error boxplots by fold

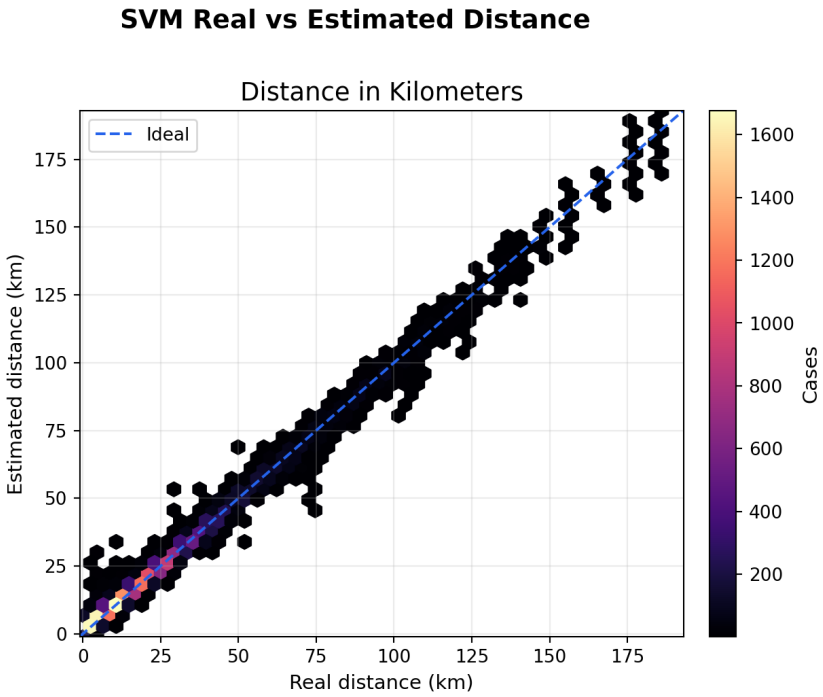


Figure 43: SVR real versus estimated distance

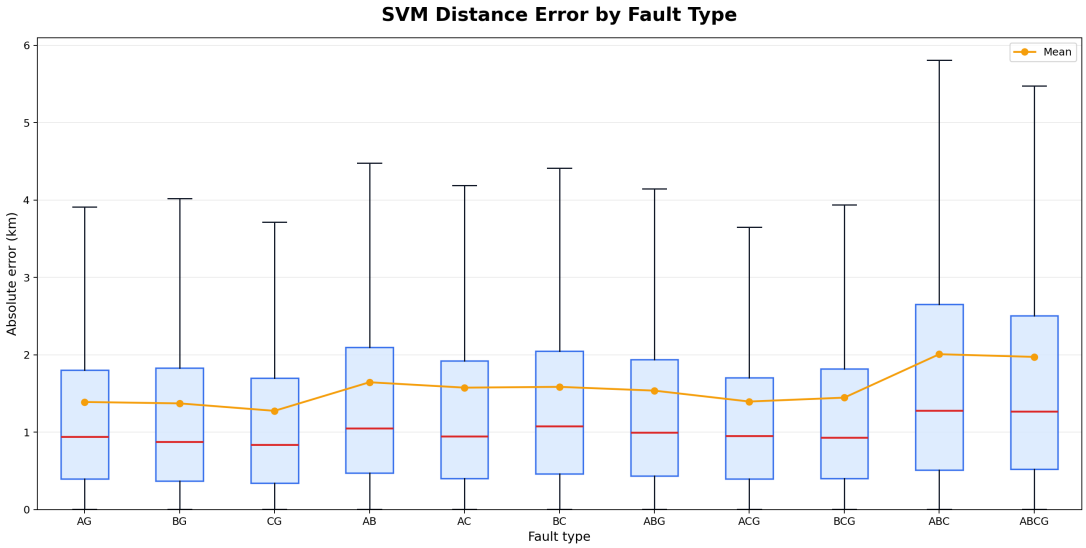


Figure 44: SVR error boxplots by fault type

SVR follows the correct distance trend, but its predictions are more dispersed than those of ANN and Random Forest. The difference between the 3.247% MAE and 4.708% RMSE points to a noticeable tail of difficult cases. The variation by fold and fault type also suggests that one global RBF mapping does not describe every operating condition equally well.

5.3.2 Fault-Classification Results

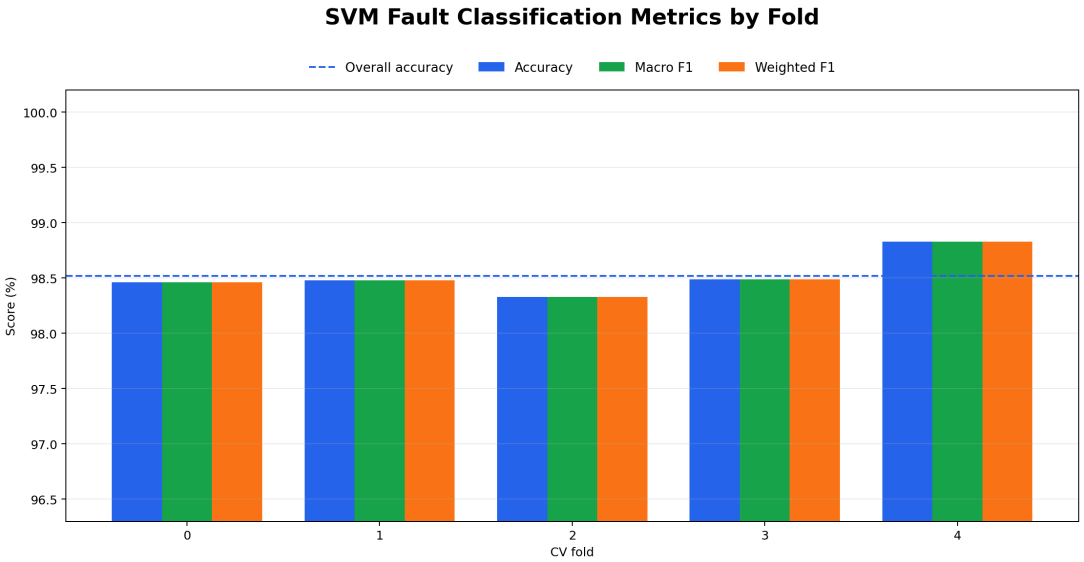


Figure 45: SVC metrics by fold

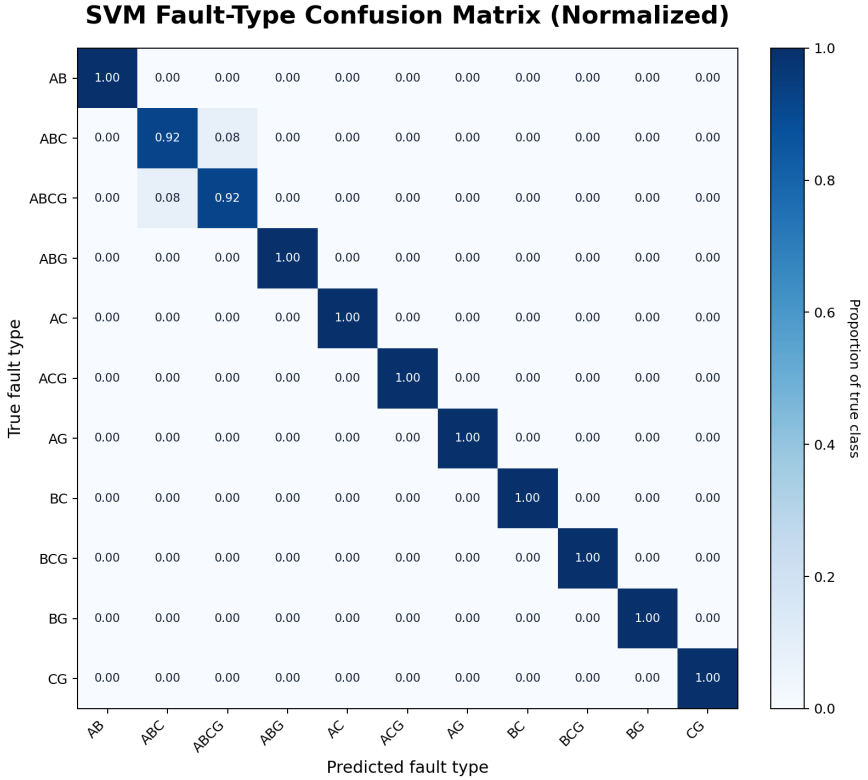


Figure 46: SVC normalized confusion matrix

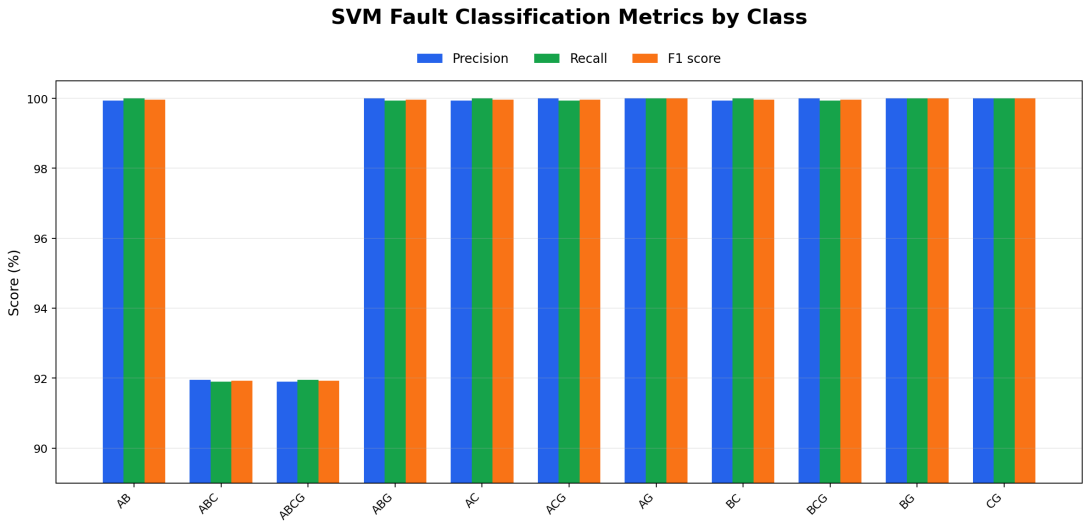


Figure 47: SVC metrics by fault type

Distance estimation is the weaker SVM task; classification remains strong at 98.494%. ABC and ABCG again account for most mistakes, while the other classes remain well separated. Comparable fold-level and class-level scores show that the result is broadly distributed across the dataset rather than being inflated by class imbalance.

5.3.3 Solver Diagnostics

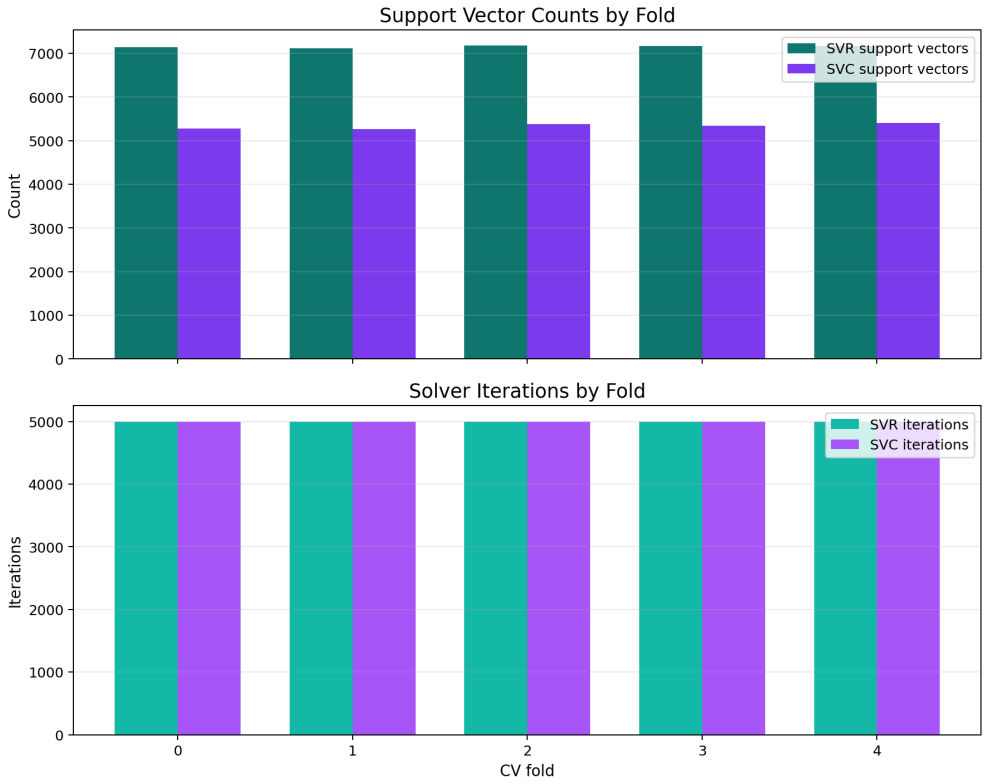


Figure 48: SVM support vectors and iterations

Many training cases remain active as support vectors. Each prediction must therefore evaluate a substantial part of the learned representation, making SVM slower than the direct analytical calculation, although still faster than the Random Forest implementation used here.

5.3.4 Bayesian Optimization Assessment

Bayesian hyperparameter optimization was also tested as an attempt to improve the SVM models. The search was nested within the existing five-fold protocol, and the outer validation fold remained unavailable during both preprocessing and parameter selection. Table 6 compares the resulting out-of-fold performance with the retained fixed configuration.

Table 6: Retained and Bayesian SVM configurations

Metric	**Retained SVM**	**Bayesian SVM**
SVR MAE (%)	3.247	3.440
SVR RMSE (%)	4.708	4.798
SVR MAE (km)	1.564	1.702
SVR RMSE (km)	2.537	2.602
SVR R^2	0.97044	0.96930
SVC accuracy (%)	98.494	98.264
SVC macro F1 (%)	98.494	98.264

The optimization did not improve either task. All principal SVR error measures increased, while SVC accuracy and macro F1 decreased slightly. The optimized parameters were therefore not adopted, and the retained fixed configuration was used in the final method comparison.

5.4 Random Forest Results

Separate Random Forest models estimate the fault distance and classify the fault type. Feature-importance values are included to identify the electrical variables most frequently used by the trees.

5.4.1 Distance-Estimation Results

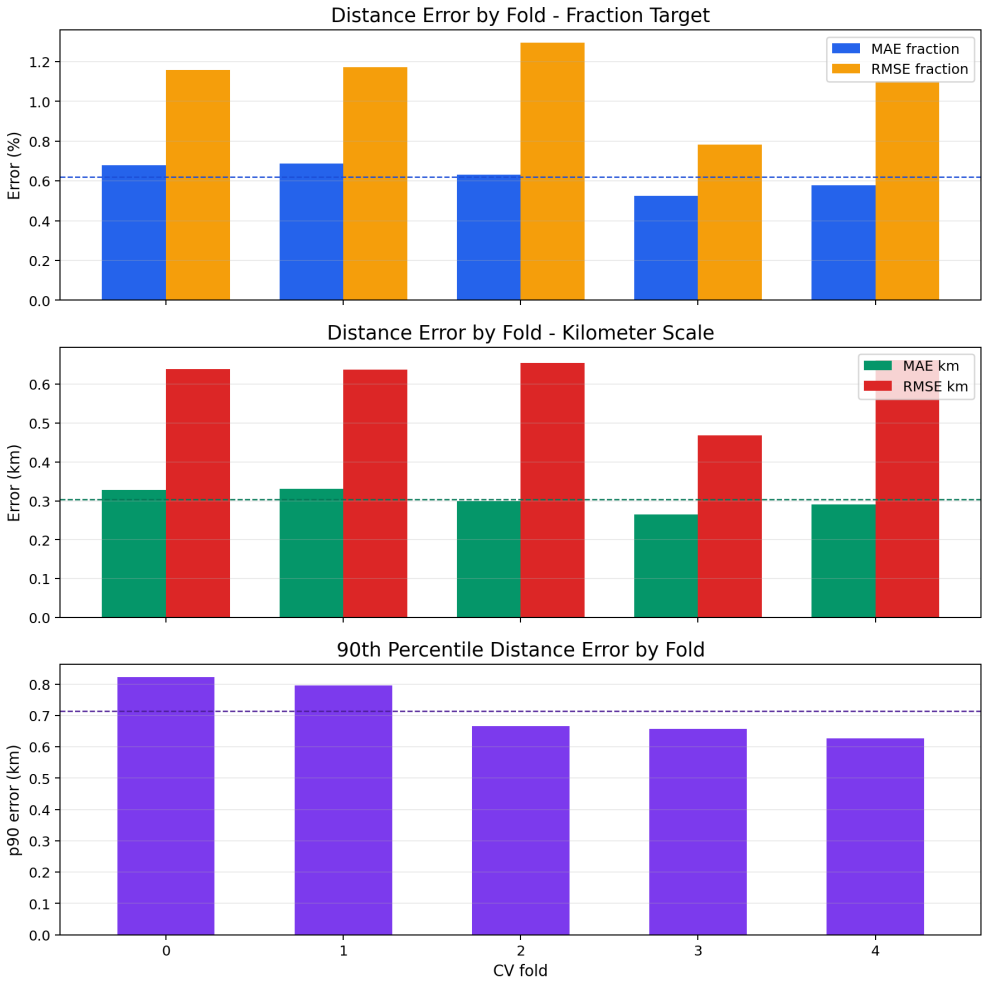


Figure 49: Random Forest distance-error boxplots by fold

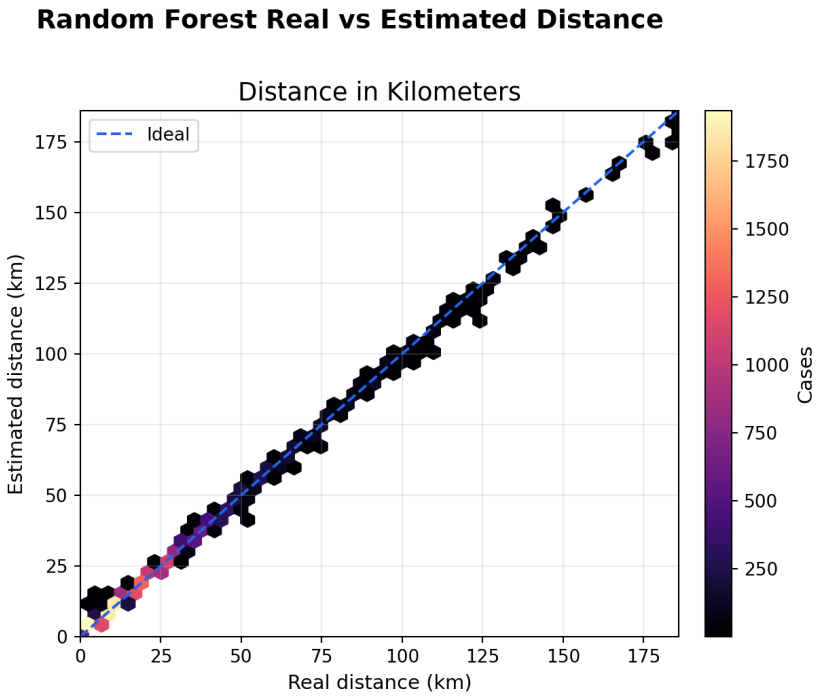


Figure 50: Random Forest real versus estimated distance

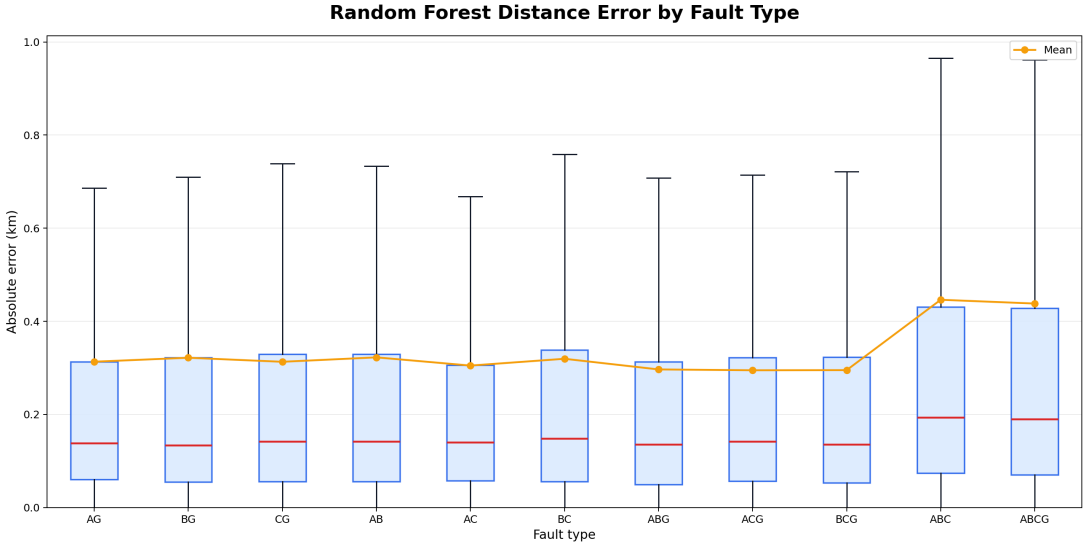


Figure 51: Random Forest error boxplots by fault type

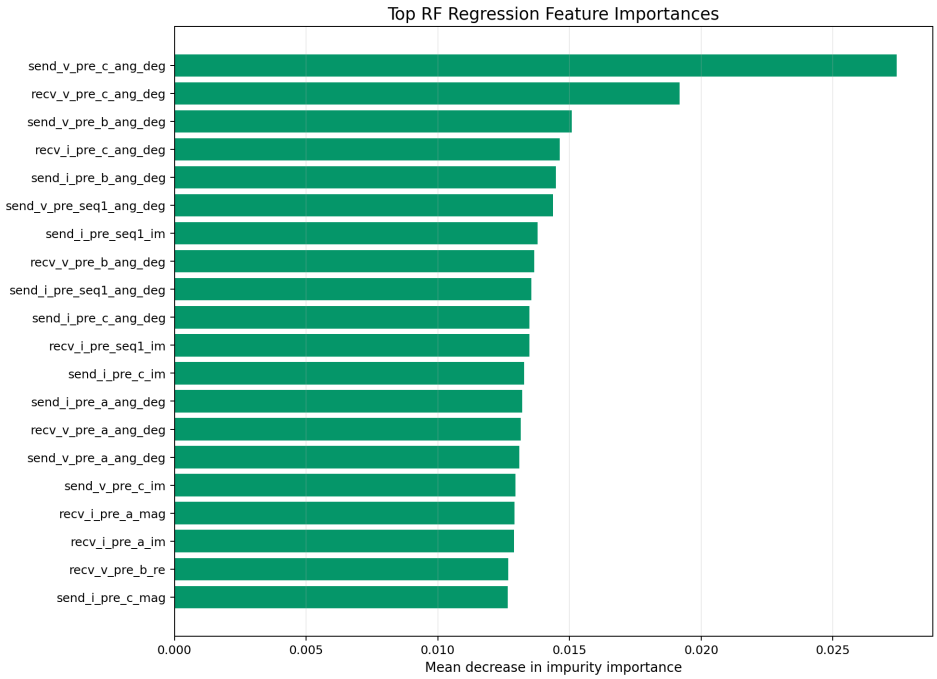


Figure 52: Random Forest regression feature importance

Random Forest gives the lowest standalone distance error: 0.664% MAE and 1.198% RMSE. The RMSE remains noticeably above the MAE because a small number of cases produce larger errors. Even so, all five folds retain low average errors. The forest also draws information from several terminal quantities instead of relying on a single dominant measurement, which matches the coupled electrical nature of fault location.

5.4.2 Fault-Classification Results

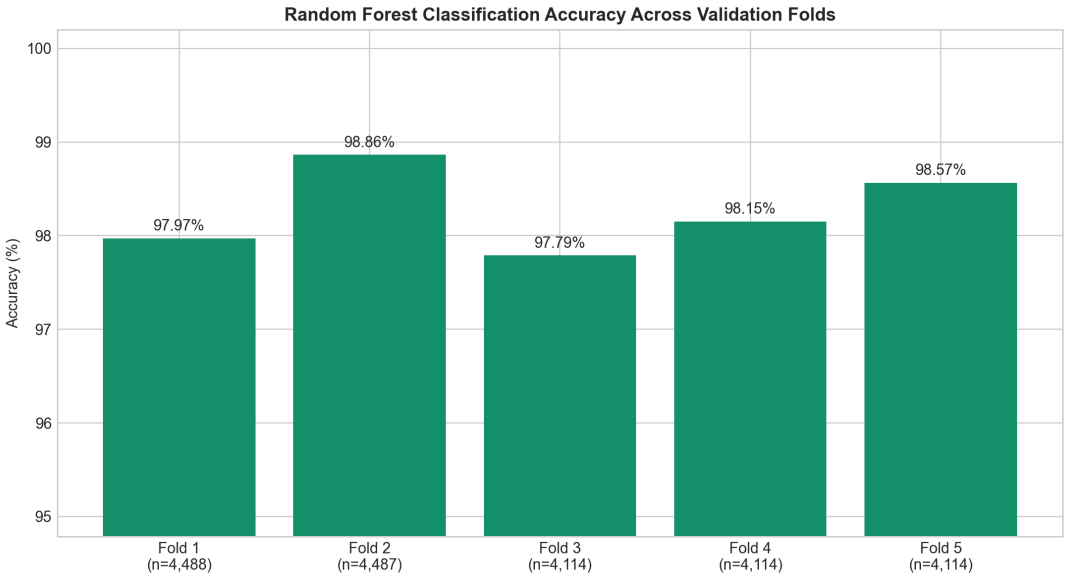


Figure 53: Random Forest classification by fold

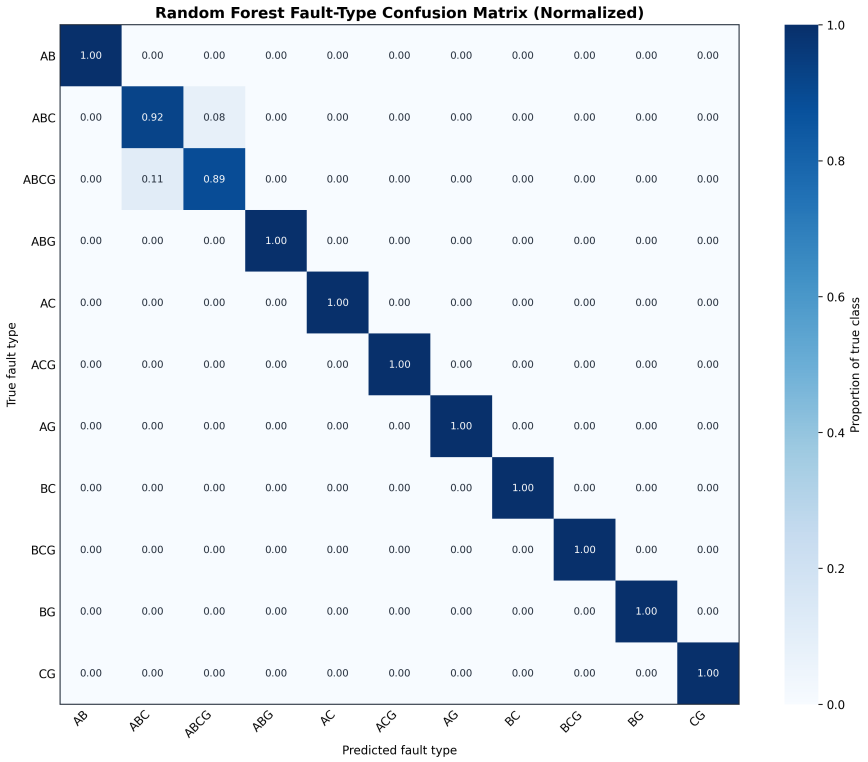


Figure 54: Random Forest normalized confusion matrix

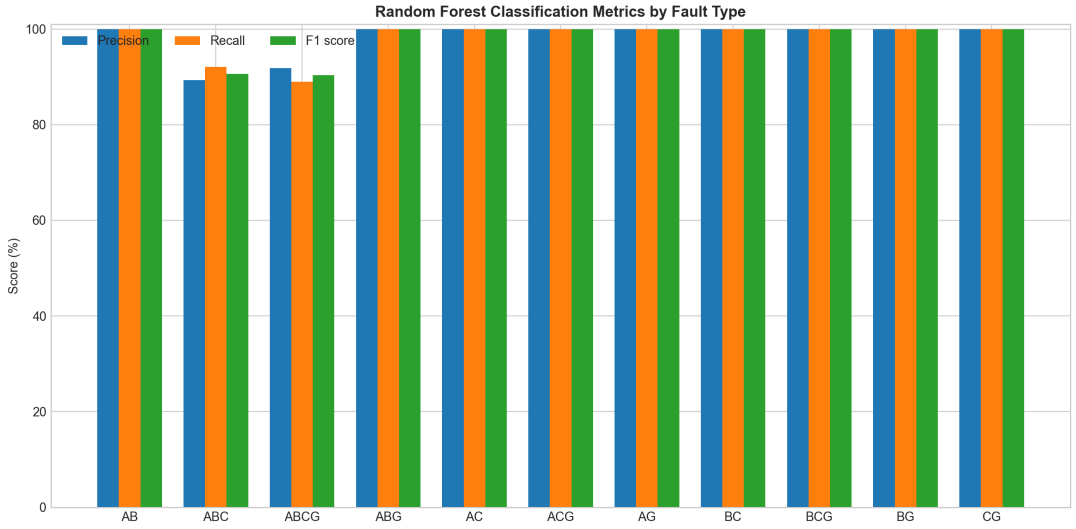


Figure 55: Random Forest metrics by fault type

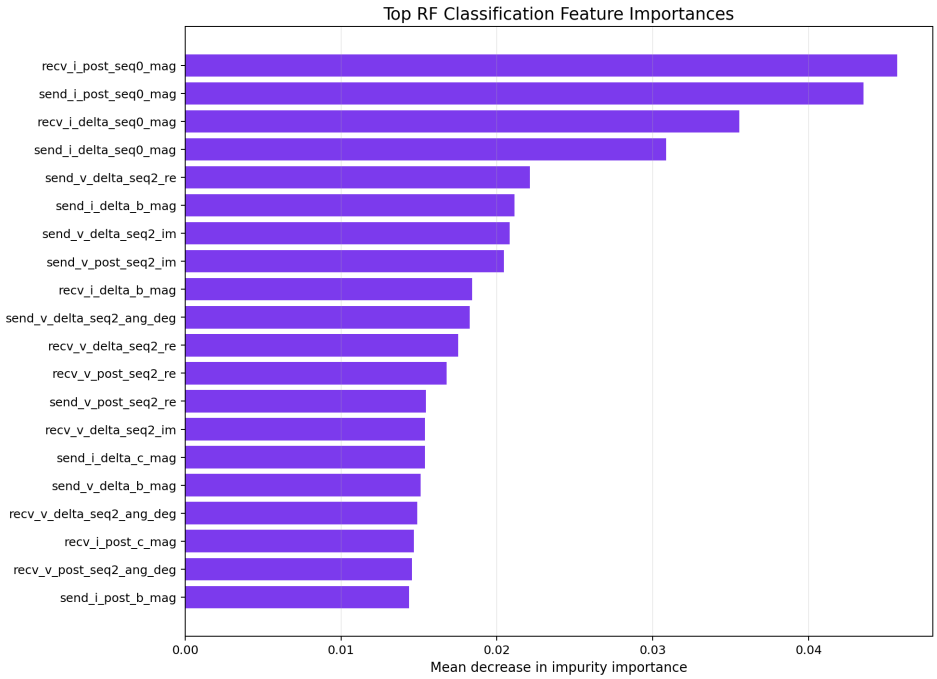


Figure 56: Random Forest classification feature importance

The classifier reaches 98.274% accuracy. Most mistakes occur in the hardest operating groups, and the unequal fold sizes make the case-level out-of-fold score more representative than a simple average of the five fold scores. Feature importance identifies variables that frequently reduce impurity, but it is not a causal electrical explanation: correlated variables can divide the apparent importance among themselves.

5.4.3 Illustrative Tree Structure

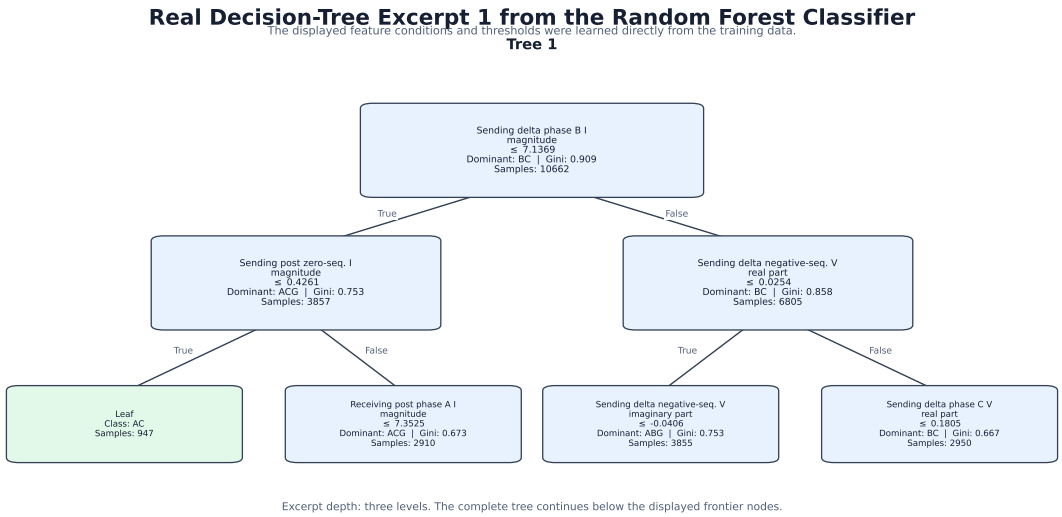


Figure 57: Random Forest tree excerpt

The tree excerpt gives a concrete example of how successive electrical thresholds divide the fault classes. It is only one small part of the model; the final classification comes from the combined vote of the complete forest.

5.5 PCA-Based Feature Reduction Assessment

Principal component analysis (PCA) is an unsupervised dimensionality-reduction technique that transforms a set of possibly correlated variables into a smaller set of mutually orthogonal components. The first component captures the largest possible variance in the input data, and each subsequent component captures the largest remaining variance under the orthogonality constraint. Retaining only the components required to explain a selected proportion of the total variance can reduce input dimensionality and correlation, although the variance preserved by PCA is not necessarily the information that is most useful for a particular prediction target [51].

PCA was assessed using the same 289 electrical-core inputs and the same five validation folds as the original ANN, SVM, and Random Forest experiments. Median imputation, standardization, and PCA fitting were performed using each training fold only, and the resulting transformations were then applied unchanged to the corresponding validation fold. Two variance thresholds were tested. Retaining 95% of the training variance produced 62 principal components in every fold, whereas retaining 99% produced 95 components. The ANN retained its joint regression and classification structure, while separate regression and classification models were used for SVM and Random Forest.

Table 7: PCA feature-reduction results

Method	Input	Dimension	MAE (%)	RMSE (%)	R^2	Accuracy (%)
ANN	Original	289	0.976	1.324	0.9977	99.099
ANN	PCA 95%	62	1.060	1.474	0.9971	98.991
ANN	PCA 99%	95	0.975	1.378	0.9975	99.043
SVM	Original	289	3.247	4.708	0.9704	98.494
SVM	PCA 95%	62	3.293	4.568	0.9722	96.482
SVM	PCA 99%	95	3.259	4.694	0.9706	97.134
Random Forest	Original	289	0.664	1.198	0.9981	98.274
Random Forest	PCA 95%	62	10.414	13.068	0.7723	96.224
Random Forest	PCA 99%	95	11.745	14.505	0.7195	95.914

PCA did not improve the three techniques consistently. Retaining 99% of the variance reduced ANN MAE only marginally, while its RMSE and classification accuracy became slightly worse. SVM gains on individual metrics were offset by losses on others. Random Forest deteriorated sharply at both PCA thresholds because PCA replaces the original electrical variables with rotated combinations, while decision trees benefit from direct threshold splits on the individual measurements.

PCA was therefore not retained for the final method comparison. Since the objective is to compare ANN, SVM, and Random Forest under the same input conditions, selecting a reduced representation that benefits one model but substantially penalizes another would make the comparison less consistent. Consequently, the final results for all three techniques are based on the same 289 electrical-core features.

5.6 Conventional Fault Location Results

The conventional method produces a fault distance estimate but does not perform fault classification. Its results are therefore presented through regression metrics and error distributions.

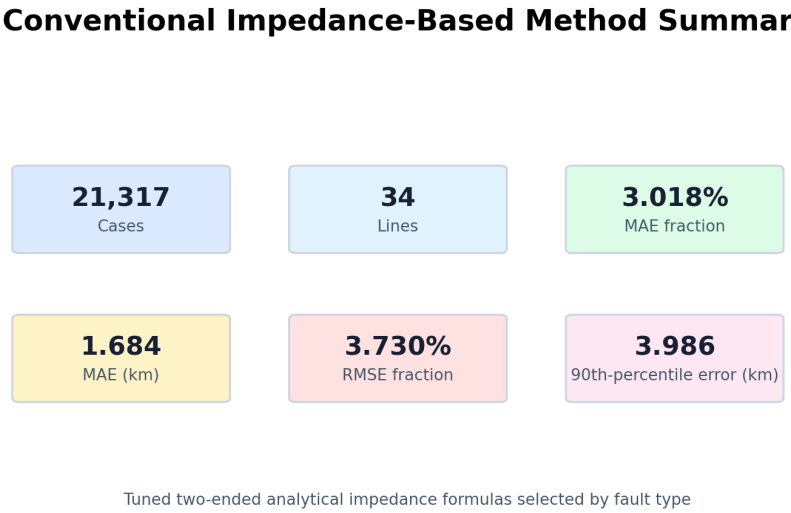


Figure 58: Conventional method metrics

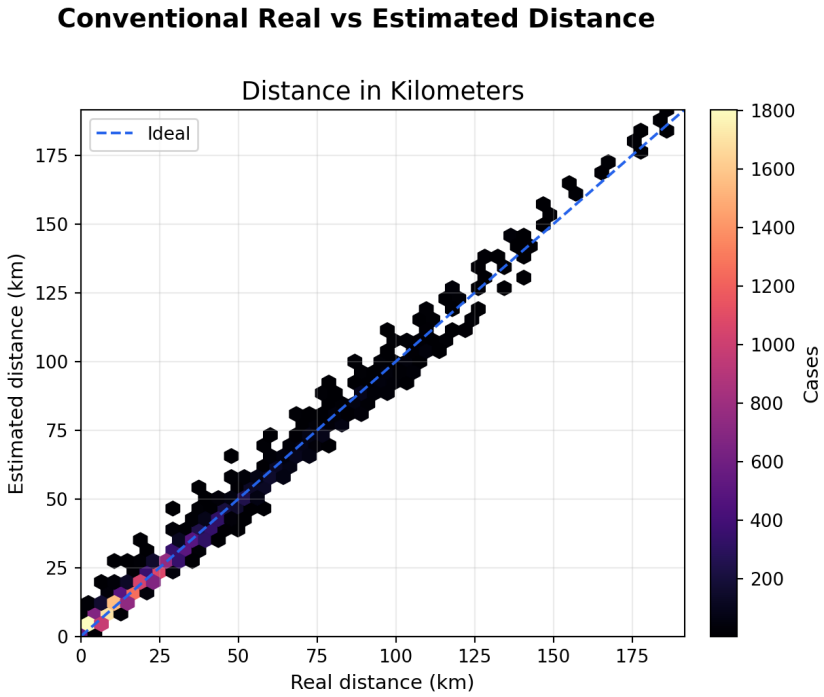


Figure 59: Conventional real versus estimated distance

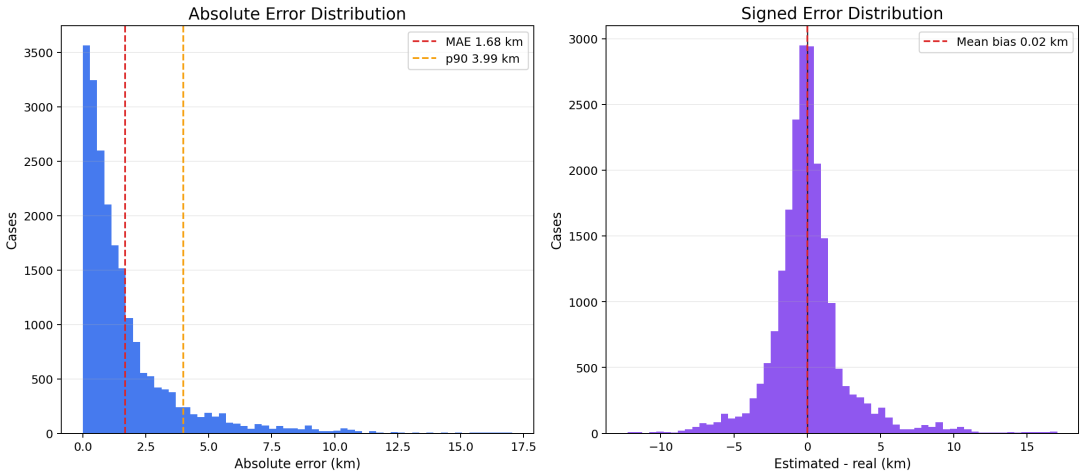


Figure 60: Conventional error distribution

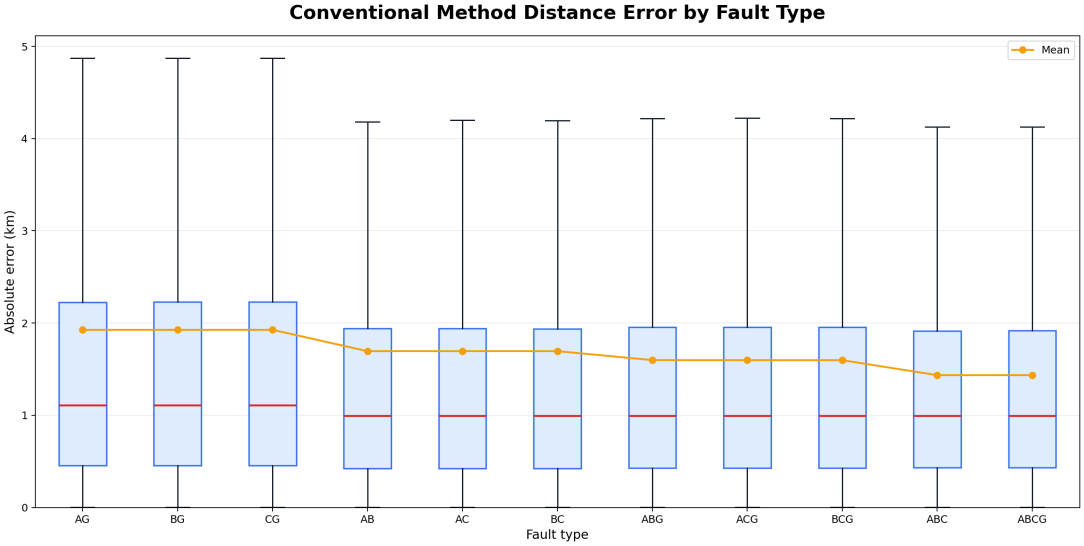


Figure 61: Conventional error boxplots by fault type



Figure 62: Conventional error boxplots by fault resistance

The conventional estimator reaches an MAE of 3.018%. Its errors vary more strongly with fault type and resistance than those of ANN and Random Forest, reflecting the

assumptions built into the impedance formulation when the fault loop departs from ideal conditions. In return, the calculation is transparent, directly linked to circuit quantities, and extremely fast.

5.7 Overall Method Comparison

The four standalone methods are compared on the main 21,317-case dataset. Renewable and hybrid results use a separate 1,140-case dataset and are therefore reported later.

The uncertainty of the predictive metrics was quantified using nonparametric bootstrap resampling of the complete case-level predictions with 4,000 replicates and a fixed random seed of 42. The reported limits are the 2.5th and 97.5th percentiles of the bootstrap distribution. Classification-accuracy limits were calculated using the Wilson interval. Values in square brackets in the following tables denote the corresponding 95% confidence interval.

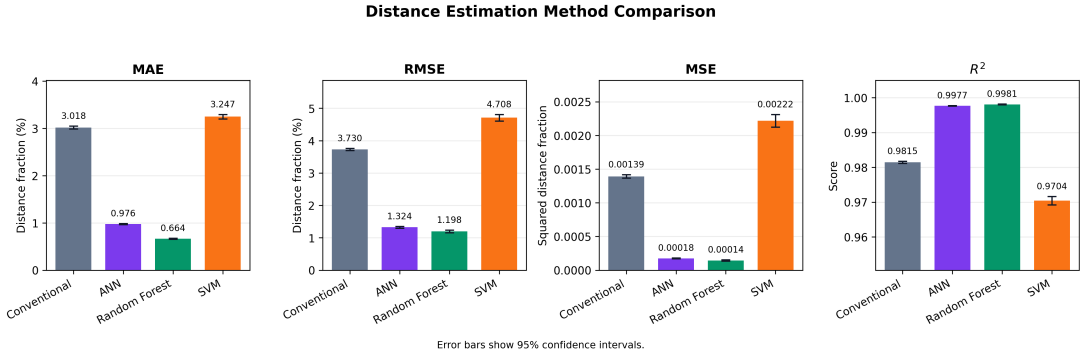


Figure 63: Standalone method comparison

Table 8: Distance errors with 95% confidence intervals

Method	MAE (%)	RMSE (%)
Conventional	3.018 [2.989, 3.048]	3.730 [3.696, 3.765]
ANN	0.976 [0.964, 0.988]	1.324 [1.298, 1.350]
Random Forest	0.664 [0.651, 0.678]	1.198 [1.161, 1.237]
SVM	3.247 [3.200, 3.291]	4.708 [4.609, 4.806]

Table 9: MSE and R^2 with 95% confidence intervals

Method	MSE	R^2
Conventional	0.001391 [0.001366, 0.001417]	0.98145 [0.98111, 0.98178]
ANN	0.000175 [0.000169, 0.000182]	0.99766 [0.99757, 0.99775]
Random Forest	0.000143 [0.000135, 0.000153]	0.99809 [0.99796, 0.99820]
SVM	0.002216 [0.002125, 0.002310]	0.97044 [0.96923, 0.97162]

Table 10: Accuracy and computation time

Method	Accuracy (%)	Preprocessing (ms)	Inference (ms)	Total (ms)
Conventional	–	0.0082	0.0049	0.0131
ANN	99.099 [98.963, 99.218]	0.0045	3.9468	3.9513
Random Forest	98.274 [98.090, 98.440]	0.0028	27.0694	27.0722
SVM	98.494 [98.322, 98.649]	0.0046	3.9010	3.9056

The MAE and RMSE values are expressed as percentages of line length, while the MSE is calculated from the normalized distance fraction. Computation time represents

the measured median preprocessing and inference time required for one distance estimate; model training time is not included, and confidence intervals are therefore not attached to the timing medians. The conventional method has no classification accuracy because it was applied only to distance estimation.

To make the comparison more concrete, Table 11 shows ten representative held-out cases from the main dataset. The entries are distances in kilometres, illustrating how the same fault case is estimated by each method before the errors are aggregated into the global metrics.

Table 11: Representative clean cases from the main dataset

Case	Line	Fault	True (km)	ANN (km)	SVM (km)	RF (km)	Conv. (km)
1	5-6	ACG	3.65	3.60	3.65	3.69	3.59
2	17-18	BG	5.14	5.31	4.43	5.17	6.40
3	26-29	BC	9.79	10.10	10.30	11.26	18.12
4	5-8	ABC	12.29	12.49	11.39	12.29	11.88
5	17-27	AG	18.97	19.69	21.09	18.83	19.46
6	2-25	ABC	22.86	22.71	22.41	22.82	21.56
7	3-4	ACG	40.02	40.64	38.51	39.69	37.91
8	1-39	BC	62.64	62.62	59.69	62.56	62.22
9	1-39	ACG	74.39	77.11	50.92	71.84	73.53
10	26-28	ACG	126.22	123.65	125.48	125.85	128.36

Random Forest is the most accurate standalone distance estimator, whereas ANN has the highest classification accuracy. SVM still classifies faults reliably, but its distance error is slightly higher than that of the conventional method. Computation time changes the practical picture: conventional estimation requires 0.0131 ms per case, ANN and SVM about 3.9 ms, and Random Forest 27.1 ms. These times are all compatible with post-fault analysis, but they expose a clear choice between maximum accuracy, physical transparency, and computational economy.

The repeated confusion between ABC and ABCG faults deserves a specific electrical interpretation. Both classes are three-phase faults, so their phase currents and voltages are dominated by a balanced positive-sequence behaviour. Adding ground involvement does not necessarily introduce a strong distinctive signature when the three phases are affected almost symmetrically. As a result, the measured feature patterns of ABC and ABCG can be very close.

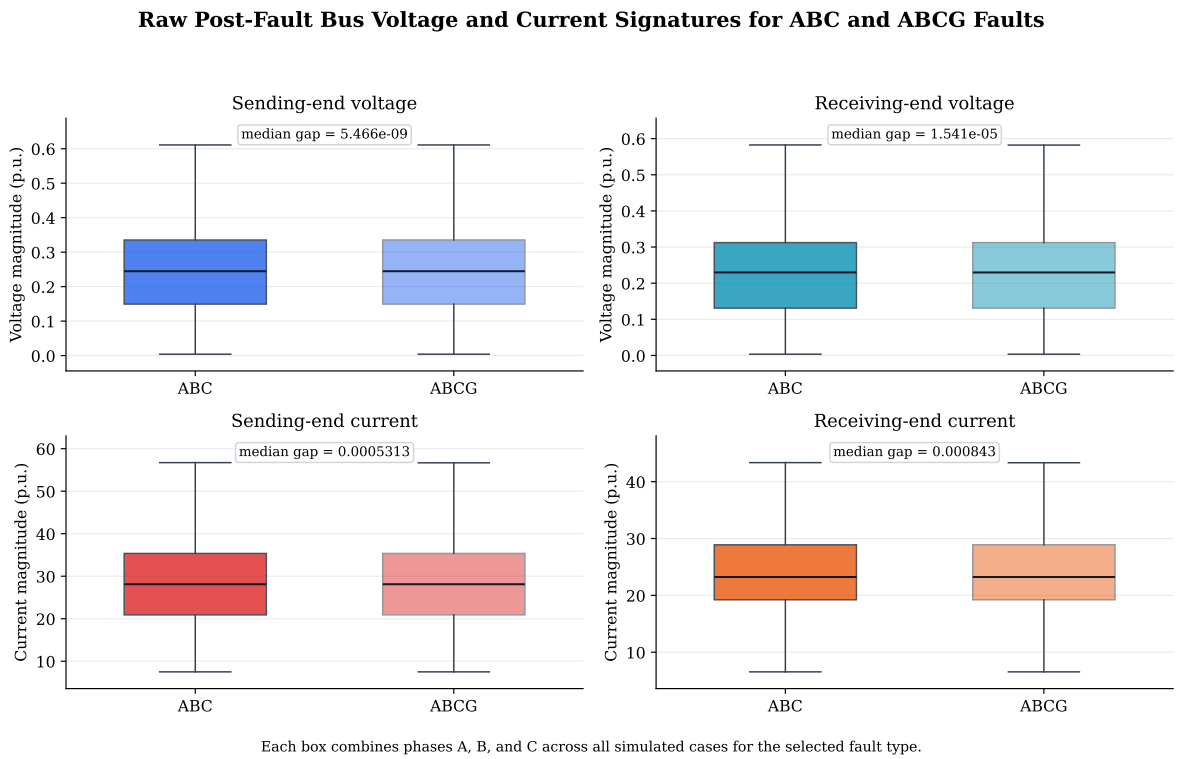


Figure 64: Raw voltage and current boxplots for ABC and ABCG faults

Figure 64 confirms this behaviour at the raw-data level. The sending-end and receiving-end voltage and current distributions of ABC and ABCG faults almost overlap, with very similar medians, interquartile ranges, and spread. This means that the additional ground connection in ABCG does not always create a clearly separable measurement signature in the selected electrical quantities. The classifiers therefore sometimes separate ABC and ABCG less clearly than unbalanced faults such as AG, AB, or ABG. This limitation has a smaller impact on distance estimation than on fault-type labelling, because both ABC and ABCG still represent severe three-phase fault conditions requiring rapid protection.

5.7.1 Robustness to Measurement Noise

A complementary robustness test was performed to examine the sensitivity of the distance estimators to perturbations in the measured electrical quantities. The noise was added to the original sending- and receiving-end voltage and current time samples, before feature extraction. For each complex phase sample X, the noisy value was generated as

$$\widetilde{X} = X + \frac{\eta |X|}{\sqrt{2}} \left(n_{\mathrm{r}} + j n_{\mathrm{i}} \right), \tag{5.1}$$

where n_{r} and n_{i} are independent standard normal variables and η is the prescribed noise ratio. Consequently, the stated percentage represents the root-mean-square magnitude of the added complex Gaussian noise relative to the magnitude of each original sample. These levels constitute a controlled sensitivity test and are not intended to represent a particular instrument-accuracy class.

After perturbation, the complete feature-extraction procedure was repeated. Thus, the pre-fault, post-fault, delta, magnitude, angle, and sequence-component quantities were all recalculated from the noisy voltage and current samples. This approach preserves the dependencies among the derived electrical features and is more representative than perturbing the final feature columns independently.

An initial examination showed that 36 features had a standard deviation below 10^{-10} in the training fold. These features mainly represented zero- and negative-sequence quantities that were practically zero under the ideal, balanced simulation conditions. Retaining them produced extremely large standardized values after even small perturbations. They were therefore removed using statistics calculated exclusively from the training fold, leaving 253 features for the three machine-learning models. The conventional estimator does not use these input features; its original line- and fault-type-specific analytical rule was retained without retuning for the noisy cases.

The test was applied to a selected set of representative cases covering different fault-distance ranges. For each noise level, several independent noise realizations were generated. The trained ANN, SVM, Random Forest, and conventional models were the same clean models used in the main study; the noisy cases were used only for testing, with no retraining, fine-tuning, or adaptation. Table 12 reports the resulting MAE as a percentage of line length.

Table 12: MAE under voltage-current noise

Noise (%)	**Conventional (%)**	**ANN MAE (%)**	**SVM MAE (%)**	**RF MAE (%)**
0.00	2.282	1.006	4.247	0.570
0.01	2.284	1.582	6.000	10.569
0.05	2.287	1.561	5.985	13.384
0.10	2.314	1.564	6.028	13.605
0.20	2.446	1.480	6.008	14.098
0.50	2.716	1.721	6.136	14.809
1.00	3.734	2.101	6.646	14.480
2.00	5.245	3.710	8.002	15.987
5.00	11.734	8.556	13.257	17.488

Table 13 gives the same idea at case level for the 1% Gaussian voltage-current noise test. The entries are absolute distance errors in kilometres, averaged over the noisy realizations for each selected case.

Table 13: Representative main-dataset cases under 1% Gaussian voltage-current noise

Case	Line	Fault	True (km)	ANN err.	SVM err.	RF err.	Conv. err.
1	5-6	ACG	3.65	0.17	0.41	0.75	1.21
2	17-18	BG	5.14	0.85	1.50	5.23	1.40
3	26-29	BC	9.79	2.32	18.57	45.69	8.35
4	5-8	ABC	12.29	0.54	0.97	1.35	0.38
5	17-27	AG	18.97	1.43	2.84	11.08	1.10
6	2-25	ABC	22.86	0.58	1.31	5.15	1.30
7	3-4	ACG	40.02	0.69	2.33	4.72	2.10
8	1-39	BC	62.64	0.61	3.32	17.18	0.77
9	1-39	ACG	74.39	2.83	24.95	21.94	0.85
10	26-28	ACG	126.22	2.67	3.10	24.97	2.13

The clean pilot cases confirm the strong nominal accuracy of Random Forest, but its error increases sharply once measurement noise is introduced. This behaviour indicates sensitivity to the precise split thresholds learned from the noise-free simulations. The conventional estimator is comparatively stable at the lower perturbation levels: its MAE changes only from 2.282% to 2.716% between clean measurements and 0.5% noise. Its error then rises to 3.734% at 1% noise and 11.734% at 5% noise. SVM degrades less abruptly than Random Forest after removal of the near-constant features, although its clean error remains higher and its performance progressively deteriorates at the larger

perturbation levels. ANN provides the most stable overall response: its MAE remains below 2.2% for noise levels up to 1%, before increasing more substantially at 2% and 5%. The experiment therefore illustrates a trade-off between nominal accuracy and robustness. Random Forest gives the lowest error for ideal measurements, the conventional method shows strong stability at low noise levels, and ANN gives the lowest noisy-case MAE over the tested range.

A second pilot was added to represent a broader measurement-channel disturbance. Gaussian noise alone models random scatter, but real voltage and current measurements can also contain coherent instrument errors. The perturbed phasor was therefore generated as

$$\widetilde{X} = (1 + \epsilon_g) X e^{j\epsilon_\theta} + \frac{\eta |X|}{\sqrt{2}} \left(n_{\mathrm{r}} + j n_{\mathrm{i}} \right), \tag{5.2}$$

where ϵ_g is a random gain error, ϵ_θ is a random phase-angle error, and the last term is the same relative complex Gaussian noise used previously. Independent gain and phase errors were drawn for the sending and receiving terminals. The combined errors were applied to the same selected test cases, and the clean trained models were evaluated directly on the perturbed measurements. Table 14 reports the same MAE measure as Table 12.

Table 14: MAE under combined measurement errors

Scenario	$\boldsymbol{\eta}$ (%)	$\boldsymbol{\sigma_g}$ (%)	$\boldsymbol{\sigma_\theta}$ (deg)	**Conventional**	**ANN**	**SVM**	**RF**
Clean	0.0	0.0	0.0	2.282	1.006	4.247	0.570
Mild	0.1	0.2	0.1	2.376	1.540	6.179	15.391
Moderate	0.5	0.5	0.5	3.458	2.340	7.101	15.519
Severe	1.0	1.0	1.0	4.553	4.993	9.010	16.380

The combined-error test is intentionally harsher than the Gaussian-only test. The conventional estimator remains stable for mild and moderate errors because the coherent gain and phase perturbations affect both terminals in a structured way and do not immediately destroy the analytical relationship. ANN is the most robust learning-based model: its MAE remains 2.340% under the moderate scenario and becomes comparable to the conventional estimator only in the severe case. SVM degrades more gradually than Random Forest but starts from a higher clean error. Random Forest is highly accurate on clean data, but its tree thresholds become sensitive when many correlated voltage-current features shift together. These results reinforce the need to evaluate both nominal accuracy and measurement robustness before considering field use.

5.8 Results with Renewable Inverter Integration

Renewable inverter integration changes the fault-current contribution seen by the line terminals. Since inverter-based resources inject fault current through control-limited power-electronic interfaces, their response can differ from the more repeatable behavior of synchronous machines and can challenge conventional line-protection assumptions [69, 70]. Random Forest and the conventional method are first examined under this operating condition, followed by the hybrid regressors and the corresponding robustness pilot.

5.8.1 Overall Method Comparison

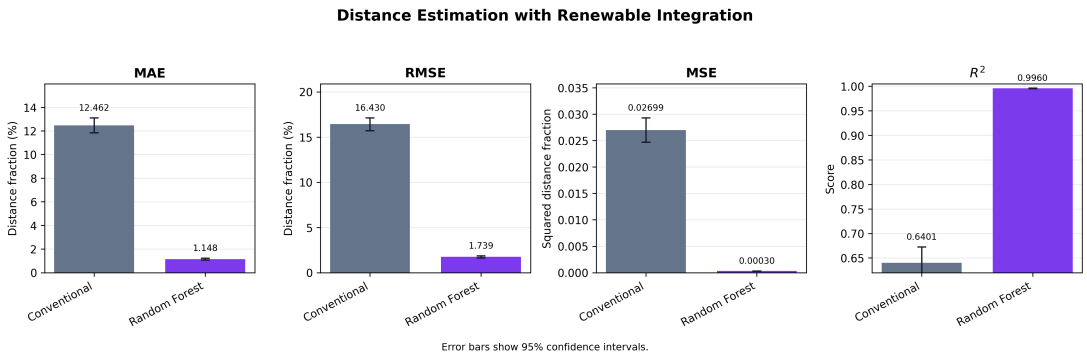


Figure 65: Renewable Random Forest and conventional metrics

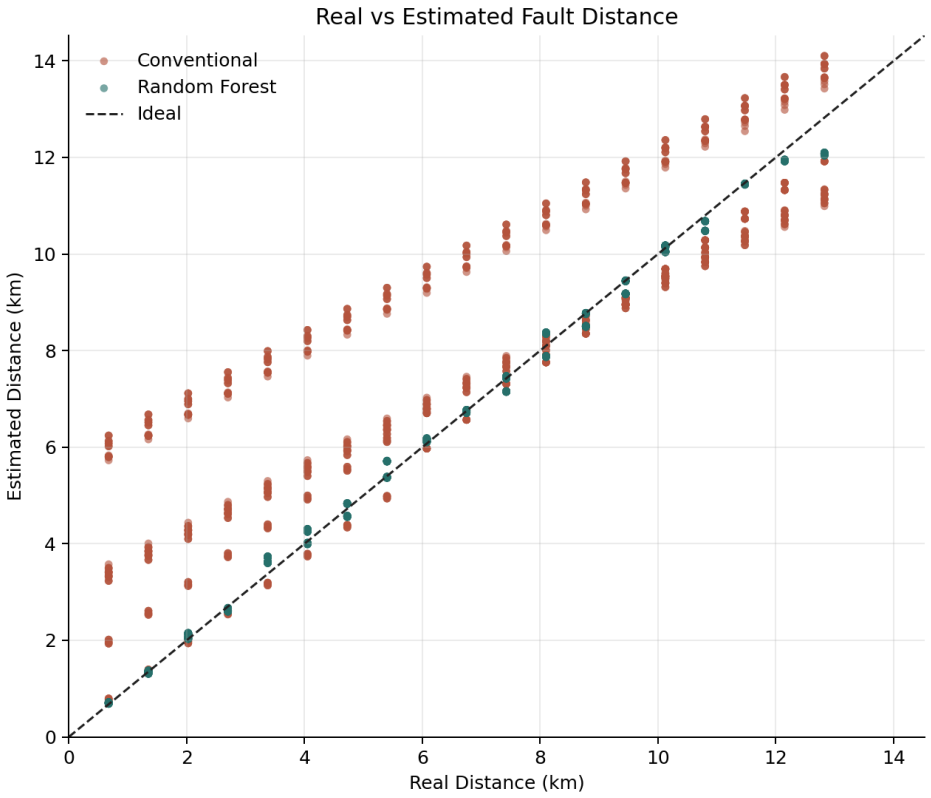


Figure 66: Renewable real versus estimated distance

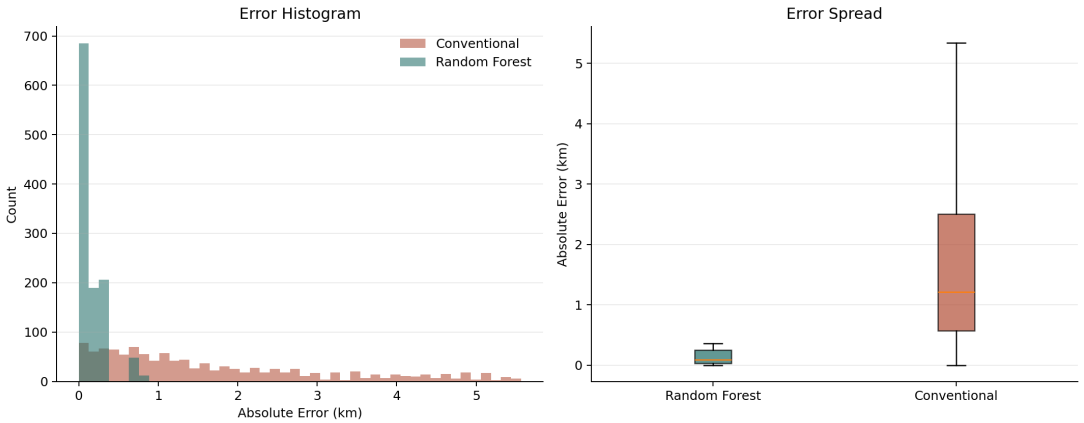


Figure 67: Renewable error distributions

The inverter affects the two estimators very differently. Conventional predictions move farther from the ideal line and develop a much wider error distribution. Random Forest remains close to the true distance because its training cases include the altered nonlinear voltage-current patterns. The analytical formulation, by contrast, is more exposed to changes in source contribution and apparent impedance.

5.8.2 Error Variation by Line and Fault Type

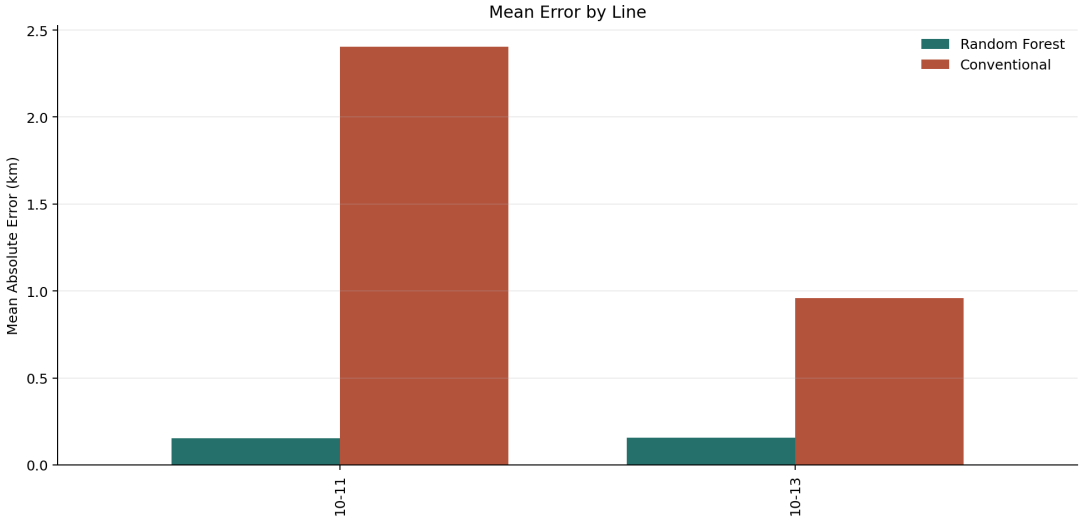


Figure 68: Renewable error boxplots by line



Figure 69: Renewable error boxplots by fault type

Errors differ across lines and fault types because inverter influence depends on network location and on the sequence components excited by the fault. A single aggregate value would therefore hide important operating differences; line-specific and fault-specific behaviour must remain part of the assessment.

5.8.3 Influence of Renewable Integration

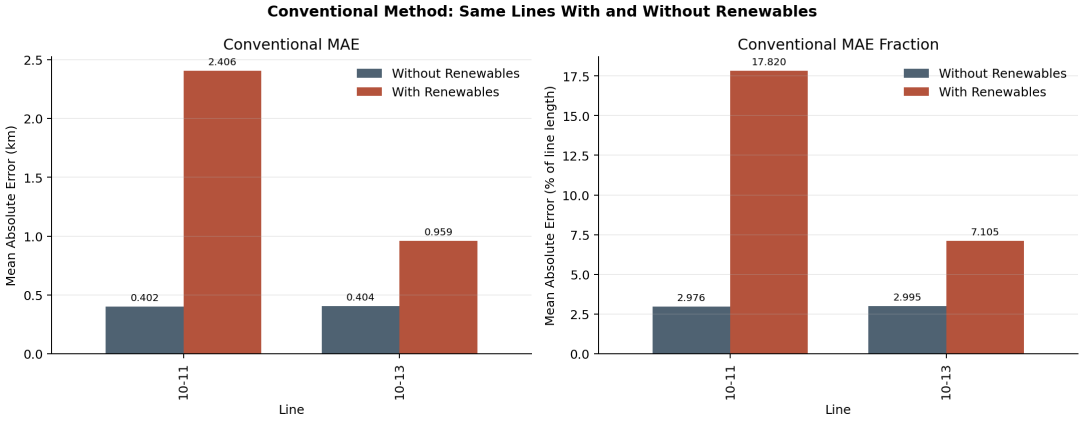


Figure 70: Conventional error boxplots with and without renewables

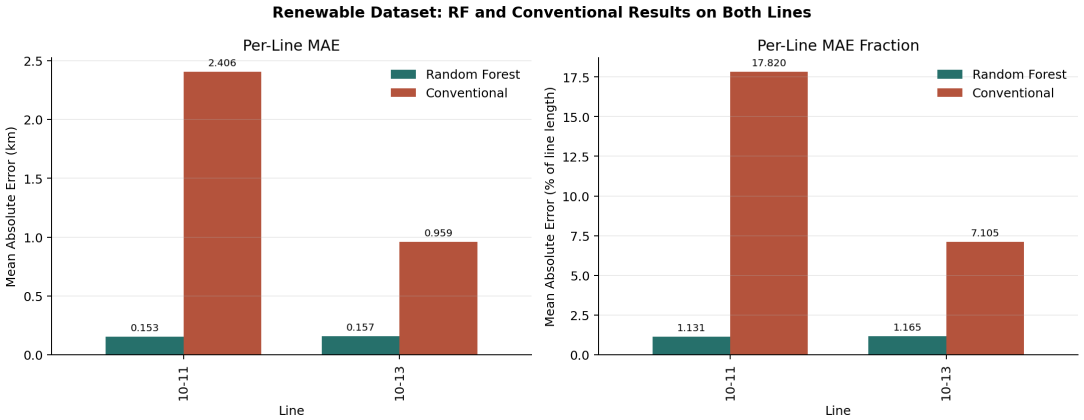


Figure 71: Renewable metrics by line

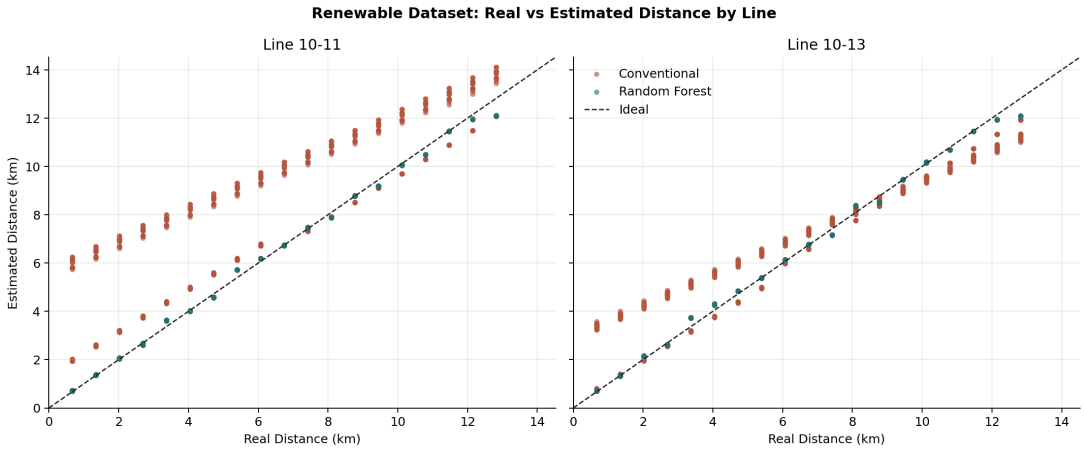


Figure 72: Renewable distance estimates by line

The conventional method deteriorates most on the line whose terminal measurements receive the stronger inverter contribution. Random Forest is more consistent across the two lines, although it also performs less accurately than on the main dataset. Fault-location performance should therefore be validated for the actual generation configuration instead of being assumed to transfer unchanged from a synchronous-source operating point.

5.8.4 Detailed Hybrid Results

Out-of-fold predictions cover all 1,140 renewable-generation cases. The conventional estimate is compared directly with the corrected Hybrid ANN and Hybrid RF distances, with physical errors reported in kilometres.

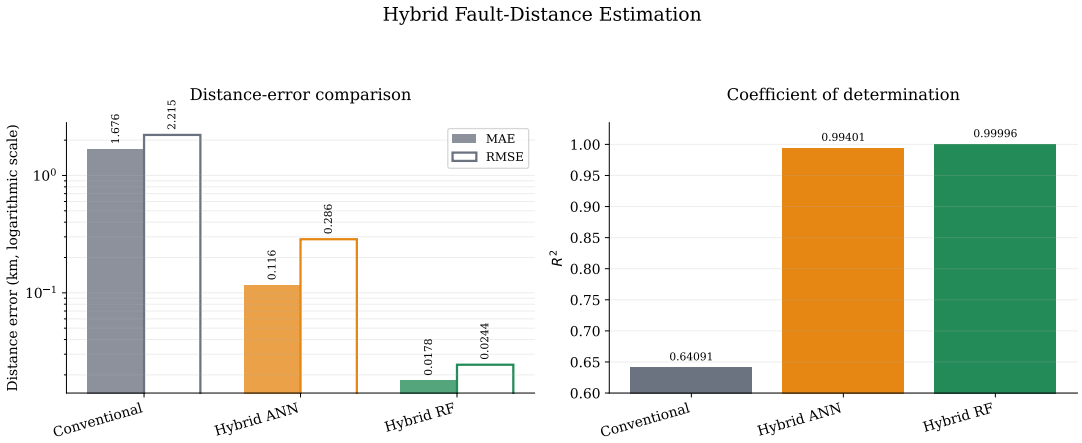


Figure 73: Hybrid method metrics

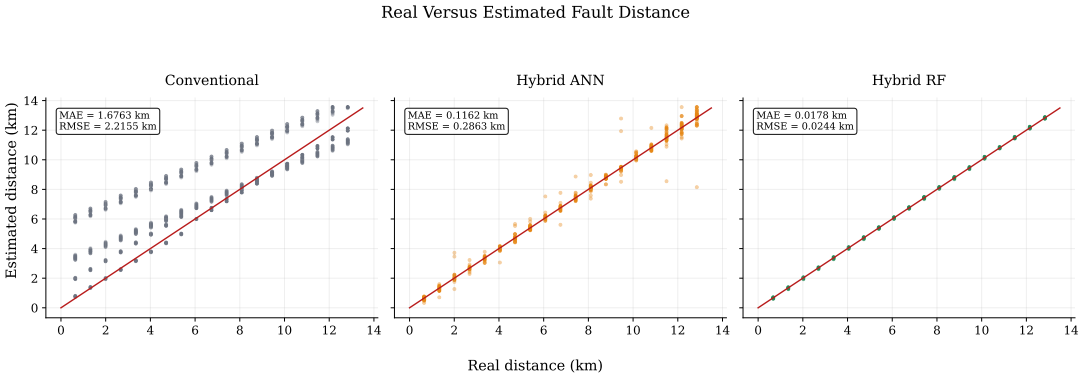


Figure 74: Hybrid real versus estimated distance

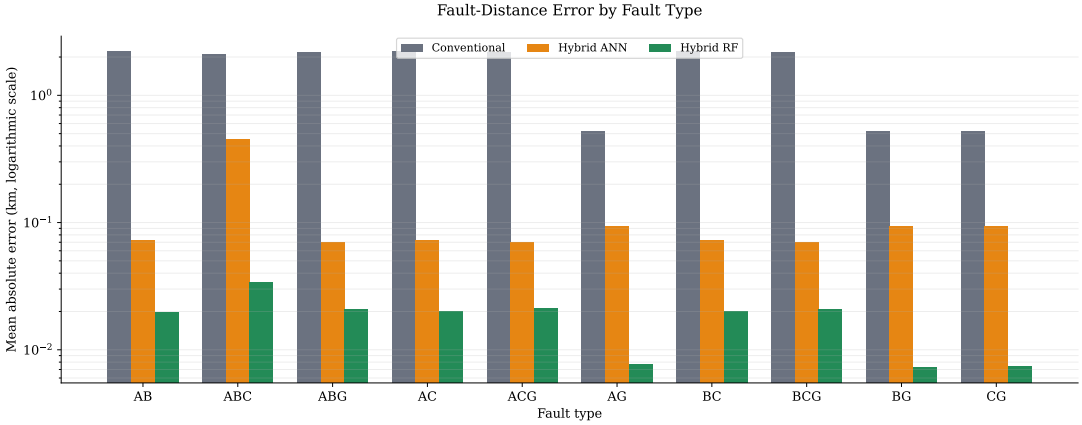


Figure 75: Hybrid error boxplots by fault type

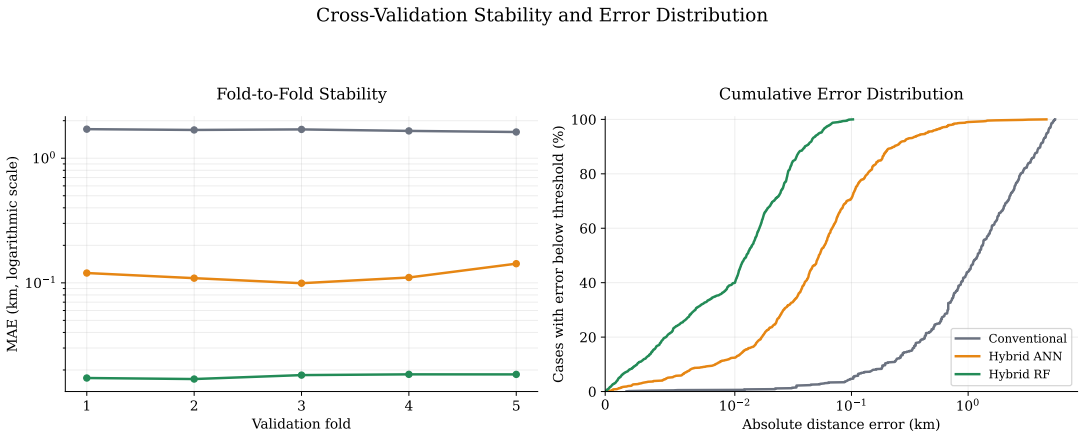


Figure 76: Hybrid fold stability and error CDF

Both hybrid models remove much of the systematic bias in the conventional estimate. Hybrid RF stays tightly concentrated around the ideal distance and is consistent across fault types and folds. Hybrid ANN lowers the average error as well, but its RMSE exceeds that of the standalone renewable Random Forest because a few residual corrections are much larger. The fold and cumulative-error plots make those occasional failures visible.

5.8.5 Numerical Summary

Table 15: Renewable distance errors with 95% confidence intervals

Method	**MAE (%)**	**RMSE (%)**
Conventional	12.462 [11.851, 13.098]	16.430 [15.718, 17.117]
Random Forest	1.148 [1.074, 1.226]	1.739 [1.624, 1.855]
Hybrid ANN	0.861 [0.755, 0.972]	2.120 [1.534, 2.699]
Hybrid RF	0.132 [0.125, 0.139]	0.181 [0.171, 0.191]

Table 16: Renewable MSE and R^2 with 95% confidence intervals

Method	**MSE**	$\boldsymbol{R^2}$
Conventional	0.026994 [0.024707, 0.029299]	0.64008 [0.60420, 0.67236]
Random Forest	0.000303 [0.000264, 0.000344]	0.99597 [0.99545, 0.99644]
Hybrid ANN	0.000450 [0.000235, 0.000728]	0.99401 [0.99022, 0.99686]
Hybrid RF	0.00000327 [0.00000292, 0.00000363]	0.99996 [0.99995, 0.99996]

Table 17: Renewable physical error and time

Method	**MAE (km)**	**RMSE (km)**	**Time (ms)**
Conventional	1.6824 [1.5999, 1.7682]	2.2180 [2.1220, 2.3108]	0.0131
Random Forest	0.1550 [0.1450, 0.1655]	0.2348 [0.2192, 0.2505]	27.0722
Hybrid ANN	0.1162 [0.1020, 0.1312]	0.2863 [0.2070, 0.3644]	5.6093
Hybrid RF	0.0178 [0.0169, 0.0188]	0.0244 [0.0231, 0.0257]	75.1204

Table 18 lists ten representative renewable-generation cases in kilometres. It shows the bias that can affect the uncorrected conventional estimate and how the RF and hybrid estimates behave on the same cases.

Table 18: Representative renewable cases

Case	Line	Fault	True (km)	Conv. (km)	RF (km)	Hybrid ANN (km)	Hybrid RF (km)
1	10-11	ABG	0.675	5.809	0.710	0.738	0.684
2	10-13	CG	2.025	1.994	2.145	2.039	2.018
3	10-13	ABG	3.375	5.068	3.726	3.332	3.340
4	10-11	ACG	4.725	8.736	4.568	4.722	4.708
5	10-11	AB	6.075	9.573	6.174	6.085	6.091
6	10-13	AC	7.425	7.660	7.148	7.429	7.440
7	10-11	ABG	8.775	11.341	8.751	8.836	8.746
8	10-11	AC	10.125	11.915	10.059	10.415	10.127
9	10-13	AG	11.475	10.729	11.447	11.296	11.507
10	10-13	BCG	12.825	11.227	12.080	13.010	12.800

The table reports out-of-fold metrics for the 1,140 renewable-generation cases. MAE and RMSE percentages are relative to line length, while MSE and R^2 are calculated from the normalized distance fraction. The bracketed values are 95% bootstrap confidence intervals calculated using the same 4,000-replicate procedure as for the main comparison. Computation time is the median per-case preprocessing and inference time. The standalone Random Forest used the stricter line-aware validation split employed in the renewable comparison plots, whereas the hybrid regressors used the existing renewable

five-fold assignments. Consequently, the table summarizes all executed renewable methods, but the RF and hybrid rows should not be interpreted as a controlled model ranking under identical folds.

Within their respective evaluation protocols, both hybrid methods improve substantially on the conventional estimate. Hybrid RF records the lowest errors and the highest R^2, while Hybrid ANN has a lower MAE than the standalone renewable Random Forest but a higher RMSE. This MAE–RMSE difference means that Hybrid ANN is accurate for most cases but produces a small number of comparatively large errors. The computation-time results show the associated cost: Hybrid RF is the slowest method, whereas Hybrid ANN offers a less expensive correction. Because the standalone and hybrid fold assignments differ, these observations describe the executed experiments and do not constitute a strictly controlled ranking between those rows.

5.8.6 Renewable Hybrid Robustness to Measurement Noise

The renewable robustness pilot was performed using the exact hybrid configuration reported in this work. The Hybrid ANN and Hybrid RF therefore received only the 27 conventional-method inputs described in Chapter 4; the 289-feature electrical-core vector was not supplied to either hybrid model. Complex Gaussian noise was added to the original sending- and receiving-end voltage and current samples before the hybrid inputs were recalculated.

The renewable noise test was applied to a selected set of representative cases covering different fault-distance ranges. For each nonzero noise level, several independent realizations were generated. The hybrid models were trained only on the original clean renewable data, and the noisy cases were used only for testing. No retraining, fine-tuning, or adaptation to noisy data was performed; the same residual-learning procedure as in the original hybrid experiment was retained.

Table 19: Renewable hybrid MAE under noise

Noise (%)	**Conventional (%)**	**Hybrid ANN (%)**	**Hybrid RF (%)**
0.00	14.119	0.570	0.116
0.01	14.119	1.970	0.128
0.05	14.123	2.390	0.280
0.10	14.136	2.572	0.450
0.20	14.146	3.241	0.826
0.50	14.340	5.049	2.106
1.00	14.688	8.915	4.089
2.00	15.826	12.876	7.241

Hybrid RF retains strong accuracy at the lower perturbation levels: its MAE changes from 0.116% without added noise to 0.128% at 0.01% noise, 0.450% at 0.1%, and 0.826% at 0.2%. Hybrid ANN is more sensitive, particularly to occasional large changes in incremental negative-sequence impedance factors, but remains more accurate than the uncorrected conventional estimate through 2% noise. The purpose of this test is therefore to show sensitivity to perturbed measurements, not to replace the nominal comparison.

The renewable cases were also tested with the more general measurement-error model used in the main robustness comparison. In this second pilot, the original voltage and current phasors were perturbed by a combination of random additive Gaussian noise, terminal-dependent gain error, and terminal-dependent phase-angle error before the 27 hybrid inputs were recalculated. The clean models were then applied directly to the perturbed test cases. No noisy samples were used for training, fine-tuning, or parameter

adjustment.

Table 20: Renewable hybrid MAE under combined errors

Scenario	η (%)	σ_g (%)	σ_θ (deg)	**Conv. (%)**	**Hybrid ANN (%)**	**Hybrid RF (%)**
Clean	0.0	0.0	0.0	14.119	0.570	0.116
Mild	0.1	0.2	0.1	14.219	3.722	1.182
Moderate	0.5	0.5	0.5	14.752	8.132	3.623
Severe	1.0	1.0	1.0	16.185	13.282	7.163

This second test is harsher than the Gaussian-only perturbation because it does not simply add sample-level random noise. Gain and phase errors shift all phasors measured at the same terminal in a coherent way, which changes the impedance quantities that feed the hybrid regressors. Hybrid RF remains the best renewable method through the severe scenario, while Hybrid ANN remains slightly more accurate than the conventional estimate at that level. This does not contradict the clean renewable results; it shows that residual correction is powerful when the measurement chain is reliable, but its advantage can shrink when systematic phasor errors distort the inputs used by the correction model.

5.9 General Conclusion

The five-fold evaluation gives a clear but balanced comparison. Random Forest achieved the lowest standalone distance error, while ANN gave the best classification accuracy. SVM remained a reliable classifier but was less competitive for distance estimation. The conventional method was not the most accurate, yet its speed and physical transparency remain important practical advantages.

Attempts to improve the models did not automatically lead to better results. PCA removed useful structure for Random Forest, and Bayesian optimization did not improve the retained SVM configuration. The noise tests also changed the ranking observed under ideal measurements: Random Forest was highly accurate on clean data but more sensitive to perturbation, whereas ANN degraded more gradually.

Renewable inverter integration had its strongest effect on the conventional estimate. Learning a residual correction from the conventional inputs reduced that error substantially, especially with Hybrid RF. The results favour a combined view of fault location: analytical methods offer interpretability, while learning methods can compensate for nonlinear effects and changing generation conditions. Direct model rankings nevertheless remain valid only when the same cases, folds, and evaluation protocol are used.

Chapter 6 Conclusion

Summary

This final chapter summarizes the main lessons of the thesis. It recalls the comparison between conventional impedance-based fault location and machine-learning methods, discusses the effect of measurement noise and renewable inverter integration, and closes with the main practical perspective: machine learning is useful, but the most credible direction is to combine it with conventional analytical methods rather than replace them completely.

This thesis studied transmission-line fault classification and distance estimation by combining conventional impedance-based calculation with machine-learning methods. The work covered the full chain from simulation to evaluation: the IEEE 39-bus New England system was modeled, operating points were selected, fault cases were generated automatically, electrical features were extracted, and all models were evaluated using five-fold out-of-fold validation.

The conventional method remained an important reference because it is fast, interpretable, and directly linked to voltage, current, sequence quantities, and line impedance. However, its accuracy can be affected by fault resistance, remote infeed, operating conditions, and renewable inverter contribution. These limits became visible when the electrical behaviour of the network changed.

The machine-learning models showed strong potential. ANN gave the best standalone classification performance, Random Forest gave the lowest standalone distance error, and SVM remained useful mainly for classification. At the same time, the PCA and Bayesian-optimization tests showed that adding more processing or tuning does not automatically improve a model. The useful method is not always the most complicated one.

The robustness tests were also important. A model can perform strongly on clean simulated signals but react differently when noise is added to the original voltage and current measurements. This means that fault-location models should not be judged only by nominal accuracy. Their behaviour under measurement perturbations must also be considered before real-world deployment.

Renewable inverter integration confirmed that modern generation can change the fault-location problem. Inverter-interfaced sources do not contribute fault current like synchronous machines, because their response depends on control limits and voltage-support behaviour. In the studied renewable cases, this affected the conventional estimate, but the hybrid approach reduced the error by using ANN or Random Forest to correct the remaining error of the analytical distance estimate.

For this reason, the main practical conclusion is that it is not realistic, for now, to rely only on machine-learning techniques in real protection applications. The models were trained and tested on simulated data from a known system, while a real grid includes sensor errors, communication delays, missing samples, uncertain parameters, topology changes, switching events, and disturbances that may not exist in the training database.

The best direction is therefore a combined one. The conventional method should remain as a transparent reference and fallback, while machine learning can be used to correct systematic errors, detect nonlinear patterns, or provide an additional estimate. If the analytical and learned estimates disagree strongly, that disagreement can itself warn the operator that the case may be unusual.

Future work should extend the renewable study to more inverter locations, penetration levels, control strategies, loading points, and network topologies. The approach should also be tested with real disturbance records and through real-time digital simulation or hardware-in-the-loop validation. More detailed robustness tests, including bias, phase-angle error, missing samples, synchronization error, current-transformer saturation, and uncertainty in line parameters, would make the evaluation closer to field conditions.

Overall, this thesis shows that machine learning can add real value to fault classification and fault-distance estimation, especially when it is used to support an analytical calculation. The strongest perspective is not to replace conventional protection knowledge, but to strengthen it with data-driven correction and uncertainty-aware decision support.

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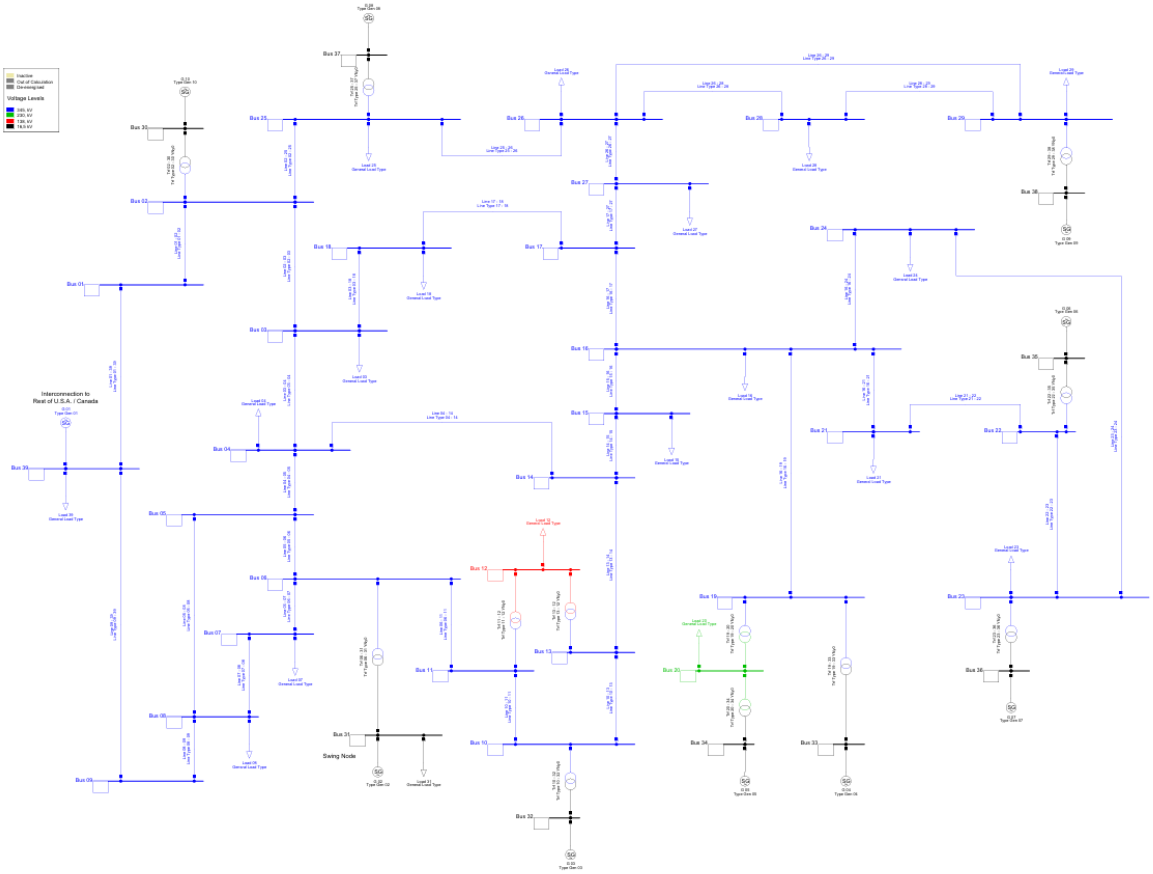
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Annexes

Annexe 1: IEEE 39-bus New England system



Annexe 2: MATLAB/Simulink system model

